

IBE and PIR from Isogenies

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Abstract. We significantly expand the reach of isogeny-based cryptography by constructing, for the first time, two fundamental primitives that have long remained out of reach from isogenies: identity-based encryption (IBE) and semi-honest single-server private information retrieval (PIR) with communication polylogarithmic in the database size. Our IBE additionally achieves anonymity, which means that the ciphertexts hide both the message and the recipient identity.

Our results are enabled by new techniques for constructing cryptographic primitives from structured isogeny computations, centered around the first construction of blind batch encryption [27, 41] from isogenies. To establish its security, we introduce a new hardness assumption, CDH with Mismatched Torsion (CDHwMT), which informally captures the hardness of a CDH-like problem [13] in the presence of structured auxiliary isogeny information. We provide evidence for its plausibility by showing that, in the Algebraic Isogeny Model [1], CDHwMT reduces to well-studied isogeny assumptions.

The same techniques yield further cryptographic applications: the first isogeny-based hierarchical IBE [51, 20, 21], laconic oblivious transfer [34], and public-key encryption simultaneously achieving high-rate leakage resilience [6, 69] and key-dependent-message/circular security [19, 12, 11]. All of our constructions can be conjectured to be post-quantum secure.

Beyond these individual primitives, our results demonstrate that isogenies can support substantially richer cryptographic functionality than previously known, and provide new tools toward building advanced post-quantum cryptography.

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1 Introduction

Identity-Based Encryption. Identity-based encryption (IBE) is a fundamental cryptographic primitive which allows users to encrypt to free-form identities, such as a name or email address, as opposed to random public keys. In more detail, in an IBE, the encrypt algorithm takes some arbitrary identity id and message μ and produces a ciphertext ct_{id} , the key generation algorithm takes some master secret and identity id' and produces a secret key sk'_{id} , and decryption of ct_{id} with sk'_{id} works to recover μ if and only if $\text{id} = \text{id}'$.

IBE as a primitive was first proposed by Shamir [73] and remained uninstigated until the seminal works by Boneh and Franklin [22] and Cocks [36], from assumptions on bilinear maps and quadratic residuosity respectively. In the post-quantum regime, constructions were provided from the learning with errors problem [50, 30, 3] and more recently from a variant of the learning parity with noise problem [41].

Private Information Retrieval. Private information retrieval (PIR) is a central cryptographic primitive that lets a client retrieve an item from a database without revealing which item it requested. More precisely, a server holds a database $z = (z_1, \dots, z_M)$, while a client holds an index $\iota \in [M]$. At the end of the protocol, the client learns z_ι , while the server's view hides ι . In this work, we focus on the *single-server* setting, where privacy must be achieved without distributing the database across multiple non-colluding servers. The trivial solution downloads the entire database, so the central efficiency goal is sublinear, ideally polylogarithmic, communication.

PIR was introduced by Chor et al. [35], who gave information-theoretic protocols using multiple non-colluding servers. In the single-server setting, however, information-theoretic privacy requires downloading the entire database, making computational assumptions necessary for nontrivial communication. Kushilevitz and Ostrovsky overcame this barrier under quadratic residuosity, giving the first single-server computational PIR protocol [63] achieving $O(M^\epsilon)$ communication for every constant ϵ . Cachin, Micali, and Stadler subsequently achieved polylogarithmic communication under the Φ -hiding assumption [29]. Later constructions obtained single-server PIR from assumptions including DCR, LWE, quadratic residuosity, DDH, and CDH [65, 28, 42, 25].

Central Primitives in Cryptography! IBE and PIR are arguably among the most central primitives in modern cryptography. IBE is not just an important and exciting primitive in its own right, it is frequently the harbinger of a new wave of advanced cryptography from a given assumption. For instance, IBE from pairings and lattices was followed by elegant constructions of stronger primitives such as attribute-based encryption [72, 56, 4, 54], broadcast encryption [23, 5, 76] and traitor tracing [24, 55] to name a few. From a technical viewpoint, it is widely accepted that the first IBE from a new assumption can indicate an overcoming of some core barrier, which in turn can open up the possibilities of using that assumption for more advanced cryptography.

Besides being a foundational primitive for private data access, PIR lies at the intersection of several central themes in cryptography and coding theory. In particular, nontrivial single-server PIR implies oblivious transfer, which is complete for general secure computation [38, 62]. PIR is also closely connected to homomorphic encryption and, in the multi-server setting, to locally decodable codes [28, 61]. From a technical viewpoint, communication-efficient single-server PIR requires computation on a hidden query while keeping communication much smaller than the database. Constructing such protocols from a new assumption can therefore reveal techniques for broader forms of communication-efficient secure computation [25].

Advanced Cryptography from Isogenies. In the post-quantum regime, lattices have enabled tremendous progress in advanced cryptography. They support fully homomorphic encryption [48, 28, 26], identity-based and attribute-based encryption [4, 54], broadcast encryption [76], traitor tracing [55], and single-server PIR with nearly optimal communication [28]. By contrast, the state of the art from code- and isogeny-based assumptions remains far more limited. In the isogeny setting, which is the focus of this work, constructions are known for only a handful of primitives. These include dual-mode public-key encryption and statistically sender-private OT [8], adaptively secure MPC [10], ring signatures [18], password-authenticated key exchange [2], blind signatures [60], and updatable encryption [64]. Against this background, two central primitives remained stubbornly out of reach: IBE and single-server PIR with polylogarithmic dependence on the database size. Recent work identifies concrete obstacles to constructing IBE from isogenies [57]. Likewise, to the best of our knowledge, no standard single-server PIR from isogeny-based assumptions has been constructed so far. Thus, a central open question is:

Can we construct IBE or single-server PIR from Isogenies?

1.1 Our Results

In this work, we answer both questions in the affirmative, marking significant progress in the field of isogeny-based cryptography. In detail, we obtain the following results:

1. *IBE from Isogenies:* We provide the first construction of IBE from isogenies (Corollary 1). As a bonus feature, our IBE satisfies *anonymity*, where ciphertexts hide their encoded identity in addition to the message.
2. *Single-server PIR from Isogenies:* We provide the first construction of semi-honest, single-server PIR in the common reference string model from isogenies (Corollary 5). In detail, for every polynomially bounded database size $M = M(\lambda)$, the total communication of our PIR protocol is at most $p(\lambda)(1 + \log M)^2$, where the fixed polynomial p is independent of M .

Our results are enabled by a new isogeny-based collision-resistant hash function supporting “shuffled-order” evaluation (i.e., its input coordinates can be processed in arbitrary order, albeit in a manner that does not violate collision resistance). We leverage this hash function to obtain the first construction of *blind batch encryption* [27, 41] from isogenies. This construction forms the common technical core of our isogeny-based anonymous IBE and single-server PIR constructions.

Security of our constructions rely on a novel assumption called the “CDH with Mismatched Torsion” (CDHwMT) assumption, which we introduce. At a high level, the assumption states the hardness of solving a CDH-like problem (similar to the CDH problem for binSIDH-hyb [13]) even when the adversary is given some additional “safe” leakage¹. We justify our assumption by showing that in the Algebraic Isogeny Model [1], the CDHwMT assumption reduces to either the well-known EndRing problem [70, 67] or to the isogeny problem with scaled torsion [13, Problem 1]. This problem, called theSSIP-A problem in [13] forms the basis of several cryptosystems [46, 16, 13, 68, 15] and has been studied as an isogeny problem with level structure in [37].

Other Applications. We leverage the same techniques to additionally obtain the first isogeny-based constructions of hierarchical IBE [51, 20, 21, 39] (an extension of IBE with additional delegation capabilities), laconic oblivious transfer [34], as well as public-key encryption that simultaneously satisfies security against high-rate key leakage [6, 69] and key-dependent message (KDM)/circular security [19, 12, 11].

All of our constructions can be conjectured to be post-quantum secure.

Perspective. We emphasize that our focus in this work is the *feasibility* of achieving new, plausibly post-quantum constructions of advanced primitives that were previously not known from isogeny-based assumptions. As discussed above, the overwhelming reliance of advanced post-quantum cryptography on lattices makes it a single point of failure – this is highly problematic and it is crucial to discover alternatives to protect against unexpected cryptanalytic breakthroughs against lattices.

The broad research agenda pursued by the present work is, therefore, to increase the cryptographic capabilities of isogeny based assumptions, without concerning ourselves, for the moment, about concrete efficiency.

1.2 Organization

The rest of the paper is organized as follows. [Section 2](#) provides an overview of our techniques. [Section 3](#) introduces notation and preliminary background material. [Section 5](#) introduces the CDHwMT hardness assumption and analyzes

¹ For the informed reader, the adversary is additionally given one of two secret isogenies, but these are of different degrees from the order of the torsion points revealed in the public keys making it hard to exploit

it in the Algebraic Isogeny Model. [Section 4](#) presents a variant of the CGL hash function that we use to build BE. [Section 6](#) presents our construction of BE from the CDHwMT assumption. Finally, [Section 7](#) presents our construction of single-server PIR from the CDHwMT assumption and one-way functions.

2 Technical Overview

Our starting point is the work of Brakerski, Lombardi, Segev, and Vaikuntanathan [27] who, building upon the work of [40, 34, 39] showed that to construct IBE, it suffices to construct a much simpler primitive called Batch Encryption (abbreviated as BE henceforth)². In fact, the authors of [27] demonstrated that BE is a versatile cryptographic gadget with applications beyond IBE, such as laconic oblivious transfer [34], as well as public-key encryption that simultaneously satisfies security against high-rate key leakage [6, 69] and key-dependent message (KDM)/circular security [19, 12, 11]. Additionally, [27] showed that a strengthening of BE, called *blind* BE, implies the stronger notion of *anonymous* IBE. In this paper, we introduce a novel blind BE scheme from isogenies, which is the common technical core of our two primary results: anonymous IBE and single-server PIR. We describe this next.

2.1 Blind Batch Encryption from Isogenies

At a high level, a BE scheme allows encryption against a hash value, say h , and a position, bit pair, say (i, c) so that decryption works iff (i) one knows the preimage $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$ of the hash h , and (ii) the i^{th} bit of the preimage is set as $x_i = c$. In more detail, a BE scheme is defined as follows:

- Setup**($1^\lambda, n$) : A setup algorithm that takes as input the security parameter λ and an input length parameter $n = \text{poly}(\lambda)$, and outputs some public parameters pp .
- Gen**($\text{pp}, \mathbf{x}$) : A projection algorithm takes as input a key pp , an input string $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$ and outputs a projection/hash value h . We remark that it is crucial that h can be represented using $\text{poly}(\lambda)$ bits for some fixed polynomial that is *independent* of the length n of the input vector $\mathbf{x}$.
- Enc**($\text{pp}, (h, i), (\mu_0, \mu_1)$) : An encryption algorithm that takes as input pp , a hash value h , an index $i \in [n]$, and a message-pair (μ_0, μ_1) , and outputs a ciphertext ct .
- Dec**($\text{pp}, (\mathbf{x}, i), \text{ct}$) : A decryption algorithm takes as input a key pp , an input $\mathbf{x}$, an index $i \in [n]$, and a ciphertext ct , and outputs a candidate message μ' .

Throughout this overview, we assume that the messages μ_0 and μ_1 are single bits (this is consistent with the treatment of BE in [27]). The extension to multi-bit messages follows via a simple hybrid argument.

² In a concurrent work, Dottling, Garg, Hajiabadi and Masny [41] introduced a primitive called Hash Encryption (HE) and showed that it also suffices to build IBE. It is folklore that BE and HE are essentially equivalent primitives.

We informally outline the correctness, succinctness, security and blindness requirements of BE below. Given a valid preimage $\mathbf{x} = (x_1, \dots, x_n)$ to the hash value h used for encryption:

- Correctness requires that the decryption procedure recovers the plaintext message μ_{x_i} .
- Succinctness requires that the size of the hash value h is upper-bounded by some fixed polynomial in the security parameter λ , that is *independent* of the input size parameter n .
- Security requires that μ_{1-x_i} remains hidden.
- The ciphertext resulting from encrypting a uniformly random plaintext message (unknown to the adversary) is indistinguishable from a uniformly random bit-string of the same length.³

We refer to [Section 3.6](#) for a more formal treatment of each of these notions.

Seeking a Suitable Projection/Hash from Isogenies. In isogeny-based cryptography, the main (and possibly the only) hash function is the CGL hash function [33]. In CGL, each bit of the input is mapped to a 2-isogeny so that hashing an input corresponds to taking a walk in the supersingular isogeny graph. The starting point is a supersingular elliptic curve E_0 , together with two points $P_{0,0}, P_{0,1}$ of order 2, which make up the hash public parameters. The computation is iterative: the first bit x_0 of the input $\mathbf{x}$ is mapped to a 2-isogeny $\phi_1 : E_0 \rightarrow E_1 := E_0 / \langle P_{0,x_0} \rangle$. On the curve E_1 , there are three points of order 2: one corresponds to the dual of ϕ_1 , which means it corresponds to the backtracking isogeny. The other two points can be deterministically labeled as $P_{1,0}, P_{1,1}$. The second bit x_1 of the input $\mathbf{x}$ is then mapped to the isogeny $\phi_2 : E_1 \rightarrow E_2 := E_1 / \langle P_{1,x_1} \rangle$; the process is then repeated until all the n bits of the input are consumed, and the end curve E_n is the output of the hash. In [44], it was shown that the collision resistance of the CGL hash function reduces to the one-endomorphism problem, which is equivalent to the endomorphism ring problem [70].

Shuffled Order Hash Evaluation. Previous constructions of BE and related primitives [40, 27, 41] rely on a structured hash function, where it is possible, in a certain specific sense, to process the input bits in a different *sequence* than their sequence in the input. In more detail, these hash functions can view the input as an unordered list $\{(1, x_1), (2, x_2), \dots, (n, x_n)\}$, and then process the elements in this list in arbitrary order. Since the input bits are tied to their position, this does not create a problem for collision resistance. As an example, the lattice-based hash from [41] is $\text{Hash}(\mathbf{x}) = \mathbf{A}\mathbf{x} \bmod q$ for a wide matrix $\mathbf{A}$ of appropriate dimensions over $\mathbb{Z}_q$, and input vector $\mathbf{x}$. It is easy to see that the hash, whether computed as $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_n\mathbf{a}_n$ or in a different sequence such

³ Following [27], we actually require a slight relaxation of the above blindness condition, where only the message-dependent part of the ciphertext is required to be uniformly random.

as $x_n \mathbf{a}_n + x_{n-1} \mathbf{a}_{n-1} + \dots + x_1 \mathbf{a}_1$, results in the same final hash so long as the bit x_i remains tied to the column $\mathbf{a}_i$. We refer to this property as “shuffled order hash evaluation”.

We remark that this property seems necessary to construct BE, where one needs to tie the ciphertext to a particular (i, c) value and allow decryption only when $x_i = c$. For instance, in the construction of BE using the above lattice hash, the ciphertext is designed as:

$$\text{ct}_0 = \mathbf{s}^\top \underbrace{\mathbf{A}_{-i}}, \quad \text{ct}_1 = \mathbf{s}^\top \underbrace{(h - c \cdot \mathbf{a}_i)} + \lfloor \frac{q}{2} \rfloor \mu$$

where $h = \mathbf{A}\mathbf{x} \bmod q$, $\mathbf{A}_{-i}$ is the matrix $\mathbf{A}$ with the i^{th} column removed, $\mathbf{s}$ is a random vector over $\mathbb{Z}_q$ of appropriate dimensions which serves as the LWE secret, μ is the message bit and $\underline{Z}$ refers to a noisy version of Z where the precise value of the noise is not important. Given a valid preimage $\mathbf{x}$ whose i^{th} bit is set as c , it is easy to see how decryption would work:

$$\begin{aligned} \text{ct}_1 - \text{ct}_0 \mathbf{x}_{-i} &= \lfloor \frac{q}{2} \rfloor \mu + \mathbf{s}^\top (\underbrace{\mathbf{A}\mathbf{x} - c \cdot \mathbf{a}_i}) - (\mathbf{s}^\top \underbrace{\mathbf{A}_{-i}}) \mathbf{x}_{-i} \\ &= \lfloor \frac{q}{2} \rfloor \mu + \mathbf{s}^\top (\underbrace{\mathbf{A}\mathbf{x} - x_i \cdot \mathbf{a}_i - \mathbf{A}_{-i} \mathbf{x}_{-i}}) \end{aligned}$$

The above allows to recover μ by checking whether the resultant value is closer to 0 or $\frac{q}{2}$ so long as the noise stays within bounds.

The crucial observation to make above is that the shuffled order hash evaluation property was essential in making the scheme work, since the contribution to h by the i^{th} position of $\mathbf{x}$ needs to be “separated out” during encryption and later “merged back” during decryption given the right preimage. Therefore, the question is whether the CGL hash can satisfy the desired property.

Building a New Hash. Unfortunately, the answer is no. Processing of the input bits while computing the CGL hash is inherently sequential and cannot be “shuffled”. To see this, note that the computation of the i -th isogeny depends on the labeling of points on the curve E_{i-1} (i.e., deciding which point is $P_{i-1,0}$ and which one is $P_{i-1,1}$). This means that skipping the i -th bit computation changes all the subsequent computations, which makes it impossible to process input bits out of order.

To overcome this limitation, we propose a new hash function. Since it is sufficient for BE, we take our hash function to have fixed input length $n = \text{poly}(\lambda)$. Our hash function is based on the CGL hash function, with one fundamental change, described below.

In the original CGL function, each bit is mapped to a 2-isogeny. In our hash function, each bit is mapped to a d_i -isogeny, with all d_i pairwise coprime. This guarantees that isogenies associated with different bits commute and thus can be computed in any order. As an added benefit, all point labeling happens on the curve E_0 . In other words, our hash function has as public parameters a curve E_0 , together with n pairs of points $\{(P_{i,0}, P_{i,1})\}_{i \in [n]}$, where each pair $(P_{i,0}, P_{i,1})$

forms a basis of $E_0[d_i]$. Given an input $\mathbf{x} = (x_1, \dots, x_n)$, the hash function thus computes the isogeny $\phi_{\mathbf{x}} : E_0 \rightarrow E_{\mathbf{x}}$ with the kernel given by

$$\ker(\phi_{\mathbf{x}}) = \oplus_{i \in [n]} \langle P_{i, x_i} \rangle$$

To make our hash function compatible with a BE, we make two further changes:

- We set all $d_i = O(2^{2\lambda})$. This is essential for the security of the BE scheme, but it also means that the points $P_{i,b}$ cannot be rational over $\mathbb{F}_{p^2}$. To ensure that these points can be efficiently represented and the isogeny $\phi_{\mathbf{x}}$ can be efficiently computed, we choose all d_i 's to be powersmooth (i.e. the product of distinct primes less than a bound B). This guarantees that each $P_{i,b}$ can be written as the sum of points of smaller order, each defined over a polynomially-sized extension field of $\mathbb{F}_{p^2}$ (see [Section 4.3](#) for a detailed discussion on the efficiency of such a representation).
- To make the hash function work within the BE framework, we need to add the image of $\phi_{\mathbf{x}}$ on a torsion basis to the hash output. We fix a basis R_0, S_0 of $E_0[N]$, with $N = O(2^{2\lambda})$ coprime to all d_i 's. The hash output is then the curve $E_{\mathbf{x}}$, together with the images $\phi_{\mathbf{x}}(R_0), \phi_{\mathbf{x}}(S_0)$. Note that revealing such information does not make finding a preimage easier, because the degree of the isogeny $\phi_{\mathbf{x}}$ is

$$\deg(\phi_{\mathbf{x}}) = \prod_{i \in [n]} d_i = O(2^{2n\lambda}),$$

which is much greater than the order $N = O(2^{2\lambda})$ of the points R_0, S_0 .

To summarize:

- Our hash function starts from a curve E_0 and a set of points $\{(P_{i,0}, P_{i,1})\}_{i \in [n]}$, where each pair $(P_{i,0}, P_{i,1})$ forms a basis of $E_0[d_i]$.
- The input $\mathbf{x} = (x_1, x_2, \dots, x_n)$ is mapped to the isogeny $\phi_{\mathbf{x}} : E_0 \rightarrow E_{\mathbf{x}}$ of kernel $\ker(\phi_{\mathbf{x}}) = \oplus_{i \in [n]} \langle P_{i, x_i} \rangle$.
- The hash output is the tuple $(E_{\mathbf{x}}, \phi_{\mathbf{x}}(R_0), \phi_{\mathbf{x}}(S_0))$.

We show that this hash function is collision resistant under the hardness of the endomorphism ring problem in [Section 4.4](#), following a similar strategy as the proof of collision resistance for the CGL hash function.

Building the Batch Encryption. We now turn to building BE. One way to view BE is as a public-key encryption scheme where a hash value plays the role of the public key and a hash preimage (with a specific bit at a prescribed location) plays the role of the secret key. Thus, to turn our new hash function into a BE scheme, we start from building a PKE where the isogeny $\phi_{\mathbf{x}}$ computed during the hash evaluation can serve as the role of a secret key.

Luckily, the binSIDH [\[13\]](#) offers precisely what we need: binSIDH is a key exchange protocol where the public parameters include a curve E_0 and a list of points $\{(P_{i,0}, P_{i,1})\}_{i \in [n]}$. A binSIDH secret key is a binary vector $\mathbf{b} = (b_0, \dots, b_n) \in \{0, 1\}^n$, whose corresponding secret isogeny has kernel generated by $\bigoplus_{i \in [n]} \langle P_{i, b_i} \rangle$.

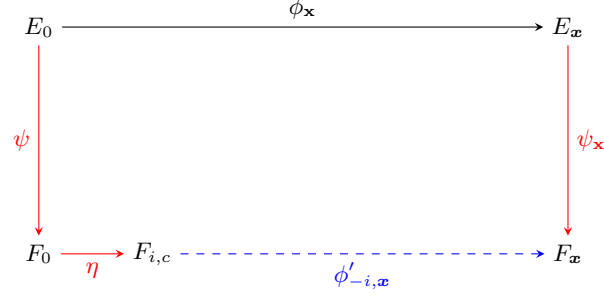


Fig. 1. A simplified overview of our isogeny-based BE scheme. The black arrow indicates the hash isogeny $\phi_{\mathbf{x}}$ (associated with the hash input $\mathbf{x}$), while the red arrows indicate isogenies computed by the encryptor. The isogeny η has degree d_i and it is what binds the ciphertext to be decryptable only by preimages where $x_i = c$. The blue dashed arrow indicates the isogeny computed by the decryptor.

To construct a PKE, we rely on the hybrid variant of binSIDH⁴ [13, Section 5] (with modified parameters). In this PKE derived from binSIDH-hyb, to encrypt a one-bit message m , and given (E_0, R_0, S_0) in the public parameters and the hash value $(E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}})$, the encryptor samples a random value $r \in [0, N - 1]$ and computes the isogenies $\psi : E_0 \rightarrow F_0$ and $\psi_{\mathbf{x}} : E_{\mathbf{x}} \rightarrow F_{\mathbf{x}}$ with kernels:

$$\ker(\psi) = \langle R_0 + [r]S_0 \rangle, \quad \ker(\psi_{\mathbf{x}}) = \langle R_{\mathbf{x}} + [r]S_{\mathbf{x}} \rangle$$

Based on this idea, a ciphertext then consists of:

- the curve F_0 ,
- the scaled images $[\alpha_{i,b}] \psi(P_{i,b})$ (where $\alpha_{i,b} \leftarrow (\mathbb{Z}/\text{ord}(P_{i,b}))^*$), and
- $\text{gl-hc}(F') \oplus m$, where gl-hc is a Goldreich-Levin hardcore predicate [53] (see Section 3.8 for a more formal exposition) and $m \in \{0, 1\}$ is the encrypted one-bit message.⁵

To decrypt, it is sufficient to recompute $F_{\mathbf{x}}$ as the codomain of the isogeny that has kernel generated by

$$\bigoplus_{i \in [n]} [\alpha_{i,b_i}] \psi(P_{i,b_i})$$

where $\mathbf{b} = (b_0, \dots, b_n)$ is the secret key.

⁴ This choice is out of simplicity, as binSIDH-hyb is a simpler protocol than the original binSIDH. Any other choice would work equally well.

⁵ We are using hardcore predicates and one-bit messages to obtain a theoretically clean construction that does not rely on the random oracle model or decisional assumptions; to encrypt a λ -bit message, it is sufficient to repeat this process λ times. A variant of PKE that relies on the ROM or a DDH-like assumption can natively support long messages.

Are We Done? While we have a PKE whose secret key is a hash preimage, we do not have a BE scheme yet: a BE scheme requires the encryptor to bind the ciphertext encrypting a pair of message-bits $(\mu_0, \mu_1) \in \{0, 1\} \times \{0, 1\}$ to a hash preimage string such that, if the i -th bit of the preimage is $c \in \{0, 1\}$, then the decryptor can only recover μ_c , while μ_{1-c} should remain hidden. To add this binding, we rely on the property that our hash function can be computed in any order: the encryptor can thus already compute the i -th step since they know the i -th bit of the preimage should be either $c = 0$ for decryption to recover μ_0 , or $c = 1$ for decryption to recover μ_1 .

To this end, the encryptor effectively creates two independently generated ciphertexts, one encrypting μ_0 and the other encrypting μ_1 . In more detail, for each $c \in \{0, 1\}$, to encrypt μ_c , after independently computing a ciphertext as in the PKE described above, the encryptor computes the isogeny $\eta : F_0 \rightarrow F_{i,c}$ of kernel $\langle \psi(P_{i,c}) \rangle$, and generates a ciphertext ct_c of the form

$$\text{ct}_c = \{F_{i,c}, \{Q_{j,b,c} = [\alpha_{j,b,c}] \eta \psi(P_{j,b})\}_{\{j \in [n] \setminus \{i\}, b \in \{0,1\}\}}, \text{gl-hc}(F_{\mathbf{x}}, c) \oplus \mu_c\}.$$

where the subscript c indicates that the corresponding components are generated independently for ct_0 and ct_1 . The final ciphertext is $\text{ct} = (\text{ct}_0, \text{ct}_1)$.

Crucially, the ciphertext ct_c does not include either F_0 or the d_i -torsion points $Q_{i,0,c}$ and $Q_{i,1,c}$, which makes it hard for anyone that does not know the right preimage $\mathbf{x} = (x_1, \dots, x_n)$ such that $x_i = c$ to compute the corresponding curve $F_{\mathbf{x},c}$. In other words, if the adversary knows a preimage $\mathbf{x}$ such that $x_i = c$, then it cannot compute $F_{\mathbf{x},1-c}$. A visual summary of the BE scheme is provided in Fig. 1. Please see Section 6 for a detailed description.

Security. Formally, the security of our BE scheme relies on a new assumption, which we call CDH with Mismatched Torsion (CDHwMT; Definition 15), that states the hardness of solving CDH even when the adversary knows an isogeny between two curves, but this is of different degree than the order of the revealed torsion points. We analyze this assumption in the Algebraic Isogeny Model [1], where we show it is equivalent to the security of the underlying PKE (binSIDH-hybrid) and it reduces to the hardness of a DLOG-type assumption (the isogeny with scaled torsion assumption, which is relied on by a wide range of cryptosystems [32, 46, 16, 13, 68, 15]). Please see Section 5.2 for more details.

Blindness. To see why our scheme satisfies blindness, note that, for each $c \in \{0, 1\}$ we can write the ciphertext ct_c of our batch encryption scheme as a tuple of sub-ciphertext components of the form $(\text{subct}_{1,c}, \text{subct}_{2,c})$ where:

- $\text{subct}_{1,c} = (F_{i,c}, \{Q_{j,b,c}\}_{\{j \in [n] \setminus \{i\}, b \in \{0,1\}\}})$ is independent of both the hash value h and the message μ_c .
- $\text{subct}_{2,c} = \text{gl-hc}(F_{\mathbf{x}}, c) \oplus \mu_c$ is a uniformly random bit whenever μ_c is uniformly random and unknown to a distinguisher.

Applications. As discussed above, the work of [27] shows that blind batch encryption suffices to obtain anonymous IBE, laconic oblivious transfer [34], as

well as public-key encryption that simultaneously satisfies security against high-rate key leakage [6, 69] and key-dependent message (KDM)/circular security [19, 12, 11]. Additionally, [39] demonstrates a generic transformation from IBE to hierarchical IBE – an extension of IBE that allows identities to form a hierarchy, where a user at any level can delegate secret keys to its descendant identities without involving the master authority [51, 20, 21]. Thus, we obtain all of these applications from the CDHwMT assumption.

2.2 Single-Server PIR from Isogenies

Our second primary result is a semi-honest single-server PIR protocol for every polynomially bounded database size. Its communication depends only polylogarithmically on the database size. The protocol is in the common reference string model and relies on CDHwMT. Our starting point is the work of Boyle, Couteau and Meyer [25], who observed that a suitable laconic private set intersection (LPSI) protocol can be transformed into a weaker primitive called *half-PIR*, which in turn can be generically bootstrapped to full PIR using random-index PIR [49]. We use the following sequence:

$$\text{Our blind BE} \implies \text{LPSI with sender security} \implies \text{half-PIR} \implies \text{PIR}.$$

The main technical challenge lies in the first implication. We adapt the Merkle-tree-based LPSI construction of Alapati et al. [7] to our isogeny-based BE scheme. This adaptation is not black-box: two properties used in [7] are unavailable for our underlying hash and BE ciphertexts, and overcoming these incompatibilities requires additional structure from our construction. We present an overview of our approach below. See Section 7 for the detailed treatment.

Laconic PSI via a Merkle Tree. We first recall the high-level idea behind the LPSI construction of [7]. The receiver holds a set S_R of M elements, while the sender holds a single element id . The receiver arranges its set at the leaves of a binary tree of depth $O(\log M)$ and recursively hashes the tree upwards, eventually sending only the succinct root digest to the sender. The sender must now produce a short message that enables the receiver to determine whether $\text{id} \in S_R$, without revealing id when it is not contained in the receiver’s set.

The sender’s response consists of a sequence of garbled circuits, one per tree level. Conceptually, evaluation starts from the root and is routed toward the unique leaf selected by comparisons involving the sender’s element id . At an internal node, the corresponding garbled circuit determines which child should be followed and outputs encryptions of the wire labels needed to evaluate the garbled circuit at that child. The encryptions are implemented using BE in its hash-encryption interpretation: the hash of the child serves as the encryption target, while a preimage of that hash allows the receiver to recover exactly one label from every wire-label pair. Repeating this process descends through the tree. At the leaf, a final garbled circuit checks whether the routed leaf actually contains id , and outputs a distinguished random value r if so. Thus, along

the correctly routed path, BE correctness and garbling correctness ensure that evaluation reaches the appropriate leaf and produces r exactly when $\text{id} \in S_R$.

A subtle issue is that the receiver can attempt to evaluate the sender’s message along *every* root-to-leaf path, not just the one selected by the sender’s comparisons. For any non-routed path, however, there is a first level at which the receiver attempts to decrypt a BE ciphertext using a preimage different from the one targeted by the sender. BE security lets us replace the corresponding complementary plaintext rows by independent random values. These values translate into invalid/random garbled labels at the next level. We prove an additional property of the point-and-permute garbling scheme—*invalid-label propagation*—showing that once such a random label enters the evaluation and syntactically influences every output, the resulting garbled output becomes pseudorandom as well. Iterating this argument down the tree ensures that every non-routed leaf produces an essentially uniform value and hence matches the distinguished tag r only with negligible probability.

Why the Construction of [7] Does Not Directly Apply. There are two important differences between our setting and the instantiation of [7]. First, the receiver-privacy argument of [7] relies on a statistical *robustness* property of the underlying hash function: hashing an input containing sufficiently many random coordinates should produce a value statistically close to a uniform bit string. Our isogeny-based hash output consists of structured algebraic objects—an elliptic curve together with torsion points—and therefore cannot be statistically close to a uniform bit string. Consequently, we do not obtain receiver-private LPSI. Fortunately, this property is not needed for our PIR application: in the half-PIR transformation, the LPSI receiver is precisely the PIR server, and the receiver’s set is derived from the server’s own database. PIR imposes no requirement that the server’s database be hidden from itself. Thus, *sender security of LPSI is sufficient for our purposes*.

The second obstacle is more technical. The sender-security proof of [7] relies on blind garbling together with the fact that the hash-encryption ciphertexts produced by the garbled circuits become pseudorandom when their plaintexts are random. Our BE ciphertexts do not have this property: they contain elliptic curves and torsion points, and are therefore efficiently recognizable as structured objects. Hence, a garbled circuit that directly outputs a complete BE ciphertext cannot itself have a pseudorandom output, preventing a direct application of blind-garbling security.

Decomposable Blindness to the Rescue. We overcome this second obstacle by exploiting additional structure already implicit in the blindness proof of our BE scheme. A ciphertext in our construction naturally decomposes into two pieces. The first is an *offline component*, consisting of the structured isogeny-based objects; crucially, this component depends only on the public parameters, the encrypted position, and fresh randomness, but is independent of both the target hash value and the encrypted plaintexts. All dependence on the target and

plaintexts is confined to a very short *online component* consisting of Goldreich–Levin masked bits. Moreover, if the two BE plaintext rows are independently uniform, then these online bits are themselves information-theoretically uniform, even conditioned on the offline component. We refer to this structural property as *decomposable blindness*.

This decomposition lets us reorganize the LPSI construction so that the garbled circuits compute *only the online components* of the BE ciphertexts. The much larger structured offline components are generated separately and transmitted in the clear. After a BE-security hybrid replaces the unused plaintext rows by independent uniform bits, the corresponding garbled-circuit outputs are therefore genuine uniform bit strings; blind garbling can now be applied exactly where needed. Importantly, the resulting sender message need not itself be pseudorandom. Instead, we show that it is computationally indistinguishable from a fixed efficiently sampleable distribution that is independent of the sender’s element. This weaker simulation-based property is sufficient for LPSI sender security and, in turn, for the PIR transformation.

From LPSI to Half-PIR and PIR. Once sender-secure LPSI is available, we follow the transformation of [25] to bootstrap LPSI into half-PIR with communication $p_{\text{half}}(\lambda)(1 + \log M)$, for a fixed polynomial p_{half} independent of the database size M . Finally, we apply the generic half-PIR-to-PIR transformation of [25], which first amplifies half-PIR through random-index PIR [49] and then de-randomizes the queried index. The resulting semi-honest single-server PIR has total communication $p(\lambda)(1 + \log M)^2$ for a fixed polynomial p independent of the database size M . This bound holds for every polynomially bounded database size M . Thus, our construction yields, to the best of our knowledge, the first single-server PIR with polylogarithmic dependence on the database size from a plausibly post-quantum isogeny-based assumption and one-way functions.

3 Preliminaries

Notation. For any positive integer n , $[n]$ denotes the set $\{1, \dots, n\}$. For any pair of integers m and n such that $m < n$, $[m, n]$ denotes the set $\{m, \dots, n\}$. Throughout, λ denotes the security parameter. Also, throughout, $\text{poly}(\lambda)$ and $\text{negl}(\lambda)$ are used to denote any generic (unspecified) polynomial and negligible function in λ , respectively. Finally, for any finite set S , $s \leftarrow S$ denotes sampling from the uniformly random distribution over the set S .

3.1 (Anonymous) Identity Based Encryption

An identity-based encryption (IBE) scheme is defined as a quadruple $\text{IBE} = (\text{Setup}, \text{KeyGen}, \text{Enc}, \text{Dec})$ of probabilistic polynomial-time algorithms:

- $\text{Setup}(1^\lambda, n) \rightarrow (\text{mpk}, \text{msk})$: The setup algorithm takes as input the (unary representation of the) security parameter λ and the identity length parameter $n = \text{poly}(\lambda)$, and outputs a master public key mpk and a master secret key msk .

- $\text{Enc}(\text{mpk}, \text{id}, \mu) \rightarrow \text{ct}$: The encryption algorithm takes as input a master public key mpk , an identity $\text{id} \in \{0, 1\}^n$, and a message bit $\mu \in \{0, 1\}$. It outputs a ciphertext ct .
- $\text{KeyGen}(\text{mpk}, \text{msk}, \text{id}) \rightarrow \text{sk}$: The key generation algorithm takes as input the master public key mpk , the master secret key msk , and an identity $\text{id} \in \{0, 1\}^n$. It outputs a private key sk .
- $\text{Dec}(\text{mpk}, \text{ct}, \text{sk}) \rightarrow \text{msg}$ or $\perp$: The decryption algorithm takes as input the master public key mpk , a ciphertext ct , a private key sk , and outputs a message $\mu' \in \{0, 1\}$ or $\perp$.

Definition 1 (Correctness of IBE). *An IBE scheme for a class of identities $\mathcal{ID}_{\lambda, n}$ is correct if for any security parameter $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, any $\text{id} \in \{0, 1\}^n$, and for any message $\mu \in \{0, 1\}$,*

$$\Pr \left[\begin{array}{l} (\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda, n), \\ \text{sk} \leftarrow \text{KeyGen}(\text{mpk}, \text{msk}, \text{id}), \\ \text{ct} \leftarrow \text{Enc}(\text{mpk}, \text{id}, \mu) : \\ \text{Dec}(\text{mpk}, \text{ct}, \text{sk}) \neq \mu \end{array} \right] \leq \text{negl}(\lambda)$$

where the probability is taken over the coins of Setup , KeyGen , and Enc .

Definition 2 (Security of IBE). *For a security parameter $\lambda \in \mathbb{N}$, a parameter $n = \text{poly}(\lambda)$, an IBE scheme $\text{IBE} = \{\text{Setup}, \text{Enc}, \text{KeyGen}, \text{Dec}\}$ for a class of identities $\mathcal{ID}_{\lambda, n}$, any $Q_0, Q_1 = \text{poly}(\lambda)$, and a stateful, probabilistic polynomial-time (PPT) adversary $\mathcal{A}$, let us define a security experiment as follows.*

1. **Setup phase:** On input the security parameter λ and the identity-length parameter $n = \text{poly}(\lambda)$, the challenger samples $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda, n)$ and sends mpk to $\mathcal{A}$.
2. **Key Query phase-1:** At this point, $\mathcal{A}$ chooses its identities $\{\text{id}_i \in \mathcal{ID}_{\lambda, n}\}_{i \in [Q_0]}$, and sends these identities to the challenger. For each key query, the challenger replies with $\text{sk}_i \leftarrow \text{KeyGen}(\text{mpk}, \text{msk}, \text{id}_i)$.
3. **Choosing the challenge identity:** Here, $\mathcal{A}$ chooses a challenge identity $\text{id}^* \in \{0, 1\}^n$ such that $\text{id}^* \neq \text{id}_i$ for any of the queried identities $\{\text{id}_i\}_{i \in [Q_0]}$, and sends id^* to the challenger.
4. **Challenge phase:** On input id^* , the challenger samples a random bit $b \leftarrow \{0, 1\}$ and computes $\text{ct} \leftarrow \text{Enc}(\text{mpk}, \text{id}^*, b)$. It gives (mpk, ct) to $\mathcal{A}$.⁶
5. **Key Query phase-2:** At this point, $\mathcal{A}$ chooses its identities $\{\text{id}_j \in \mathcal{ID}_{\lambda, n}\}_{j \in [Q_1]}$, such that $\text{id}_j \neq \text{id}^*$ and sends these identities to the challenger. For each key query, the challenger replies with $\text{sk}_j \leftarrow \text{KeyGen}(\text{mpk}, \text{msk}, \text{id}_j)$.
6. **Output phase:** $\mathcal{A}$ outputs a guess bit $b' \in \{0, 1\}$ as the output of the experiment.

⁶ Since the message space is $\{0, 1\}$, this is equivalent to the standard security definition of IBE where the adversary $\mathcal{A}$ chooses its own challenge messages; we simply assume without loss of generality that the challenge messages are $\mu_0 = 0$ and $\mu_1 = 1$.

We define the advantage $\text{Adv}_{\text{IBE}, \mathcal{A}}^{\text{ind}}(1^\lambda, n)$ of the adversary $\mathcal{A}$ in the above security experiment as

$$\text{Adv}_{\text{IBE}, \mathcal{A}}^{\text{ind}}(1^\lambda, n) = |\Pr[\mathcal{A} \text{ outputs } 1 \mid b = 0] - \Pr[\mathcal{A} \text{ outputs } 1 \mid b = 1]|.$$

The IBE scheme IBE is said to satisfy **ind** security if for any security parameter $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, any $Q_0, Q_1 = \text{poly}(\lambda)$, and any stateful, PPT adversary $\mathcal{A}$, we have

$$\text{Adv}_{\text{IBE}, \mathcal{A}}^{\text{ind}}(1^\lambda, n) \leq \text{negl}(\lambda).$$

Definition 3 (Anonymous Security of IBE). For a security parameter $\lambda \in \mathbb{N}$, a parameter $n = \text{poly}(\lambda)$, an IBE scheme $\text{IBE} = \{\text{Setup}, \text{Enc}, \text{KeyGen}, \text{Dec}\}$ for a class of identities $\mathcal{ID}_{\lambda, n}$, any $Q_0, Q_1 = \text{poly}(\lambda)$, and a stateful, probabilistic polynomial-time (PPT) adversary $\mathcal{A}$, let us define another security experiment as follows.

1. **Setup phase:** On input the security parameter λ and the identity-length parameter $n = \text{poly}(\lambda)$, the challenger samples $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda, n)$ and sends mpk to $\mathcal{A}$.
2. **Key Query phase-1:** At this point, $\mathcal{A}$ chooses its identities $\{\text{id}_i \in \mathcal{ID}_{\lambda, n}\}_{i \in [Q_0]}$, and sends these identities to the challenger. For each key query, the challenger replies with $\text{sk}_i \leftarrow \text{KeyGen}(\text{mpk}, \text{msk}, \text{id}_i)$.
3. **Choosing the challenge identity:** Here, $\mathcal{A}$ chooses a pair of challenge identities $\text{id}_0^*, \text{id}_1^* \in \{0, 1\}^n$ such that $\text{id}_0^* \neq \text{id}_i$ and $\text{id}_1^* \neq \text{id}_i$ for any of the queried identities $\{\text{id}_i\}_{i \in [Q_0]}$, and sends $(\text{id}_0^*, \text{id}_1^*)$ to the challenger.
4. **Challenge phase:** On input $(\text{id}_0^*, \text{id}_1^*)$, the challenger samples a random bit $b \leftarrow \{0, 1\}$ and computes $\text{ct} \leftarrow \text{Enc}(\text{mpk}, \text{id}_b^*, b)$. It gives (mpk, ct) to $\mathcal{A}$.⁷
5. **Key Query phase-2:** At this point, $\mathcal{A}$ chooses its identities $\{\text{id}_j \in \mathcal{ID}_{\lambda, n}\}_{j \in [Q_1]}$, such that $\text{id}_j \neq \text{id}_0^*$ and $\text{id}_j \neq \text{id}_1^*$, and sends these identities to the challenger. For each key query, the challenger replies with $\text{sk}_j \leftarrow \text{KeyGen}(\text{mpk}, \text{msk}, \text{id}_j)$.
6. **Output phase:** $\mathcal{A}$ outputs a guess bit $b' \in \{0, 1\}$ as the output of the experiment.

We define the advantage $\text{Adv}_{\text{IBE}, \mathcal{A}}^{\text{anon-ind}}(1^\lambda, n)$ of the adversary $\mathcal{A}$ in the above security experiment as

$$\text{Adv}_{\text{IBE}, \mathcal{A}}^{\text{anon-ind}}(1^\lambda, n) = |\Pr[\mathcal{A} \text{ outputs } 1 \mid b = 0] - \Pr[\mathcal{A} \text{ outputs } 1 \mid b = 1]|.$$

The IBE scheme IBE is said to satisfy **anonymous-ind** (or **anon-**) security if for any security parameter $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, any $Q_0, Q_1 = \text{poly}(\lambda)$, and any stateful, PPT adversary $\mathcal{A}$, we have

$$\text{Adv}_{\text{IBE}, \mathcal{A}}^{\text{anon-ind}}(1^\lambda, n) \leq \text{negl}(\lambda).$$

⁷ Again, since the message space is $\{0, 1\}$, this is equivalent to the adversary $\mathcal{A}$ choosing its own challenge messages.

3.2 Private Information Retrieval and Variants

We first recall the standard notion of (single-server) PIR [35, 63], formulated as an interactive protocol in the Common Reference String (CRS) model.

Definition 4 (Private Information Retrieval). *A PIR protocol is an interactive protocol between a server S holding a database $z \in \{0, 1\}^M$ and a client C holding an index $\iota \in [M]$, both additionally receiving a common reference string $\text{crs} \leftarrow \text{GenCRS}(1^\lambda)$, at the end of which the client produces an output bit. We require:*

- **Correctness:** *for every $z \in \{0, 1\}^M$ and $\iota \in [M]$, the client outputs z_ι with probability at least $1 - \text{negl}(\lambda)$, over crs and the coins of both parties.*
- **Client privacy (semi-honest):** *for every $z \in \{0, 1\}^M$ and every $\iota_0, \iota_1 \in [M]$, the view of the server (its input, coins, crs , and all messages received) when the client holds ι_0 is computationally indistinguishable from its view when the client holds ι_1 .*
- **Communication:** *the total communication is at most $\text{poly}(\lambda, \log M)$.*

Half-PIR. We also recall the notion of half-PIR, which was introduced as a building block for realizing standard PIR in [25].

Definition 5 (Half-PIR [25, Definition 25]). *A half-PIR protocol has the syntax of Definition 4, except that the client outputs a pair $(v, \text{flag}) \in \{0, 1\} \times \{\text{success}, \text{fail}\}$. We require the following.*

- **Correctness with noticeable success:** *for every $z \in \{0, 1\}^M$ and $\iota \in [M]$,*

$$\Pr[\text{Out}_C(1^\lambda, z, \iota) = (z_\iota, \text{success})] \geq \frac{1}{2} - \text{negl}(\lambda).$$

- **Correctness on success:** *for every $z \in \{0, 1\}^M$ and $\iota \in [M]$,*

$$\Pr[v \neq z_\iota \wedge \text{flag} = \text{success}] \leq \text{negl}(\lambda),$$

where $(v, \text{flag}) \leftarrow \text{Out}_C(1^\lambda, z, \iota)$.

- **Client privacy on success:** *for every $z \in \{0, 1\}^M$ and $\iota_0, \iota_1 \in [M]$, the server's views in the two executions are computationally indistinguishable conditioned on the corresponding client output having flag success.*

Remark 1. In the above definition of half-PIR, the success flag is visible to the client; this is essential for the repetition used in the half-PIR-to-PIR transformation. The correctness-on-success requirement additionally ensures that repetitions declared successful return the requested database bit except with negligible probability.

The following bootstrapping theorem of [25] converts half-PIR into full PIR by first constructing random-index PIR [49] and then de-randomizing the index. The transformation is generic and preserves polylogarithmic communication.

Theorem 1 (Imported from [25, Section 5.3, Lemma 30], using [49]).

If there exists a half-PIR protocol as per Definition 5 (including correctness on success) with communication $c(\lambda, M)$, round complexity $r(\lambda, M)$, and semi-honest security, then there exists a semi-honest PIR protocol as per Definition 4 with parameters

$$c'(\lambda, M) = O(\lambda c(\lambda, M)(1 + \log M)), \quad r'(\lambda, M) = O(r(\lambda, M)(1 + \log M)).$$

In particular, the above half-PIR-to-PIR transformation preserves polylogarithmic communication.

3.3 Leakage Resilient Public-key Encryption

Informally, a public-key encryption (PKE) scheme is said to be leakage resilient [6, 69] if it retains semantic/IND-CPA security against a PPT adversary that has access to some bounded leakage about the secret key. The formal definition is as follows.

Definition 6 (Leakage Resilient Public Key Encryption [6, 69]). *For a security parameter $\lambda \in \mathbb{N}$, a public-key encryption scheme $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$, a leakage-rate parameter $L(\cdot)$ and a stateful PPT adversary $\mathcal{A}$, consider the following security experiment.*

1. **Setup phase:** *On input the security parameter λ , the challenger generates $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$ and provides pk to $\mathcal{A}$.*
2. **Leakage phase:** *$\mathcal{A}$ outputs a leakage function $f : \{0, 1\}^{|\text{sk}|} \rightarrow \{0, 1\}^{L(|\text{sk}|)}$. The challenger provides $f(\text{sk})$ to $\mathcal{A}$.*
3. **Challenge phase:** *$\mathcal{A}$ outputs a message pair (μ_0, μ_1) . The challenger samples a random bit $b \leftarrow \{0, 1\}$ and provides to $\mathcal{A}$ the challenge ciphertext ct generated as:*

$$\text{ct} \leftarrow \text{Enc}(\text{pk}, \mu_b)$$

4. **Output phase:** *$\mathcal{A}$ outputs a bit $b' \in \{0, 1\}$ as the output of the experiment.*

We define the advantage $\text{Adv}_{\text{PKE}, \mathcal{A}}^{\text{lr-ind}}(1^\lambda)$ of the adversary $\mathcal{A}$ in the above security experiment as

$$\text{Adv}_{\text{PKE}, \mathcal{A}}^{\text{lr-ind}}(1^\lambda) = |\Pr[\mathcal{A} \text{ outputs } 1 | b = 0] - \Pr[\mathcal{A} \text{ outputs } 1 | b = 1]|.$$

The PKE scheme PKE is said to satisfy $L(\cdot)$ -leakage resilient IND-CPA security (or, equivalently, $L(\cdot)$ -lr-ind security) if for any security parameter $\lambda \in \mathbb{N}$ and any stateful, PPT adversary $\mathcal{A}$, we have

$$\text{Adv}_{\text{PKE}, \mathcal{A}}^{\text{lr-ind}}(1^\lambda) \leq \text{negl}(\lambda).$$

3.4 KDM Secure Public-key Encryption

Informally, a PKE scheme with message space $\{0,1\}$ is said to satisfy key-dependent message (KDM) security [19, 12, 11] if it retains semantic/IND-CPA security even when a PPT adversary obtains encryptions of bit values computed as (adversarially chosen) functions of the secret key. More precisely, the adversary should be unable to distinguish between encryptions of zero and encryptions of these secret-key-dependent function outputs. This notion is strictly stronger than standard semantic/IND-CPA security; in particular, standard semantic/IND-CPA security is subsumed as a special case of KDM security by allowing the adversary to choose the constant “1” function in the security experiment. The formal definition is as follows.

Definition 7 (KDM Secure Public Key Encryption [19, 12, 11]). *For a security parameter $\lambda \in \mathbb{N}$, a public-key encryption scheme $\text{PKE} = (\text{Gen}, \text{Enc}, \text{Dec})$ with binary message space $\{0,1\}$, a circuit size bound S , any $Q = \text{poly}(\lambda)$, and a stateful PPT adversary $\mathcal{A}$, consider the following security experiment.*

1. **Setup phase:** *On input the security parameter λ , the challenger generates $(\text{pk}, \text{sk}) \leftarrow \text{Gen}(1^\lambda)$ and provides pk to $\mathcal{A}$.*
2. **Sampling challenge bit:** *The challenger samples a random bit $b \leftarrow \{0,1\}$.*
3. **Challenge Query phase:** *At this point, $\mathcal{A}$ (adaptively) chooses functions $\{f_i : \{0,1\}^{|\text{sk}|} \rightarrow \{0,1\}\}_{i \in [Q]}$, subject to the restriction that each f_i is a circuit of size $\leq S$, and sends these functions to the challenger. For each such query, the challenger replies with a ciphertext ct_i , where:*
 - *If $b = 0$, $\text{ct}_i \leftarrow \text{Enc}(\text{pk}, 0)$.*
 - *If $b = 1$, $\text{ct}_i \leftarrow \text{Enc}(\text{pk}, f_i(\text{sk}))$.*
4. **Output phase:** *$\mathcal{A}$ outputs a bit $b' \in \{0,1\}$ as the output of the experiment.*

We define the advantage $\text{Adv}_{\text{PKE}, \mathcal{A}}^{\text{kdm}}(1^\lambda, S)$ of the adversary $\mathcal{A}$ in the above security experiment as

$$\text{Adv}_{\text{PKE}, \mathcal{A}}^{\text{kdm}}(1^\lambda, S) = |\Pr[\mathcal{A} \text{ outputs } 1 | b = 0] - \Pr[\mathcal{A} \text{ outputs } 1 | b = 1]|.$$

The PKE scheme PKE is said to satisfy kdm-security with respect to functions (of the secret key) representable as circuits of size $\leq S$ if, for any security parameter $\lambda \in \mathbb{N}$, any $Q = \text{poly}(\lambda)$, and any stateful, PPT adversary $\mathcal{A}$, we have

$$\text{Adv}_{\text{PKE}, \mathcal{A}}^{\text{kdm}}(1^\lambda, S) \leq \text{negl}(\lambda).$$

In particular, when each f_i in the above security game is required to be *affine* (over $\mathbb{F}_2$), the PKE scheme PKE is said to be KDM-secure with respect to affine functions of the secret key. Note that, by prior works [12, 11], any PKE scheme that satisfies KDM security with respect to affine functions of the secret key can additionally be used to build a PKE scheme satisfying KDM security with respect to *arbitrary* functions of the secret key that can be represented via circuits with a priori bounded size.

3.5 Laconic Oblivious Transfer

We recall the formal definition of laconic oblivious transfer (OT) from [34] below.

Definition 8 (Laconic OT [34]). A laconic OT (ℓ OT) scheme syntactically consists of four algorithms crsGen , Hash , Send and Receive .

- $\text{crs} \leftarrow \text{crsGen}(1^\lambda)$. It takes as input the security parameter 1^λ and outputs a common reference string crs .
- $(\text{digest}, \hat{D}) \leftarrow \text{Hash}(\text{crs}, D)$. It takes as input a common reference string crs and a database $D \in \{0, 1\}^*$ and outputs a digest digest of the database and a state $\hat{D}$.
- $e \leftarrow \text{Send}(\text{crs}, \text{digest}, L, m_0, m_1)$. It takes as input a common reference string crs , a digest digest , a database location $L \in \mathbb{N}$ and two messages m_0 and m_1 of length λ , and outputs a ciphertext e .
- $m \leftarrow \text{Receive}^{\hat{D}}(\text{crs}, e, L)$. This is a RAM algorithm with random read access to $\hat{D}$. It takes as input a common reference string crs , a ciphertext e , and a database location $L \in \mathbb{N}$. It outputs a message m .

We require the following properties of an ℓ OT scheme $(\text{crsGen}, \text{Hash}, \text{Send}, \text{Receive})$.

- **Correctness:** We require that it holds for any database D of size at most $M = \text{poly}(\lambda)$ for any polynomial function $\text{poly}(\cdot)$, any memory location $L \in [M]$, and any pair of messages $(m_0, m_1) \in \{0, 1\}^\lambda \times \{0, 1\}^\lambda$ that

$$\Pr \left[m = m_{D[L]} \mid \begin{array}{l} \text{crs} \leftarrow \text{crsGen}(1^\lambda) \\ (\text{digest}, \hat{D}) \leftarrow \text{Hash}(\text{crs}, D) \\ e \leftarrow \text{Send}(\text{crs}, \text{digest}, L, m_0, m_1) \\ m \leftarrow \text{Receive}^{\hat{D}}(\text{crs}, e, L) \end{array} \right] = 1,$$

where the probability is taken over the random choices made by crsGen and Send .

- **Sender Privacy Against Semi-Honest Receivers:** There exists a PPT simulator ℓOTSim such that the following holds. For any database D of size at most $M = \text{poly}(\lambda)$ for any polynomial function $\text{poly}(\cdot)$, any memory location $L \in [M]$, and any pair of messages $(m_0, m_1) \in \{0, 1\}^\lambda \times \{0, 1\}^\lambda$, let $\text{crs} \leftarrow \text{crsGen}(1^\lambda)$ and $\text{digest} \leftarrow \text{Hash}(\text{crs}, D)$. Then it holds that

$$(\text{crs}, \text{Send}(\text{crs}, \text{digest}, L, m_0, m_1)) \stackrel{c}{\approx} (\text{crs}, \ell\text{OTSim}(D, L, m_{D[L]})).$$

- **Efficiency Requirement:** The length of digest is a fixed polynomial in λ independent of the size of the database; we will assume for simplicity that $|\text{digest}| = \lambda$. Moreover, the algorithm Hash runs in time $|D| \cdot \text{poly}(\log |D|, \lambda)$, Send and Receive run in time $\text{poly}(\log |D|, \lambda)$.

Receiver Privacy. The above definition of laconic OT does not require receiver privacy, unlike standard OT. As discussed in [34], receiver privacy can be added to laconic OT in a straightforward way using standard two-message OT and garbled circuits. In this paper, we focus on laconic OT without receiver privacy; as noted in [34], receiver privacy is not crucial for many natural applications of laconic OT.

Multiple executions of Send can share the same digest. Since the common reference string is public, the sender can invoke the `Send` function multiple times using the same digest; security can be argued from the above definition (for the case of single execution) via a standard hybrid argument. We refer to [34] for a formal exposition.

3.6 (Blind) Batch Encryption

Here we provide the definition of a (blind) batch encryption (BE) scheme from [27]. The original definition of BE in [27] is parameterized by a *block-size* parameter B . For ease of exposition, we recall a simpler definition of BE from [9], which is equivalent to the definition in [27] for $B = 2$. This definition of BE suffices to yield all of the applications outlined in [27].

Syntax of BE. A BE scheme consists of four algorithms (`Setup`, `Gen`, `Enc`, `Dec`) with the following syntax.

`Setup`($1^\lambda, n$) $\rightarrow$ `pp`. The generation algorithm takes as input the security parameter λ and the input parameter $n = \text{poly}(\lambda)$, and outputs the public parameter `pp`.

`Gen`(`pp`, $\mathbf{x}$) $\rightarrow$ h . The key generation algorithm takes as input the public parameters `pp` and an input string $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, and deterministically hashes (or projects) $\mathbf{x}$ to a hash value h .

`Enc`(`pp`, (h, i) , (μ_0, μ_1)) $\rightarrow$ `ct` = (`subct`₁, `subct`₂). The encryption algorithm takes as input a key `pp`, a hash value h , an index $i \in [n]$, and a message pair $(\mu_0, \mu_1) \in \{0, 1\} \times \{0, 1\}$, and outputs a ciphertext `ct` = (`subct`₁, `subct`₂).

`Dec`(`pp`, $(\mathbf{x}, i)$, `ct`) $\rightarrow$ μ' . The decryption algorithm takes as input a key `pp`, an input string $\mathbf{x} \in \{0, 1\}^n$, an index $i \in [n]$, and a ciphertext `ct` and outputs $\mu' \in \{0, 1\}^* \cup \{\perp\}$.

Definition 9 (Correctness of BE). A BE scheme is said to be correct if for any $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, any `pp` $\leftarrow$ `Setup`($1^\lambda, n$), any input string $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, any index $i \in [n]$, and any message-pair $(\mu_0, \mu_1) \in \{0, 1\} \times \{0, 1\}$, letting `ct` $\leftarrow$ `Enc`(`pp`, (`Gen`(`pp`, $\mathbf{x}$), i), (μ_0, μ_1)), we have:

$$\text{Dec}(\text{pp}, (\mathbf{x}, i), \text{ct}) = \mu_{x_i}.$$

Definition 10 (Succinctness of BE). A BE scheme is said to be succinct if for any $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, any `pp` $\leftarrow$ `Setup`($1^\lambda, n$), and any input string $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, letting $h = \text{Gen}(\text{pp}, \mathbf{x})$, we have $|h| \leq \text{poly}(\lambda)$ for some fixed polynomial in λ that is independent of the size parameter n .

In other words, the bit-length of the hash value h should be independent of the length of the input string $\mathbf{x} \in \{0, 1\}^n$.

Definition 11 (Security of BE). For a BE scheme and an adversary $\mathcal{A}$, we consider the following indistinguishability experiment $\text{Expt}_{\beta, \mathcal{A}}^{\text{BE-IND}}(1^\lambda, n)$ for $\beta \in \{0, 1\}$.

1. $\mathcal{A}$ chooses a string $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$ and an index $i \in [n]$, and sends $(\mathbf{x}, i)$ to the challenger.
2. The challenger generates $\mathbf{pp} \leftarrow \text{Setup}(1^\lambda, n)$ and sends $\mathbf{pp}$ to $\mathcal{A}$.
3. $\mathcal{A}$ outputs two message pairs $(\mu_{0,0}, \mu_{0,1}), (\mu_{1,0}, \mu_{1,1}) \in \{0, 1\} \times \{0, 1\}$ subject to the restriction that $\mu_{0,x_i} = \mu_{1,x_i}$.
4. The challenger returns $\mathbf{ct}_\beta \leftarrow \text{Enc}(\mathbf{pp}, (\text{Gen}(\mathbf{pp}, \mathbf{x}), i), (\mu_{\beta,0}, \mu_{\beta,1}))$ to $\mathcal{A}$.
5. $\mathcal{A}$ outputs a guess bit β' as the output of the experiment.

We define the advantage $\text{Adv}_{\mathcal{A}}^{\text{BE-IND}}(\lambda, n)$ of $\mathcal{A}$ in the above game as

$$\text{Adv}_{\mathcal{A}}^{\text{BE-IND}}(\lambda, n) := \left| \Pr[\text{Expt}_{0,\mathcal{A}}^{\text{BE-IND}}(1^\lambda, n) = 1] - \Pr[\text{Expt}_{1,\mathcal{A}}^{\text{BE-IND}}(1^\lambda, n) = 1] \right|.$$

A BE scheme is said to be secure if for any security parameter $\lambda \in \mathbb{N}$, any input parameter $n = \text{poly}(\lambda)$, and any PPT adversary $\mathcal{A}$, we have

$$\text{Adv}_{\mathcal{A}}^{\text{BE-IND}}(\lambda, n) \leq \text{negl}(\lambda)$$

Definition 12 (Blindness of BE). BE = (Setup, Gen, Enc, Dec) be a batch encryption scheme such that the encryption algorithm admits a decomposition of the form

$$\text{Enc}(\mathbf{pp}, (h, i), (\mu_0, \mu_1); r) = (\text{Enc}_1(\mathbf{pp}, i; r), \text{Enc}_2(\mathbf{pp}, h, i, (\mu_0, \mu_1); r)),$$

where the output of the first component Enc_1 is independent of both the hash value h and the message pair (μ_0, μ_1) . We write the resulting ciphertext as $\mathbf{ct} = (\text{subct}_1, \text{subct}_2)$. For such a BE scheme and an adversary $\mathcal{A}$, we consider the following blindness experiment $\text{Expt}_{\beta,\mathcal{A}}^{\text{BE-BLIND}}(1^\lambda, n)$ for $\beta \in \{0, 1\}$.

1. $\mathcal{A}$ chooses a string $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$ and an index $i \in [n]$, and sends $(\mathbf{x}, i)$ to the challenger.
2. The challenger generates $\mathbf{pp} \leftarrow \text{Setup}(1^\lambda, n)$ and sends $\mathbf{pp}$ to $\mathcal{A}$.
3. The challenger samples a uniformly random message pair $(\mu_0, \mu_1) \leftarrow \{0, 1\} \times \{0, 1\}$ and sets $\mathbf{ct} = (\text{subct}_1, \text{subct}_{2,0}) \leftarrow \text{Enc}(\mathbf{pp}, (\text{Gen}(\mathbf{pp}, \mathbf{x}), i), (\mu_0, \mu_1))$. The challenger additionally samples $\text{subct}_{2,1} \leftarrow \{0, 1\}^{|\text{subct}_{2,0}|}$.
4. The challenger returns $\mathbf{ct}_\beta = (\text{subct}_1, \text{subct}_{2,\beta})$ to $\mathcal{A}$.
5. $\mathcal{A}$ outputs a guess bit β' as the output of the experiment.

We define the advantage $\text{Adv}_{\mathcal{A}}^{\text{BE-BLIND}}(\lambda, n)$ of $\mathcal{A}$ in the above game as

$$\text{Adv}_{\mathcal{A}}^{\text{BE-BLIND}}(\lambda, n) := \left| \Pr[\text{Expt}_{0,\mathcal{A}}^{\text{BE-BLIND}}(1^\lambda, n) = 1] - \Pr[\text{Expt}_{1,\mathcal{A}}^{\text{BE-BLIND}}(1^\lambda, n) = 1] \right|.$$

A BE scheme is said to be blind if for any security parameter $\lambda \in \mathbb{N}$, any input parameter $n = \text{poly}(\lambda)$, and any PPT adversary $\mathcal{A}$, we have

$$\text{Adv}_{\mathcal{A}}^{\text{BE-BLIND}}(\lambda, n) \leq \text{negl}(\lambda)$$

Applications of BE. In this paper, we leverage the following known implications of (blind) BE from [27].

Theorem 2 (Blind BE implies Anonymous IBE [27]). *Suppose there exists a BE scheme satisfying correctness as in Definition 9, succinctness as in Definition 10, security as in Definition 11, and blindness as in Definition 12. Then there exists an anonymous IBE scheme satisfying Definition 3.*

Theorem 3 (BE implies Leakage-resilient and KDM-secure PKE [27]). *Suppose there exists a BE scheme satisfying correctness as in Definition 9, succinctness as in Definition 10, and security as in Definition 11. Then there exists a public-key encryption scheme that simultaneously satisfies: (i) leakage resilience as in Definition 6 for leakage rate $(1 - o(1))$, and (ii) KDM security as in Definition 7 with respect to affine functions of the secret key.*

Further, by prior works [12, 11], BE also implies the existence of a PKE scheme satisfying KDM security as in Definition 7 with respect to a much larger class of *arbitrary* functions of the secret key that can be represented via circuits with a priori bounded size.

Theorem 4 (BE implies Laconic OT [27]). *Suppose there exists a BE scheme satisfying correctness as in Definition 9, succinctness as in Definition 10, and security as in Definition 11. Then there exists a laconic oblivious transfer (OT) scheme satisfying Definition 8.*

Technically, some of the above implications additionally require (blind) garbled circuits. However, since blind garbled circuits are implied by *any* one-way function (and hence by BE), we do not state this as a separate assumption.

3.7 Collision-Resistant Hash

Here we provide the definition of a collision-resistant hash function family that supports inputs of a fixed (polynomial) size.

Setup($1^\lambda, n$): Given the security parameter λ and the bit-length of inputs $n = \text{poly}(\lambda)$, the setup algorithm outputs a public hash key pp .

Hash($\text{pp}, \mathbf{x}$): On input pp and an input $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, the hashing algorithm outputs a deterministically computed hash value $h_{\mathbf{x}}$.

Definition 13 (Collision-Resistance). *A collision-resistant hash function family (Setup, Hash) is said to be collision-resistant if, for any security parameter $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, any $\text{pp} \leftarrow \text{Setup}(1^\lambda, n)$, and for any PPT adversary Adv , we have*

$$\Pr[\text{Adv}(\text{pp}) = (\mathbf{x}_0, \mathbf{x}_1) \wedge \mathbf{x}_0 \neq \mathbf{x}_1 \wedge \text{Hash}(\text{pp}, \mathbf{x}_0) = \text{Hash}(\text{pp}, \mathbf{x}_1)] \leq \text{negl}(\lambda)$$

3.8 One-Time Encryption via Goldreich–Levin Hardcore Predicate

We adopt from [27] a simple one-time encryption (OTE) mechanism derived from the Goldreich–Levin (abbreviated as GL henceforth) hard-core predicate [53].

Definition 14 (GL-OTE [53, 27]). *The GL-OTE scheme consists of a pair of algorithms (gl-enc, gl-dec) defined as follows:*

- **gl-enc**(x, μ): a randomized algorithm that takes as input $x \in \{0, 1\}^\ell$ and a message bit $\mu \in \{0, 1\}$, samples a uniformly random vector $\alpha \leftarrow \{0, 1\}^\ell$ and outputs $(\alpha, \sigma = \langle \alpha, x \rangle \oplus \mu)$, where the inner product is over the binary field.
- **gl-dec**($x, (\alpha, \sigma)$): a deterministic algorithm that takes as input $x \in \{0, 1\}^\ell$, $\alpha \in \{0, 1\}^\ell$ and $\sigma \in \{0, 1\}$, and outputs $\mu' = \sigma \oplus \langle \alpha, x \rangle$.

Correctness. Correctness follows immediately: for any $x \in \{0, 1\}^\ell$ and any message-bit $\mu \in \{0, 1\}$, we have $\text{gl-dec}(x, \text{gl-enc}(x, \mu)) = \mu$.

Security. The security of this construction is a direct consequence of the Goldreich–Levin theorem [53]. Specifically, for security parameter λ , let $\{(X_\lambda, Z_\lambda)\}_\lambda$ be any ensemble of jointly distributed random variables such that recovering X_λ from Z_λ is computationally infeasible, i.e., for every PPT adversary $\mathcal{A}$,

$$\Pr_{(x, z) \leftarrow (X_\lambda, Z_\lambda)} [\mathcal{A}(1^\lambda, z) = x] \leq \text{negl}(\lambda).$$

Then, for any (possibly z -dependent) message bit $\mu \in \{0, 1\}$, the following distributions are computationally indistinguishable

$$(z, (\alpha, \sigma) \leftarrow \text{gl-enc}(x, \mu)) \approx_c (z, (\alpha', \sigma') \leftarrow \{0, 1\}^\ell \times \{0, 1\})$$

An additional observation from [27] that we use is the following: if the message bit μ itself is chosen uniformly at random and remains hidden from the adversary $\mathcal{A}$, then the ciphertext produced by $\text{gl-enc}(x, \mu)$ is *information-theoretically* uniform from $\mathcal{A}$'s point of view, independent of x .

3.9 Elliptic Curves and Isogenies

Supersingular Elliptic Curves. All supersingular elliptic curves are isomorphic to a curve defined over $\mathbb{F}_{p^2}$ [74, Theorem V.3.1 (a)], so we can restrict ourselves to $\mathbb{F}_{p^2}$. Any supersingular curve $E(\mathbb{F}_{p^2})$ has $(p \pm 1)^2$ points, where the choice of plus or minus depends on the choice of twist. In this work, we consider only curves that have $(p + 1)^2$ points.

The group $E(\mathbb{F}_{p^2})$ is thus isomorphic to $\mathbb{Z}/(p + 1)\mathbb{Z} \times \mathbb{Z}/(p + 1)\mathbb{Z}$. Hence, if $N \mid p + 1$, the full N -torsion subgroup $E[N]$ is rational over $\mathbb{F}_{p^2}$. It is possible to sample a point of order N or a basis of $E[N]$ (i.e., two linearly independent points of order N) in polynomial time [59, Section 4.1]. We write $\langle P \rangle$ for the cyclic subgroup generated by P and $\langle P, Q \rangle$ or $\langle P \rangle \oplus \langle Q \rangle$ for the bicyclic group

generated by P and Q (i.e., containing all points of the form $[\alpha]P + [\beta]Q$ for $\alpha, \beta \in \mathbb{Z}$).

If the order of a point $P \in E$ does not divide $p + 1$, the point is not defined over $\mathbb{F}_{p^2}$ (i.e., $P \notin E(\mathbb{F}_{p^2})$), but over an extension field of index $O(\text{ord } P)$ (i.e., $P \notin E(\mathbb{F}_{p^{O(\text{ord } P)}})$) [45]. In this work, we will work with points of B -powersmooth⁸ order defined over large extension fields. However, they can be efficiently represented (and worked with) because they have powersmooth order: let P be a point of order $\prod_k^L \ell_k$. The point P can be expressed as the sum $P = \sum_k^L P_k$, where each P_k of order ℓ_k . Since $\ell_k < B$, each P_k is defined over a small extension field $\mathbb{F}_{p^{O(B)}}$. This technique is widely used to work with large extension fields [45, 44].

Isogeny Maps. An *isogeny* is a rational map between elliptic curves that sends the identity to the identity. We write $\phi : E \rightarrow E'$ for an isogeny ϕ between E and E' . Every isogeny is a group homomorphism, i.e. $\phi([a]P + [\beta]Q) = [a]\phi(P) + [\beta]\phi(Q)$, and it has an associated quantity called its *degree*. If the isogeny is separable (all isogenies in this work will be separable), the degree represents the size of its kernel. Moreover, if the degree of an isogeny ϕ is coprime with the order of a point P , the isogeny preserves the order of the point, i.e. $\text{ord } \phi(P) = \text{ord } P$. The degree is multiplicative, i.e. $\deg(\psi \circ \phi) = \deg \psi \deg \phi$, and whenever the degree of an isogeny is composite, the isogeny can be decomposed into multiple isogenies. In particular, if the degree of an isogeny is $q_1 q_2$, it can be written as the composition of two isogenies of degree q_1 and q_2 . We call an isogeny *cyclic* if its kernel is cyclic: this implies that the isogeny does not factor through scalar multiplication $[m]$ for any $m \neq 1$.

Isogeny Problems. Below, we describe certain isogeny-based hard problems (widely conjectured to be post-quantum hard [70]) that our construction will rely on.

Problem 1 (OneEnd_p). Given a uniformly random supersingular elliptic curve E defined over $\mathbb{F}_{p^2}$, find a non-trivial endomorphism $\theta \in \text{End}(E)/\mathbb{Z}$.

Problem 2 (EndRing_p). Given a uniformly random supersingular elliptic curve E defined over $\mathbb{F}_{p^2}$, find a basis of its endomorphism ring $\text{End}(E)$.

Theorem 5 ([70, Thm. 1.1]). *The OneEnd_p and EndRing_p problems are equivalent, under probabilistic polynomial time reductions.*

Our construction is also based on binSIDH and binSIDH-hyb [13], which relies on the conjectured (post-quantum) hardness of the following problems (we later adapt the binSIDH-hyb DLOG problem to our parameters in Problem 5).

Problem 3 (binSIDH-hyb DLOG_B). Fix integers N and d_i for $i \in [n]$ such that $\prod_{i \in [n]} d_i \gg N$, let p be a prime of the form $p = (fN \prod_{i \in [n]} d_i) - 1$, and let E_0 be a supersingular curve over $\mathbb{F}_{p^2}$. Let $\{P_{i,0}, P_{i,1}\}_{i \in [n]}$ be a list of bases of $E_0[d_i]$, and let R_0, S_0 be a basis of $E_0[N]$.

⁸ A number n is B -powersmooth if all the prime factors of n are less or equal than B .

Let $\psi : E_0 \rightarrow E_B$ be a random N -isogeny. Given

$$\left(E_0, \{P_{i,b}\}_{\substack{i \in [n], \\ b \in \{0,1\}}}, E_B, \{[\alpha_{i,b}]\psi(P_{i,b})\}_{\substack{i \in [n], \\ b \in \{0,1\}}} \right),$$

where $\alpha_{i,b} \leftarrow_R (\mathbb{Z}/d_i)^*$, compute the isogeny ψ .

Problem 4 (binSIDH-hyb CDH). With the same notation as [Problem 3](#), let $\phi : E_0 \rightarrow E_A$ be an isogeny with kernel $\ker \phi = \langle \bigoplus_{i \in [n]} P_{i,b_i} \rangle$, where $b = (b_1, \dots, b_n) \in \{0,1\}^n$ is a random binary string, and let $\psi : E_0 \rightarrow E_B$ be a random N -isogeny. Given

$$(E_0, \{P_{i,b}\}_{\substack{i \in [n], \\ b \in \{0,1\}}}, (R_0, S_0), E_A, \phi(R_0), \phi(S_0), E_B, \{[\alpha_{i,b}]\psi(P_{i,b})\}_{\substack{i \in [n], \\ b \in \{0,1\}}}),$$

where $\alpha_{i,b} \leftarrow_R (\mathbb{Z}/d_i)^*$, compute the codomain of $[\phi]_*\psi$ (or, equivalently, the codomain of $[\psi]_*\phi$).

4 A Collision-Resistant Hash Function Family

In this section, we present a new collision-resistant hash function (CRHF) family over supersingular isogeny graphs. This CRHF family forms the basis for our batch encryption construction, and is a variant of the well-known Charles-Goren-Lauter (CGL) hash function [\[33\]](#). As we illustrate in [Section 4.5](#), beyond the usual collision-resistance property, the proposed CRHF family also satisfies a *shuffled order hash evaluation* property, which the original CGL hash function does not. We use this property crucially in our construction of batch encryption.

4.1 Notations and Parameters

Let λ be a security parameter, and let $n = \text{poly}(\lambda)$ denote the (fixed) bit-length of inputs accepted by the hash function family. Let $N = 2^{2\lambda}$ and let p be a prime of the form $p = Nf - 1$, where f is a small co-factor needed for primality.

Let $D = \{d_1, \dots, d_n\}$ be a set of odd integers such that each $d_i = O(2^{2\lambda})$ is B -powersmooth, for some polynomial $B = \text{poly}(\lambda)$, and all d_i are pairwise coprime. Concretely, for each $j \in [n]$, we have

$$d_j = \prod_{k=1}^{L_j} \ell_{j,k},$$

where each $\ell_{j,k} < B$ is a distinct prime, and each L_j is chosen to ensure that $d_j = O(2^{2\lambda})$.

To determine a smoothness bound B that makes this possible, let us choose D such that the primes $\ell_{i,k}$ are as small as possible. Then, the product $\prod_{i=1}^n d_i \approx 2^{2n\lambda}$ is the product of all the odd primes less than B . By the Prime Number Theorem, we have $B \approx 2n\lambda$. Since $n = \text{poly}(\lambda)$, this gives $B = \text{poly}(\lambda)$. Thus, choosing such a smoothness bound guarantees the existence of n coprime integers d_i with powersmooth bound B .

Algorithm 1 DIC: Decomposable Isogeny Computation

Input: An elliptic curve $E_0/\mathbb{F}_{p^2}$
 A list of bases $\{(P_{i,0}, P_{i,1})\}_{i \in [n]}$ where each $P_{i,b}$ is represented as $P_{i,b} = \sum_{k=1}^{L_i} P_{i,b,k}$, with $\text{ord}(P_{i,b}) = d_i$ and $\text{ord}(P_{i,b,k}) = \ell_{i,k}$
 An input $X = \{(i, \mathbf{x}_i)\}_{i \in [n]}$ (corresponding to the string $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$)
 Two points $R_0, S_0 \in E_0[N]$

Output: The codomain $E_{\mathbf{x}}$ and the images $R_{\mathbf{x}} := \phi_{\mathbf{x}}(R)$ and $S_{\mathbf{x}} := \phi_{\mathbf{x}}(S)$, where $\phi_{\mathbf{x}} : E_0 \rightarrow E_{\mathbf{x}}$ is the isogeny with kernel generated by $K_{\mathbf{x}} = \bigoplus_{j \in [n]} \langle P_{j,x_j} \rangle$

```

1: Initialize:  $E = E_0$ ,  $R = R_0$  and  $S = S_0$ .
2: for  $i \in [n]$  do
3:   for  $k \in [L_i]$  do  $\triangleright$  This loop computes the isogeny with kernel  $P_{i,\mathbf{x}_i}$ 
4:     Compute the  $\ell_{i,k}$ -isogeny  $\phi : E \rightarrow E'$  with kernel  $\langle P_{i,\mathbf{x}_i,k} \rangle$ 
5:     Finda an isomorphism  $\theta$  from  $E'$  to an isomorphic curve defined over  $\mathbb{F}_{p^2}$ 
6:     Update alla points  $P_{i',b,k'} = \theta\phi(P_{i',b,k'})$  for  $i' \in [n]$ ,  $b \in \{0, 1\}$ ,  $k' \in [L_{i'}]$ 
7:     Update the curve  $E \leftarrow \theta(E')$ 
8:     Update the points  $R \leftarrow \theta\phi(R)$ ,  $S \leftarrow \theta\phi(S)$ 
9: return  $E_{\mathbf{x}} := E$ ,  $R_{\mathbf{x}} := R$ ,  $S_{\mathbf{x}} := S$ 

```

4.2 Construction

We now present our hash function family, which is a tuple of algorithms (Setup, Hash) as described below:

Setup($1^\lambda, n$): Given the security parameter λ and the bit-length of inputs $n = \text{poly}(\lambda)$, the setup algorithm proceeds as follows:

- Set the parameters $(p, N, B, \{\ell_{j,k}\}_{j \in [n], k \in [L_j]})$ as described in [Section 4.1](#).
- Sample a uniformly random supersingular elliptic curve E_0 over $\mathbb{F}_{p^2}$ by computing a long random walk in the supersingular isogeny graph⁹.
- Sample R_0, S_0 to be a torsion basis of $E_0[N]$.
- For each $j \in [n]$, do the following in order:
 - Set $d_j = \prod_{k=1}^{L_j} \ell_{j,k}$ as in [Section 4.1](#).
 - Generate a basis $P_{j,0}, P_{j,1}$ of $E_0[d_j]$. Since d_j is B -powersmooth (as explained in [Section 4.1](#)), each point $P_{j,b}$ for $b \in \{0, 1\}$ can be decomposed as $P_{j,b} = \sum_{k=1}^{L_j} P_{j,b,k}$, where each $P_{j,b,k}$ has order $\ell_{j,k}$ and is thus defined over an extension field of $\mathbb{F}_{p^2}$ of index $O(\ell_{j,k}) < O(B) \approx O(2n\lambda)$, and has an efficient representation.

^a We canonically normalize the marked triple (E', R', S') , rather than only E' : after fixing a canonical representative E^{can} , we choose the $\mathbb{F}_{p^2}$ -isomorphism $\theta' \rightarrow E^{\text{can}}$ minimizing a fixed encoding of $(\theta(R'), \theta(S'))$, and apply θ to all tracked points. The choice is unique since the N -torsion basis (R', S') , for $N > 2$, has trivial stabilizer under $\text{Aut}(E^{\text{can}})$. This ensures that [Algorithm 1](#) is a deterministic function.

^a It is sufficient to update the points whose (i, k) index has not been processed yet.

⁹ To be precise, this strategy samples the curve E_0 from a distribution whose statistical distance to the uniform one is negligible.

- Set $\text{par} = (p, N, B)$ and output the public hash key

$$\text{pp} = (\text{par}, (E_0, R_0, S_0), \{\ell_{j,k}\}_{j \in [n], k \in [L_j]}, \{P_{j,b}\}_{j \in [n], b \in \{0,1\}}).$$

where, throughout the rest of the paper, each $P_{j,b}$ is (implicitly) represented efficiently as the tuple $\{P_{j,b,k}\}_{k \in [L_j]}$.

Hash($\text{pp}, \mathbf{x} = (x_1, \dots, x_n)$): On input pp as described above, and an input $\mathbf{x} = (x_1, \dots, x_n) \in \{0,1\}^n$, proceed as follows:

- Let $K_{\mathbf{x}} \subset E_0$ be the composite kernel of order $d = \prod_{i=1}^n d_i$ computed as the direct sum: $K_{\mathbf{x}} = \bigoplus_{i \in [n]} \langle P_{i,x_i} \rangle$.
- Compute $(E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}}) \leftarrow \text{DIC}(\text{pp}, (E_0, R_0, S_0), ((j, x_j))_{j \in [n]})$ using [Algorithm 1](#), such that:
 - The isogeny $\phi_{\mathbf{x}} : E \rightarrow E_{\mathbf{x}}$ of composite degree $d = \prod_{j \in [n]} d_j$ has kernel $K_{\mathbf{x}}$.
 - $R_{\mathbf{x}} = \phi_{\mathbf{x}}(R_0)$ and $S_{\mathbf{x}} = \phi_{\mathbf{x}}(S_0)$.
- Output the hash value $h_{\mathbf{x}} = (E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}})$.

4.3 Efficiency

We now analyze the efficiency of the **Setup** and **Hash** algorithms. We report all computational costs in terms of field operations (multiplication and inversion) in $\mathbb{F}_{p^2}$. Recall from [Section 4.1](#) that $\log p = O(\lambda)$. Hence, each $\mathbb{F}_{p^2}$ (multiplication/inversion) operation incurs $O(\text{polylog}(p)) = \tilde{O}(\lambda)$ cost in terms of bit operations (assuming efficient multiplication).

Lemma 1 (Efficiency of Setup). *For any security parameter $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, $\text{Setup}(1^\lambda, n)$ runs in time $\text{poly}(\lambda)$.*

Proof. The proof of the lemma follows from the following observations:

- Both $N = 2^{2\lambda}$ and $p = Nf - 1$ for a small co-factor f require $\text{poly}(\lambda)$ bits to represent, and hence, can be chosen efficiently.
- Since $B = \text{poly}(\lambda)$, each (odd) prime less than or equal to B takes $O(\log B) = O(\log \lambda)$ bits to represent. Hence, enumerating all odd primes that are less than or equal to B takes time at most $\text{poly}(\lambda)$. Thus, sampling $\{\ell_{j,k}\}_{j \in [n], k \in [L_j]}$ takes time at most $\text{poly}(\lambda)$, and is efficient.
- Sampling a uniformly random E_0 in the $\mathbb{F}_{p^2}$ isogeny graph requires a random, non-backtracking walk of length at most $\text{poly}(\lambda)$ (this follows from the expander property of the $\mathbb{F}_{p^2}$ isogeny graph). Thus, E_0 can be efficiently sampled.
- Given E_0 , sampling a basis R_0, S_0 of $E_0[N]$ is efficient [[59](#), Section 4.1].
- For each $i \in [n]$, computing $d_i = \prod_{k \in [L_i]} \ell_{i,k}$ takes time at most $\text{poly}(\lambda)$ as each $\ell_{j,k} < B = \text{poly}(\lambda)$, and L_i is chosen such that each $d_i = O(2^{2\lambda})$.
- For each $j \in [n]$, each $b \in \{0,1\}$ and each $k \in [L_j]$, sampling $P_{j,b,k}$ of order $\ell_{j,k}$ is efficient because $P_{j,b,k}$ is defined over an extension field of index $O(\ell_{j,k}) < O(B) = O(n\lambda) = \text{poly}(\lambda)$.

- Finally, for each $j \in [n]$ and each $b \in \{0, 1\}$, $P_{i,b}$ is represented efficiently as the tuple $\{P_{j,b,k}\}_{k \in [L_j]}$.

Lemma 2 (Efficiency of Hashing). *For any security parameter $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, and any arbitrary input $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, letting $\text{pp} \leftarrow \text{Setup}(1^\lambda, n)$, the algorithm $\text{Hash}(\text{pp}, \mathbf{x})$ runs in time $\text{poly}(\lambda)$.*

Proof. The proof follows from the following observations:

- **Cost per Step:** Computing an isogeny of degree $\ell \leq B$ and evaluating it on a fixed number of points requires $\tilde{O}(\sqrt{\ell})$ field operations using velusqrt formulas [17]. Since ℓ is bounded by $B \approx O(n\lambda) = \text{poly}(\lambda)$, the cost per step is $C_{\text{step}} = \tilde{O}(\sqrt{n\lambda}) \mathbb{F}_{p^2}$ operations, and is therefore polynomial in λ . Once the isogeny has been computed, evaluating it on points (similarly defined over an extension field of polynomial index) is efficient [44, Lemma 1].
- **Total Steps:** The hash function processes n blocks. With the parameter choice from Section 4.1, the selected small prime factors have total logarithmic mass $O(n\lambda)$ and are bounded by $B \approx 2n\lambda$. Hence, the total number of selected prime factors, and therefore the total number of isogeny steps, is

$$N_{\text{total}} = \sum_{i=1}^n L_i = O\left(\frac{n\lambda}{\log B}\right) \leq n \cdot \tilde{O}(\lambda) = \text{poly}(\lambda).$$

4.4 Collision Resistance

Theorem 6 (Collision Resistance). *For any security parameter $\lambda \in \mathbb{N}$ and any $n = \text{poly}(\lambda)$, assuming that OneEnd_p (or equivalently, EndRing_p) is computationally hard, the above hash function family satisfies collision resistance as in Definition 13.*

Proof. Suppose a PPT adversary $\mathcal{A}$ that, given (λ, N, pp) where $\text{pp} \leftarrow \text{Setup}(1^\lambda, n)$, outputs with non-negligible probability $\mathbf{x}, \mathbf{x}'$ such that $\mathbf{x} \neq \mathbf{x}'$ but $H(\text{pp}, \mathbf{x}) = H(\text{pp}, \mathbf{x}')$. We construct a PPT algorithm $\mathcal{B}$ that breaks the OneEnd problem as follows:

- $\mathcal{B}$ receives as input OneEnd challenge (p, E_0) where $p = O(2^{c\lambda})$.
- $\mathcal{B}$ picks some $n = \text{poly}(\lambda)$, sets up the hash key pp as in Setup using p and E_0 , where $\text{pp} = ((p, N, B, E_0, R_0, S_0), \{\ell_{i,k}\}_{i \in [n], k \in [L_i]}, \{P_{i,b}\}_{i \in [n], b \in \{0,1\}})$, such that p and E_0 are identical to $\mathcal{B}$'s OneEnd challenge, while everything else in pp is generated by $\mathcal{B}$ exactly as in Setup. Also, each $j \in [n]$ and $b \in \{0, 1\}$, let $d_i = \prod_{k=1}^{L_i} \ell_{i,k}$.
- Next, $\mathcal{B}$ runs $(\mathbf{x}, \mathbf{x}') \leftarrow \mathcal{A}(1^\lambda, n, \text{pp})$.
- If $\mathbf{x} = \mathbf{x}'$ or $\text{Hash}(\text{pp}, \mathbf{x}) \neq \text{Hash}(\text{pp}, \mathbf{x}')$, $\mathcal{B}$ aborts (this happens with negligible probability by assumption).
- Otherwise $\mathcal{B}$ outputs the isogeny $\phi^* = \hat{\phi}_{\mathbf{x}'} \circ \phi_{\mathbf{x}}$, where $\hat{\phi}_{\mathbf{x}'} : E^* \rightarrow E_0$ is the dual of $\phi_{\mathbf{x}'}$.

We now claim that ϕ^* output by $\mathcal{B}$ is a valid solution to the **OneEnd** problem. Let $H(\mathbf{x}) = H(\mathbf{x}') = (E^*, R^*, S^*)$. Also, let $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{x}' = (x'_1, \dots, x'_n)$. This implies that $\mathcal{B}$ obtains two isogenies $\phi_{\mathbf{x}} : E_0 \rightarrow E^*$ and $\phi_{\mathbf{x}'} : E_0 \rightarrow E^*$ such that the respective kernels $\ker(\phi_{\mathbf{x}})$ and $\ker(\phi_{\mathbf{x}'})$ are as follows:

$$\ker(\phi_{\mathbf{x}}) = \bigoplus_{j \in [n]} \langle P_{j, x_j} \rangle, \quad \ker(\phi_{\mathbf{x}'}) = \bigoplus_{j \in [n]} \langle P_{j, x'_j} \rangle$$

Now, consider the dual isogeny $\hat{\phi}_{\mathbf{x}'} : E^* \rightarrow E_0$, and the composition $\phi^* = \hat{\phi}_{\mathbf{x}'} \circ \phi_{\mathbf{x}}$. Clearly, ϕ^* is an endomorphism of E_0 . Further, since $\mathbf{x} \neq \mathbf{x}'$, there is at least one index $i \in [n]$ such that $x_i \neq x'_i$ (i.e., $x'_i = 1 - x_i$). This implies that the kernel of $\phi_{\mathbf{x}}$ includes a subgroup of order d_i generated by P_{i, x_i} , while the kernel of $\phi_{\mathbf{x}'}$ includes a subgroup of order d_i generated by $P_{i, x'_i} = P_{i, 1-x_i}$. Since these subgroups are distinct and d_i is coprime to all other d_j for $j \in [n] \setminus \{i\}$, the kernels $\ker(\phi_{\mathbf{x}})$ and $\ker(\phi_{\mathbf{x}'})$ are distinct.

Therefore, the isogeny $\phi^* = \hat{\phi}_{\mathbf{x}'} \circ \phi_{\mathbf{x}}$ is not a scalar multiplication. Indeed, writing $D = \prod_{j \in [n]} d_j$, we have $\deg(\phi^*) = D^2$, so if $\phi^* \in \mathbb{Z}$ then necessarily $\phi^* = [\pm D]$. On the other hand, the distinct d_i -primary kernel components of $\phi_{\mathbf{x}}$ and $\phi_{\mathbf{x}'}$ at any index i for which $x_i \neq x'_i$ imply that $\hat{\phi}_{\mathbf{x}'} \circ \phi_{\mathbf{x}}$ does not act as scalar multiplication by $\pm D$ on the corresponding d_i -torsion. Hence $\phi^* \notin \mathbb{Z}$, and so it defines a non-trivial element of $\text{End}(E_0)/\mathbb{Z}$, as required for a valid solution to the **OneEnd** challenge.

4.5 Shuffled Order Hash Evaluation

In this section, we show that there are two (indeed, multiple) equivalent ways to compute our hash function. In more detail, our hash algorithm essentially processes the input $\mathbf{x} = (x_1, \dots, x_n)$, as an unordered set of tuples $(i, x_i)_{i \in [n]}$ so that the order of processing does not matter. We emphasize that this does not break collision-resistance because the bit x_i is tied to the position i . Looking ahead, this property will be crucial in using the hash function to design a batch encryption scheme. Below, $\text{pp} \leftarrow \text{Setup}(1^\lambda, n)$ as in the construction above.

Lemma 3 (Shuffled Order Hash Evaluation). *For any security parameter $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, and any input $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, let $\text{pp} \leftarrow \text{Setup}(1^\lambda, n)$. For every permutation σ of $[n]$, [Algorithm 1](#) produces the same hash value $h_{\mathbf{x}} = \text{Hash}(\text{pp}, \mathbf{x})$ when its outer loop processes the indices in the order $\sigma(1), \dots, \sigma(n)$.*

Proof. For every ordering, the composite isogeny computed by [Algorithm 1](#) has kernel $K_x = \bigoplus_{j \in [n]} \langle P_{j, x_j} \rangle$. Indeed, the subgroups $\langle P_{j, x_j} \rangle$ have pairwise coprime orders d_j , so successively quotienting by their pushforwards yields the same quotient isogeny up to a unique compatible codomain isomorphism. In particular, any two orderings produce isomorphic marked triples $(E_x, \phi_x(R_0), \phi_x(S_0))$. By the canonicalization convention of [Algorithm 1](#), these marked triples are mapped to the same canonical representative. Hence every ordering returns exactly the same hash value $h_x = \text{Hash}(\text{pp}, x)$, as desired.

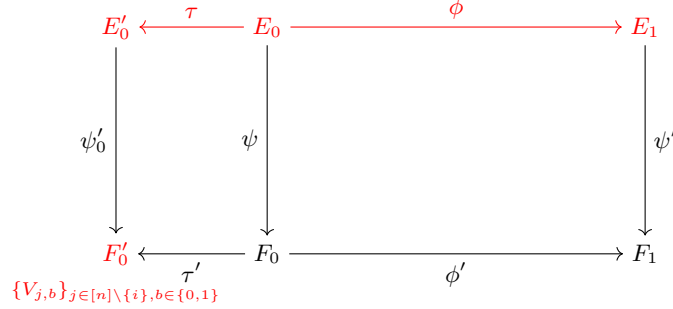


Fig. 2. Representation of the CDHwMT assumption. The items marked in red are provided to the PPT adversary $\mathcal{A}$ by the challenger in the CDHwMT game. The adversary’s goal is to compute $j(F_1)$. Intuitively, this is hard because the adversary does not know the (scaled) torsion points $V_{i,0}$ and $V_{i,1}$.

5 The CDHwMT Assumption

In this section, we introduce a plausibly post-quantum isogeny-based hardness assumption, which we call the CDH with Mismatched Torsion (CDHwMT) assumption. This assumption forms the basis of the batch encryption and PIR constructions presented in this paper. A pictorial depiction of the assumption appears in Fig. 2. In Section 5.2, we formally analyze the hardness of the CDHwMT assumption in the Algebraic Isogeny Model (AIM) [1]. Concretely, we show that, in the AIM, the hardness of CDHwMT reduces to the hardness of the isogeny with scaled torsion assumption that underlies binSIDH [13] and many other cryptosystems [32, 46, 16, 68, 15].

5.1 The Assumption

Let λ be a security parameter and let $n = \text{poly}(\lambda)$. Let $N = 2^{2\lambda}$ and let p be a prime of the form $p = Nf - 1$, where f is a small co-factor needed for primality. Let $D = \{d_1, \dots, d_n\}$ be a set of odd integers such that each $d_i = O(2^{2\lambda})$ is B -powersmooth, for a smoothness bound $B \approx 2n\lambda = \text{poly}(\lambda)$, and all d_i are pairwise coprime. Concretely, for each $i \in [n]$, we have $d_i = \prod_{k=1}^{L_i} \ell_{i,k}$, where each $\ell_{i,k} < B$ is a distinct prime, and each L_i is chosen to ensure that $d_i = O(2^{2\lambda})$. Consider the following game $\text{G}_{\text{CDHwMT}}(\lambda, n, \text{Adv})$ between a challenger $\mathcal{C}$ and a probabilistic polynomial-time (PPT) adversary $\mathcal{A}$.

Game $\text{G}_{\text{CDHwMT}}(\lambda, n, \text{Adv})$:

The challenger does the following:

- Sample a uniformly random¹⁰ supersingular elliptic curve E_0 over $\mathbb{F}_{p^2}$ by computing a long random walk in the supersingular isogeny graph.

¹⁰ To be precise, the game samples the curve E_0 from a distribution whose statistical distance to the uniform one is negligible.

- For each $j \in [n]$, generate a basis $U_{j,0}, U_{j,1}$ of $E_0[d_j]$. Since d_j is B -powersmooth (as explained in [Section 4.1](#)), each point $U_{j,b}$ for $b \in \{0,1\}$ can be decomposed as $U_{j,b} = \sum_{k=1}^{L_j} U_{j,b,k}$, where each $U_{j,b,k}$ has order $\ell_{j,k}$ and is thus defined over an extension field of $\mathbb{F}_{p^2}$ of index $O(\ell_{j,k}) < O(B) \approx O(2n\lambda)$, which allows it to have an efficient representation.
- Sample R_0, S_0 to be a torsion basis of $E_0[N]$, and set

$$\text{pp} = ((p, N, B, E_0, (R_0, S_0)), \{\ell_{j,k}\}_{j \in [n], k \in [L_j]}, \{U_{j,b}\}_{j \in [n], b \in \{0,1\}})$$

where each $U_{j,b}$ is represented as the tuple $\{U_{j,b,k}\}_{k \in [L_j]}$.

- Compute the kernel $K_\phi = \oplus_{i \in [n]} \langle U_{i,0} \rangle$. Compute the isogeny $\phi : E_0 \rightarrow E_1$ of degree $D = \prod_{i=1}^n d_i$ with kernel K_ϕ . Compute $R_1 = \phi(R_0)$ and $S_1 = \phi(S_0)$. Canonically normalize the marked triple (E_1, R_1, S_1) according to the convention of [Algorithm 1](#), applying the resulting isomorphism also to the codomain of ϕ .
- Sample uniformly random $r \leftarrow_R [0, N-1]$ and compute the kernel $K_\psi = \langle R_0 + [r]S_0 \rangle$. Compute the isogeny $\psi : E_0 \rightarrow F_0$ of degree N .
- Compute the pushforward isogeny $\psi' = [\phi]_* \psi : E_1 \rightarrow F_1$ as the isogeny with kernel $\ker \psi' = \langle \phi(R_0) + [r]\phi(S_0) \rangle$.
- Pick a uniformly random $i \leftarrow_R [n]$ and compute the isogeny $\tau : E_0 \rightarrow E'_0$ of degree d_i with kernel $\langle U_{i,1} \rangle$.
- Compute its pushforward $\tau' : F_0 \rightarrow F'_0$ under ψ by computing the isogeny with kernel $\ker(\tau') = \langle \psi(U_{i,1}) \rangle$.
- For every $j \in [n] \setminus \{i\}$ and every $b \in \{0,1\}$, compute the scaled action image of ψ on the d_j -torsion by sampling random scalars $\alpha_j, \beta_j \leftarrow_R (\mathbb{Z}/d_j\mathbb{Z})^\times$ and computing the points

$$V_{j,0} = [\alpha_j]\tau'\psi(U_{j,0}), \quad V_{j,1} = [\beta_j]\tau'\psi(U_{j,1}).$$

The challenger provides the following tuple to the adversary $\mathcal{A}$

$$(\text{pp}, i, (E_1, R_1, S_1), F'_0, \{V_{j,b}\}_{j \in [n] \setminus \{i\}, b \in \{0,1\}})$$

Eventually, the adversary $\mathcal{A}$ outputs y . If $y = j(F_1)$, the game outputs 1. Else, the game outputs 0.

Definition 15 (The CDHwMT Assumption). *We say that the CDHwMT assumption holds if for any security parameter $\lambda \in \mathbb{N}$, any $n = \text{poly}(\lambda)$, and any PPT adversary $\mathcal{A}$, we have*

$$\Pr[\text{G}_{\text{CDHwMT}}(\lambda, n, \text{Adv}) = 1] \leq \text{negl}(\lambda)$$

Remark 2. We remark that, even though the challenger does not explicitly provide the elliptic curve E'_0 and the isogeny $\tau : E_0 \rightarrow E'_0$ to the adversary $\mathcal{A}$ in the CDHwMT game, these are implicitly known to $\mathcal{A}$, since $\ker \tau = \langle U_{i,1} \rangle$, and $U_{i,1}$ is provided to $\mathcal{A}$ as part of pp .

On the Hardness of CDHwMT. We provide here an intuitive explanation for why the CDHwMT hardness assumption holds. Observe that the CDHwMT game effectively requires $\mathcal{A}$ to solve a CDH instance of the form (E_0, E_1, F_0, F_1) along with the corresponding (scaled) torsion points. This is similar to the CDH-style assumption underlying the security of binSIDH [13], albeit with two differences:

- $\mathcal{A}$ knows the isogeny $\phi : E_0 \rightarrow E_1$, which it would not know in a typical CDH game. However, the isogeny ϕ has degree that is different from the order of the scaled torsion points $\{V_{j,b}\}_{j \neq i}$ on the curve F'_0 that are provided to $\mathcal{A}$.
- Crucially, $\mathcal{A}$ knows neither the curve F_0 nor the scaled d_i -torsion points $(V_{i,0}, V_{i,1})$.

As a result, in order to compute F_1 , $\mathcal{A}$ effectively needs to compute one of the following (see Fig. 2 for an illustration): (i) either compute $\psi' : E_1 \rightarrow F_1$, where ψ' is the pushforward of the secret vertical isogeny ψ under ϕ ; or (ii) compute $\rho : F'_0 \rightarrow F_1$, where ρ is the pushforward of the isogeny $\phi' : E'_0 \rightarrow E_1$ under the secret vertical map ψ . Intuitively, both of these strategies are computationally hard without knowing either the secret vertical isogeny ψ or the scaled d_i -torsion points $(V_{i,0}, V_{i,1})$ (neither of which are known to $\mathcal{A}$ in the CDHwMT game). See Section 5.2 for a formal analysis in the Algebraic Isogeny Model.

5.2 Justifying CDHwMT in the Algebraic Isogeny Model

In this section, we analyze the hardness of the CDHwMT problem against algebraic adversaries. More concretely, we show that, in the Algebraic Isogeny Model [1], an algebraic solver for the CDH with Mismatched Torsion (CDHwMT) problem in Definition 15 yields a solver for either the EndRing problem or the isogeny problem with scaled torsion, which we restate here in a form adapted to our setting. This problem has appeared in the literature under the names SSIP-A [13, Problem 1] and isogeny problem with level structure [37], and it is the basis of several cryptosystems [32, 46, 16, 13, 68, 15].

Problem 5 (The (N, d) -Isogeny with Scaled Torsion (IwST) Problem). Let $N = 2^{2\lambda}$ and p a prime of the form $p = Nf - 1$, where f is a cofactor needed for primality. Let E_0 be a uniformly random supersingular elliptic curve defined over $\mathbb{F}_{p^2}$, and let $\phi : E_0 \rightarrow E_1$ be a random N -isogeny. Let P_0, Q_0 be a basis of $E_0[d]$, and define

$$P_1 = [\alpha]\phi(P_0), \quad Q_1 = [\beta]\phi(Q_0),$$

where α, β are random values in $(\mathbb{Z}/d)^*$.

Given the degree N , the curve E_0 with points P_0, Q_0 , and the curve E_1 with points P_1, Q_1 , compute ϕ .

The reduction from the IwST problem incurs a superpolynomial loss factor.¹¹ More specifically, we first introduce an easier variant of the IwST problem, which

¹¹ The loss is $2^{\Theta(\lambda/(C \log \lambda))}$. Increasing the constant C decreases the hidden constant in the exponent, but for every fixed C the loss remains superpolynomial.

we call the Pushforward with Scaled Torsion (PwST) problem. In PwST, the setup is the same as in the IwST problem, but the adversary is asked to compute a pushforward under the secret isogeny rather than recover the secret isogeny itself. While the two problems are generally equivalent, our parameters include isogeny degrees that are highly composite; this means that the IwST problem reduces to the PwST problem only with a superpolynomial reduction. We then show that a CDHwMT solver yields, in polynomial time, a solver for either EndRing or PwST; combining this with the previous reduction gives the corresponding result from EndRing or IwST.

Problem 6 (The (N, d, m) -Pushforward with Scaled Torsion (PwST) Problem). Let $(E_0, P_0, Q_0, F_0, P_1, Q_1)$ be an IwST instance as in [Problem 5](#), and denote its secret isogeny by $\psi : E_0 \rightarrow F_0$. Sample a cyclic isogeny $\nu : E_0 \rightarrow E_\nu$ of degree m . Given the IwST instance and an efficient representation of ν , compute the pushforward $\nu' = [\psi]_* \nu : F_0 \rightarrow F_\nu$.

The Algebraic Isogeny Model (AIM). The Algebraic Isogeny Model (AIM), introduced in [\[1\]](#), is an algebraic model for isogenies and supersingular elliptic curves, akin to the Algebraic Group Model (AGM) [\[47\]](#) in the group setting. It allows one to analyze the generic hardness of isogeny-based problems over supersingular elliptic curves against algebraic adversaries. The AIM works with arbitrary isogenies over $\mathbb{F}_{p^2}$ (including over non-oriented supersingular elliptic curves), and is significantly more general than the Algebraic Group Action Model (AGAM) [\[43\]](#), which was restricted to oriented supersingular elliptic curves.

Analyzing CDHwMT in the AIM. We state and prove the following theorem.

Theorem 7. *In the Algebraic Isogeny Model, for any algebraic adversary $\mathcal{A}$ that solves the CDHwMT game ([Section 5.1](#)), there exists an algebraic algorithm $\mathcal{B}$ that solves either the Pushforward with Scaled Torsion problem or the OneEnd_p problem such that*

$$\text{Adv}_{\text{CDHwMT}}(\mathcal{A}) \leq 9 \cdot (\text{Adv}_{\text{OneEnd}}(\mathcal{B}) + \text{Adv}_{\text{PwST}}(\mathcal{B})),$$

where $\text{Adv}_{\text{CDHwMT}}(\mathcal{A})$ is $\mathcal{A}$'s advantage in the CDH with Mismatched Torsion game ([Section 5.1](#)) with parameters (N, D) . The quantity $\text{Adv}_{\text{PwST}}(\mathcal{B})$ is $\mathcal{B}$'s advantage in solving the Pushforward with Scaled Torsion problem ([Problem 6](#)) with parameters $(N, D/d_i, d_i^2)$, averaged over the same uniformly random index $i \in [n]$ as in the CDHwMT game.

Proof. The adversary $\mathcal{A}$ is algebraic (since we are in the Algebraic Isogeny Model), which means that when $\mathcal{A}$ produces the solution F_1 , they have to “explain” it by providing (an efficient representation of) an isogeny to F_1 from one of the input curves (E_0, E'_0, E_1, F'_0) or a curve E_{end} of known endomorphism ring.

The proof uses a similar approach as the proof of [\[1, Theorem 6.7\]](#). The reduction first guesses which explanation the adversary is going to provide (i.e. the domain of the explanation isogeny). Then, the reduction also guesses whether the explanation isogeny ϕ_{expl} is compatible, which we define to mean:

- If the explanation isogeny ϕ_{expl} has domain E_1 , ϕ_{expl} is compatible if it is of the form $\phi_{\text{expl}} = [\alpha]\psi'$, for some integer α .
- If ϕ_{expl} has domain E_0 , it is compatible if it is of the form $\phi_{\text{expl}} = [\alpha]\psi'\phi$, for some integer α .
- If ϕ_{expl} has domain E'_0 , it is compatible if it is of the form $\phi_{\text{expl}} = [\alpha]\psi'\phi\hat{\tau}$, for some integer α .
- If ϕ_{expl} has domain F'_0 , it is compatible if it is of the form $\phi_{\text{expl}} = [\alpha]\phi'\hat{\tau}'$, for some integer α .
- If ϕ_{expl} has domain E_{end} , it is always compatible.

If the guess is incorrect, the reduction aborts; this explains the reduction loss of 9 since there are $2 \times 4 + 1$ cases (five possible domains of explanation, for four of which the explanation can be compatible or not) among which the reduction needs to guess. The algorithm $\mathcal{B}$ then handles the cases separately.

The explanation isogeny is not compatible. In this case, the adversary $\mathcal{A}$ can be used to solve the **OneEnd** problem. Let E_{chlg} be the challenge curve where the reduction needs to find a non-trivial endomorphism. Given E_{chlg} , the reduction samples a CDHwMT challenge where $E_0 = E_{\text{chlg}}$.¹² When the adversary responds with F_1 and an explanation isogeny ϕ_{expl} , it is possible to extract a non-trivial endomorphism. For instance, say the explanation isogeny ϕ_{expl} has domain E_1 . Thus, $\hat{\phi}_{\text{expl}} \circ \psi'$ is an endomorphism of E_1 (the reduction knows ψ' because it generated the CDHwMT challenge itself), and this endomorphism is non-trivial because by assumption the explanation is not compatible, which means that $\psi' \neq [\alpha]\phi_{\text{expl}}$. For the other domains, the same argument gives the endomorphisms

$$\hat{\phi}_{\text{expl}} \circ \psi' \circ \phi, \quad \hat{\phi}_{\text{expl}} \circ \psi' \circ \phi \circ \hat{\tau}, \quad \hat{\phi}_{\text{expl}} \circ \phi' \circ \hat{\tau}',$$

on E_0 , E'_0 , and F'_0 , respectively. These endomorphisms are non-trivial for the same reason, and can be transported to E_{chlg} using the isogenies computed by the reduction when generating the CDHwMT instance.

The explanation isogeny is compatible. For the cases with domain E_1 , E_0 , E'_0 , or F'_0 , we use the following PwST embedding. Let

$$(E'_0, P_0, Q_0, F'_0, P_1, Q_1, \nu)$$

be a PwST instance with parameters $(N, D/d_i, d_i^2)$ for a uniformly random $i \in [n]$, secret isogeny $\psi'_0 : E'_0 \rightarrow F'_0$, and $\deg(\nu) = d_i^2$. The reduction uses this same index i when constructing the CDHwMT challenge, so this embedding incurs no

¹² It is possible to randomize E_0 by computing E_0 as the codomain of a long isogeny starting from E_{chlg} . In this way, E_0 is guaranteed to be (statistically close to) uniformly random; given a non-trivial endomorphism on E_0 , it is then efficient to translate this into a non-trivial endomorphism on E_{chlg} .

additional $1/n$ loss. For each $j \neq i$, let $P_{j,0}, P_{j,1}$ be the d_j -order components¹³ of P_0, Q_0 , respectively, and let $V_{j,0}, V_{j,1}$ be the corresponding components of P_1, Q_1 . Factor $\nu = \nu_2 \circ \nu_1$, where both factors have degree d_i , use the codomain of ν_1 as E_0 , and set $\hat{\tau} = \nu_1$ and $\tau = \hat{\nu}_1$. For $j \neq i$, define

$$U_{j,b} = [d_i^{-1}] \hat{\tau}(P_{j,b}),$$

where the inverse is taken modulo d_j ; then $\tau(U_{j,b}) = P_{j,b}$. Let $U_{i,1}$ generate $\ker(\tau)$ and $U_{i,0}$ generate $\ker(\nu_2)$. Since ν is cyclic, $U_{i,0}, U_{i,1}$ form a basis of $E_0[d_i]$.

Let κ be the (D/d_i) -isogeny with kernel $\langle P_0 \rangle$ and define

$$\phi = ([\nu]_* \kappa) \circ \nu_2,$$

using the codomain of ϕ as E_1 . Then $\ker(\phi) = \bigoplus_j \langle U_{j,0} \rangle$ and

$$\phi \circ \hat{\tau} = ([\nu]_* \kappa) \circ \nu.$$

The pullback square satisfies

$$\tau' \psi(U_{j,b}) = \psi'_0 \tau(U_{j,b}) = \psi'_0(P_{j,b}),$$

so the target components $V_{j,b}$ have the required distribution: a uniform scaling modulo D/d_i gives independent uniform scalings modulo the pairwise coprime d_j 's. The same pullback square with the secret isogeny ψ'_0 defines the hidden curves and isogenies F_0, ψ , and τ' , which the reduction does not need to compute. Finally, the reduction samples a basis R_0, S_0 of $E_0[N]$ and sets $R_1 = \phi(R_0)$ and $S_1 = \phi(S_0)$; the remaining public values are sampled as in the CDHwMT game.

The reduction now uses the explanation as follows.

- **The explanation isogeny has domain E_1 .** Here $\phi_{\text{expl}} = [\alpha] \psi'$. Composing with $\phi \circ \hat{\tau}$ gives

$$\phi_{\text{expl}} \circ \phi \circ \hat{\tau} = [\alpha] \psi' \circ \phi \circ \hat{\tau} = [\alpha] \phi' \circ \hat{\tau}' \circ \psi'_0.$$

After removing the scalar part, the reduction separates the coprime N - and $d_i D$ -degree factors and obtains $\phi' \circ \hat{\tau}'$. Its d_i^2 -degree factor is exactly $[\psi'_0]_* \nu$, which solves the PwST instance.¹⁴

- **The explanation isogeny has domain E_0 or E'_0 .** If the domain is E_0 , the reduction composes ϕ_{expl} with $\hat{\tau}$; if the domain is E'_0 , no additional composition is needed. In both cases it obtains $[\alpha] \phi' \circ \hat{\tau}' \circ \psi'_0$ and proceeds exactly as in the previous case.

¹³ More precisely, for a point P , P_j is the d_j -order component of P if d_j divides $\text{ord}(P)$ and P_j is defined as $P_j = [(\text{ord}(P)/d_j)^{-1} \bmod d_j][\text{ord}(P)/d_j]P$. This implies that $\text{ord}(P_j) = d_j$ and $\sum_{j \neq i} P_j = P$.

¹⁴ In this case, and similarly when the explanation has domain E_0 or E'_0 , the same factorization also reveals the N -isogeny ψ'_0 . Thus, a direct reduction to IwST is possible; we use PwST to keep the proof uniform.

- **The explanation isogeny has domain F'_0 .** Here

$$\phi_{\text{expl}} = [\alpha]\phi' \circ \hat{\tau}'.$$

After removing the scalar part, the reduction extracts the d_i^2 -degree factor $[\psi'_0]_*\nu$ and solves the PwST instance.

- **The explanation isogeny has domain E_{end} .** In this case, the adversary $\mathcal{A}$ can be used to solve the EndRing problem (and thus also the OneEnd problem). The reduction receives a challenge curve E_{chlg} and samples a CDHwMT challenge where $E_0 = E_{\text{chlg}}$, which is passed to the adversary $\mathcal{A}$. When $\mathcal{A}$ responds with F_1 and produces an isogeny $\phi_{\text{expl}} : E_{\text{end}} \rightarrow F_1$, the reduction computes a connecting isogeny between E_{chlg} and E_{end} as $\hat{\phi}_{\text{expl}} \circ \psi' \circ \phi$. Transporting a non-scalar endomorphism of E_{end} along this isogeny gives a non-trivial endomorphism of E_{chlg} .

Lemma 4. *Let $d_i = \prod_{k=1}^{L_i} \ell_{i,k}$, where all $\ell_{i,k}$ are distinct odd primes, and assume that $d_i^4/N \geq 2^{\Omega(\lambda)}$. For any adversary $\mathcal{A}$ solving the $(N, D/d_i, d_i^2)$ -PwST problem, there exists an algorithm $\mathcal{B}$ solving the $(N, D/d_i)$ -IwST problem such that*

$$\text{Adv}_{\text{PwST}}(\mathcal{A}) \leq (2^{L_i} \text{Adv}_{\text{IwST}}(\mathcal{B}) + \text{negl}(\lambda))^{1/3}.$$

In particular, if $L_i = \Theta(\lambda/(C \log \lambda))$, then the reduction loss is $2^{\Theta(\lambda/(C \log \lambda))}$.

Proof. Let $\epsilon = \text{Adv}_{\text{PwST}}(\mathcal{A})$, and let $\psi'_0 : E'_0 \rightarrow F'_0$ be the secret isogeny in the IwST instance. The reduction samples three independent random points $T_1, T_2, T_3 \in E'_0[d_i^2]$ of order d_i^2 , lets ν_r be the cyclic d_i^2 -isogeny with kernel $\langle T_r \rangle$, and invokes $\mathcal{A}$ on the three corresponding PwST instances. Conditioned on a fixed IwST instance X , these three PwST instances are independent. Hence, if p_X is the success probability of $\mathcal{A}$ on X , the probability that all three answers are correct is

$$\mathbb{E}_X[p_X^3] \geq \mathbb{E}_X[p_X]^3 = \epsilon^3.$$

Assume from now on that the three answers are correct. They give the three image subgroups

$$\psi'_0(\langle T_1 \rangle), \quad \psi'_0(\langle T_2 \rangle), \quad \psi'_0(\langle T_3 \rangle).$$

For each prime $\ell \mid d_i$, look at the ℓ -order subgroups in $\langle T_1 \rangle$, $\langle T_2 \rangle$, and $\langle T_3 \rangle$. If these points are pairwise distinct, the argument of [14, Lemma 1] recovers the action of ψ'_0 on $E'_0[\ell^2]$ up to a common scalar. If they are not pairwise distinct, we simply discard the ℓ^2 -component. The probability that ℓ is discarded is at most $3/\ell$. It follows that, except with negligible probability, the discarded part has size only $2^{o(\lambda)}$. Thus we recover the action of ψ'_0 on $E'_0[m_{\text{good}}]$ up to a common scalar, where $m_{\text{good}} = d_i^2/2^{o(\lambda)}$. The assumption $d_i^4/N \geq 2^{\Omega(\lambda)}$ is guaranteed by our choice of parameters. Thus, with overwhelming probability, we have the image of ψ'_0 on torsion points of order m_{good} , up to a common scalar.

If the recovered action is $[\gamma]\psi'_0$, then the Weil pairing determines $\gamma^2 \bmod m_{\text{good}}$. For each good prime $\ell \mid d_i$ there are two possible square roots modulo ℓ^2 ,

and hence at most 2^{L_i} possible values of γ . Guessing the correct one gives the exact action of ψ'_0 on $E'_0[m_{\text{good}}]$. Since $m_{\text{good}}^2 > N$ and m_{good} is powersmooth, the SIDH attacks recover ψ'_0 [31, 66, 71].

6 Blind Batch Encryption from CDHwMT

In this section, we construct a blind BE scheme using the commutative hash function defined in the previous section, and argue its security based on the CDHwMT assumption from Section 5. Concretely, we prove the following theorem.

Theorem 8 (Blind BE from CDHwMT). *Under the CDHwMT assumption (Definition 15), there exists a batch encryption scheme satisfying correctness as in Definition 9, succinctness as in Definition 10, security as in Definition 11, and blindness as in Definition 12.*

6.1 Construction of Blind Batch Encryption

Let (Setup, Hash) denote our hash function from Section 4. Also, let (gl-enc, gl-dec) be a simple one-time encryption (OTE) mechanism [27] derived from the Goldreich-Levin (GL) hard-core predicate [53] (see Definition 14 for the formal description). The construction is described below.

BE.Setup($1^\lambda, n$) : Given λ $n = \text{poly}(\lambda)$, sample and output $\text{pp} \leftarrow \text{Setup}(1^\lambda, n)$ as described in Section 4.2, where

$$\text{pp} = ((p, N, B, E_0, R_0, S_0), \{\ell_{i,k}\}_{i \in [n], k \in [L_i]}, \{P_{i,b}\}_{i \in [n], b \in \{0,1\}})$$

BE.Gen($\text{pp}, \mathbf{x}$) : On input the hash key pp as described above and an input $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, output $h_{\mathbf{x}} = \text{Hash}(\text{pp}, \mathbf{x})$ as described in Section 4.2.

BE.Enc($\text{pp}, (h_{\mathbf{x}}, i), (\mu_0, \mu_1)$) : Given as input pp as described above, a hash value $h_{\mathbf{x}} = (E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}})$ for some unknown pre-image string $\mathbf{x} \in \{0, 1\}^n$, an index $i \in [n]$, and a message-pair $(\mu_0, \mu_1) \in \{0, 1\} \times \{0, 1\}$, compute the following for each $j \in [n]$: $d_j = \prod_{k=1}^{L_j} \ell_{j,k}$. Next, for each $c \in \{0, 1\}$:

- Sample a uniformly random scalar $r_c \leftarrow_R [0, N-1]$. Compute the secret kernel $K_{\psi,c} = \langle R_0 + [r_c]S_0 \rangle$. Compute the following isogeny with kernel $K_{\psi,c}$: $\psi_c : E_0 \rightarrow F_{0,c} = E_0/K_{\psi,c}$.
- Compute the image point $T_{i,c} = \psi_c(P_{i,c})$ on the curve $F_{0,c}$, and compute the following isogeny with kernel $\langle T_{i,c} \rangle$ following the same strategy as in Lines 3-8 in Algorithm 1: $\eta_c : F_{0,c} \rightarrow F_{i,c}$.
- For each index $j \in [n] \setminus \{i\}$ and bit $b \in \{0, 1\}$, do the following:
 - Compute the point $Q_{j,b,c}^* = \eta_c \psi_c(P_{j,b})$ on the curve $F_{i,c}$.
 - Sample a uniformly random blinding scalar $\alpha_{j,b,c} \leftarrow_R (\mathbb{Z}/d_j\mathbb{Z})^\times$.
 - Compute a scaled¹⁵ point $Q_{j,b,c} = [\alpha_{j,b,c}]Q_{j,b,c}^*$ on the curve $F_{i,c}$.

¹⁵ The scaling serves to prevent known torsion-point attacks [31, 66, 71].

- Compute the pushforward kernel on E_x as $K_{\mathbf{x},c} = \langle R_{\mathbf{x}} + [r_c]S_{\mathbf{x}} \rangle$, and compute the pushforward isogeny $\psi_{\mathbf{x},c} : E_{\mathbf{x}} \rightarrow F_{\mathbf{x},c}$ with kernel $K_{\mathbf{x},c}$.
- Encrypt the message μ_c using the j-invariant $j(F_{\mathbf{x},c})$ via the GL-OTE from Definition 14 as ¹⁶: $\mathbf{e}_c = \text{gl-enc}(j(F_{\mathbf{x},c}), \mu_c)$.
- Set $\text{subct}_{1,c} = (F_{i,c}, \{Q_{j,b,c}\}_{j \in [n] \setminus \{i\}, b \in \{0,1\}})$ and $\text{subct}_{2,c} = \mathbf{e}_c$.

Finally, output the ciphertext:

$$\text{ct} = (\text{subct}_1 = (\text{subct}_{1,0}, \text{subct}_{1,1}), \text{subct}_2 = (\text{subct}_{2,0}, \text{subct}_{2,1}))$$

BE.Dec(pp, (x, i), ct): Given pp as described above, an input $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, an index $i \in [n]$, and a ciphertext ct as described above, do the following:

- Construct the kernel $K'_{-i, \mathbf{x}, x_i} = \bigoplus_{j \in [n], j \neq i} \langle Q_{j, x_j, x_i} \rangle$.
- Compute the isogeny $\rho'_{-i, \mathbf{x}, x_i} : F_{i, x_i} \rightarrow F'_{\mathbf{x}, x_i}$ with kernel $K'_{-i, \mathbf{x}, x_i}$ following the same strategy as in Algorithm 1.
- Output $\mu' = \text{gl-dec}(j(F'_{\mathbf{x}, x_i}), \mathbf{e}_{x_i})$.

Efficiency. Each $P_{i,b}$ in the public parameter pp is represented efficiently as the tuple $\{P_{i,b,k}\}_{k \in [L_i]}$, as described in Section 4.2. This allows setup and hashing of our BE construction to run in time $\text{poly}(\lambda)$ by Lemma 1 and Lemma 2, respectively. Similarly, each $Q_{j,b,c}$ for $c \in \{0, 1\}$ in the ciphertext ct is (implicitly) represented efficiently as the tuple $\{Q_{j,b,c,k}\}_{k \in [L_j]}$. Hence, following analogous arguments to those in the proofs of Lemma 1 and Lemma 2, encryption and decryption of our BE construction also run in time $\text{poly}(\lambda)$.

6.2 Correctness

We state and prove the following theorem for the correctness of our BE scheme. The proof of this theorem crucially relies on the proof of Lemma 3 for equivalent hash computation. For a visual representation of why the correctness theorem holds, see Fig. 3.

Theorem 9 (Correctness of BE). *Our BE construction satisfies correctness as per Definition 9. More concretely, for any security parameter $\lambda \in \mathbb{N}$, any input length parameter $n = \text{poly}(\lambda)$, any $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$, any index $i \in [n]$, and any pair of message-bits $(\mu_0, \mu_1) \in \{0, 1\} \times \{0, 1\}$, let*

$$\text{pp} \leftarrow \text{Setup}(1^\lambda, n), \quad h_{\mathbf{x}} = \text{Gen}(\text{pp}, \mathbf{x}), \quad \text{ct} \leftarrow \text{Enc}(\text{pp}, (h_{\mathbf{x}}, i), (\mu_0, \mu_1))$$

Then, the following holds

$$\text{Dec}(\text{pp}, (\mathbf{x}, i), \text{ct}) = \mu_{x_i}$$

¹⁶ For simplicity of exposition, we assume that the j-invariant $j(F_{\mathbf{x},c})$ is canonically encoded as a bit-string of appropriate length that allows it to be fed as an input to gl-enc.

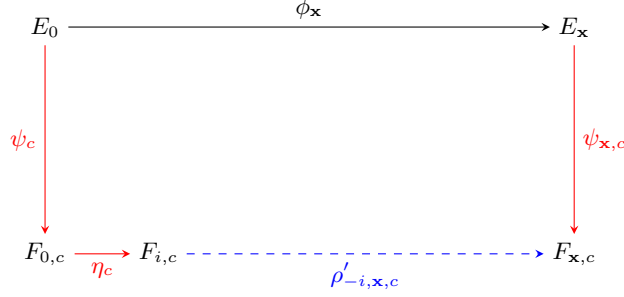


Fig. 3. Our blind BE construction. The black arrow indicates the hash isogeny $\phi_{\mathbf{x}}$ (associated with the hash input $\mathbf{x}$), while the red arrows indicate isogenies computed by the encryptor. The isogeny η_c has degree d_i and it is what binds the ciphertext to be decryptable only by a pre-image string $\mathbf{x} = (x_1, \dots, x_n)$ such that $x_i = c$. The blue dashed arrow indicates the isogeny computed by the decryptor, subject to $x_i = c$.

Proof. Let $h_{\mathbf{x}} = (E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}}) = \text{Gen}(\text{pp}, \mathbf{x}) = \text{Hash}(\text{pp}, \mathbf{x})$, and let $\phi_{\mathbf{x}} : E_0 \rightarrow E_{\mathbf{x}}$ be the corresponding isogeny. Define

$$K_{-i,\mathbf{x}} = \bigoplus_{j \in [n] \setminus \{i\}} \langle P_{j,x_j} \rangle \subseteq E_0,$$

let $\chi : E_0 \rightarrow E_{i,x_i}$ have kernel $\langle P_{i,x_i} \rangle$, and let $\phi_{-i,\mathbf{x}} : E_{i,x_i} \rightarrow E_{\mathbf{x}}$ have kernel

$$\chi(K_{-i,\mathbf{x}}) = \bigoplus_{j \in [n] \setminus \{i\}} \langle \chi(P_{j,x_j}) \rangle.$$

Since the orders d_j are pairwise coprime, the composition $\phi_{-i,\mathbf{x}} \circ \chi$ has the same kernel as $\phi_{\mathbf{x}}$. Hence, by [Lemma 3](#) and the canonicalization convention of [Algorithm 1](#), it yields the same canonical marked tuple (E_x, R_x, S_x) as ϕ_x .

Next, set $c = x_i$. The isogeny $\eta_{x_i} : F_{0,x_i} \rightarrow F_{i,x_i}$ is the pushforward of χ under $\psi_{x_i} : E_0 \rightarrow F_{0,x_i}$. Thus there is an isogeny $\psi_{i,x_i} : E_{i,x_i} \rightarrow F_{i,x_i}$ such that

$$\eta_{x_i} \circ \psi_{x_i} = \psi_{i,x_i} \circ \chi.$$

For every $j \neq i$, the blinding scalar α_{j,x_j,x_i} is a unit modulo d_j , and hence

$$\begin{aligned} \langle Q_{j,x_j,x_i} \rangle &= \langle \eta_{x_i} \psi_{x_i}(P_{j,x_j}) \rangle \\ &= \langle \psi_{i,x_i} \chi(P_{j,x_j}) \rangle. \end{aligned}$$

It follows that the decryptor's isogeny $\rho'_{-i,\mathbf{x},x_i} : F_{i,x_i} \rightarrow F'_{\mathbf{x},x_i}$ has kernel

$$\ker(\rho'_{-i,\mathbf{x},x_i}) = \bigoplus_{j \in [n] \setminus \{i\}} \langle Q_{j,x_j,x_i} \rangle = \psi_{i,x_i}(\ker(\phi_{-i,\mathbf{x}})).$$

Therefore, $\rho'_{-i,\mathbf{x},x_i}$ is the pushforward of $\phi_{-i,\mathbf{x}}$ under ψ_{i,x_i} . Pasting the two pushforward squares shows that

$$\rho'_{-i,\mathbf{x},x_i} \circ \eta_{x_i}$$

is the pushforward of $\phi_{\mathbf{x}} = \phi_{-i, \mathbf{x}} \circ \chi$ under ψ_{x_i} .

The encryptor computes $\psi_{\mathbf{x}, x_i} : E_{\mathbf{x}} \rightarrow F_{\mathbf{x}, x_i}$ from the same pushforward square. Hence

$$F'_{\mathbf{x}, x_i} = F_{\mathbf{x}, x_i}.$$

Consequently,

$$\mu' = \text{gl-dec}(j(F'_{\mathbf{x}, x_i}), \mathbf{e}_{x_i}) = \text{gl-dec}(j(F_{\mathbf{x}, x_i}), \mathbf{e}_{x_i}) = \mu_{x_i},$$

which proves [Theorem 9](#).

6.3 Succinctness

Theorem 10 (Succinctness of BE). *Our BE construction satisfies succinctness as per [Definition 10](#).*

Proof. Note that, in our BE construction, each sub-component of the hash output $h_{\mathbf{x}} = (E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}})$ has a fixed $\text{poly}(\lambda)$ -sized representation that only depends on the choice of p and N , both of which are parameters chosen independently of the bit length n of the input string $\mathbf{x}$. This completes the proof of [Theorem 10](#) for the succinctness of our BE scheme.

6.4 Security of Our Batch Encryption Scheme

We argue the security of our BE scheme based on the CDHwMT assumption from [Section 5](#). Concretely, we state and prove the following theorem.

Theorem 11 (Security of BE). *Assume that the CDHwMT assumption ([Definition 15](#)) holds. Then our BE construction satisfies security as per [Definition 11](#).*

Proof. To prove the above theorem, first consider the following auxiliary experiment (parameterized by λ and n) between a challenger and a stateful, PPT adversary $\mathcal{A}$ with respect to our BE construction.

1. $\mathcal{A}$ chooses a string $\mathbf{x} = (x_1, \dots, x_n) \in \{0, 1\}^n$ and an index $i \in [n]$, and sends $(\mathbf{x}, i)$ to the challenger.
2. The challenger generates $\text{pp} \leftarrow \text{BE.Setup}(1^\lambda, n)$ as per our BE construction, where

$$\text{pp} = ((p, N, B, E_0, R_0, S_0), \{\ell_{i,k}\}_{i \in [n], k \in [L_i]}, \{P_{i,b}\}_{i \in [n], b \in \{0,1\}})$$

and sends pp to $\mathcal{A}$.

3. The challenger internally generates the following hash as per our BE construction

$$h_{\mathbf{x}} = (E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}}) = \text{BE.Gen}(\text{pp}, \mathbf{x})$$

4. Next, the challenger generates the following (hash and message-independent) sub-ciphertext components exactly as described in **BE.Enc**:

$$\text{subct}_{1,0} = (F_{i,0}, \{Q_{j,b,0}\}_{j \in [n] \setminus \{i\}, b \in \{0,1\}})$$

$$\text{subct}_{1,1} = (F_{i,1}, \{Q_{j,b,1}\}_{j \in [n] \setminus \{i\}, b \in \{0,1\}})$$

Additionally, the challenger also uses $h_{\mathbf{x}}$ to internally generate $F_{\mathbf{x},0}$ and $F_{\mathbf{x},1}$ exactly as described in **BE.Enc** using $h_{\mathbf{x}}$. However, the challenger *does not* use these to generate the other, message-dependent sub-ciphertext components $\text{subct}_{2,0}$ and $\text{subct}_{2,1}$.

5. The challenger now provides $(\text{subct}_{1,0}, \text{subct}_{1,1})$ to $\mathcal{A}$.
6. Eventually $\mathcal{A}$ outputs a value γ . $\mathcal{A}$ wins the experiment if

$$\gamma = j(F_{\mathbf{x},1-x_i})$$

We now observe the following: suppose we prove that no PPT adversary can win the above experiment with non-negligible probability. Then, by the Goldreich-Levin theorem underlying the security of (gl-enc, gl-dec) (see [Section 3.8](#)), no PPT adversary can break the security of our BE scheme with non-negligible advantage. Hence, the proof of [Theorem 11](#) follows immediately from the proof of the following lemma.

Lemma 5. *Assume that the CDHwMT assumption ([Definition 15](#)) holds. Then, for any security parameter $\lambda \in \mathbb{N}$ and any input length parameter $n = \text{poly}(\lambda)$, any PPT adversary $\mathcal{A}$ wins the above auxiliary experiment with probability at most negligible in λ .*

Proof. Suppose there exists a PPT adversary $\mathcal{A}$ that wins the above auxiliary experiment with non-negligible probability. We construct a PPT algorithm $\mathcal{B}$ that breaks the CDHwMT assumption with non-negligible probability. More concretely, $\mathcal{B}$ does the following:

- $\mathcal{B}$ receives from the challenger in the CDHwMT security game the following tuple

$$(\text{pp}, i, (E_1, R_1, S_1), F'_0, \{V_{j,b}\}_{j \in [n] \setminus \{i\}, b \in \{0,1\}})$$

where we have

$$\text{pp} = ((p, N, B, E_0, (R_0, S_0)), \{\ell_{i,k}\}_{i \in [n], k \in [L_i]}, \{U_{i,b}\}_{i \in [n], b \in \{0,1\}})$$

- $\mathcal{B}$ poses as the challenger in the above auxiliary experiment and interacts with the adversary $\mathcal{A}$, as described below.
- Eventually, $\mathcal{B}$ outputs $j(F_1)$ with non-negligible probability.

$\mathcal{B}$ proceeds as follows:

- $\mathcal{B}$ receives from the adversary $\mathcal{A}$ in the above auxiliary experiment a bit vector $\mathbf{x} = (x_1, \dots, x_n) \in \{0,1\}^n$ and an index $i^* \in [n]$. If $i \neq i^*$, $\mathcal{B}$ aborts.

- Otherwise, for each $j \in [n]$, $\mathcal{B}$ sets

$$P_{j,x_j} = U_{j,0}, \quad P_{j,1-x_j} = U_{j,1}$$

- $\mathcal{B}$ outputs to the adversary $\mathcal{A}$ the hash key

$$\mathbf{pp} = ((p, N, B, E_0, (R_0, S_0)), \{\ell_{i,k}\}_{i \in [n], k \in [L_i]}, \{P_{j,b}\}_{j \in [n], b \in \{0,1\}})$$

- Next, $\mathcal{B}$ generates the following sub-ciphertext component:

$$\mathbf{subct}_{1,x_i} = (F_{i,x_i}, \{Q_{j,b,x_i}\}_{j \in [n] \setminus \{i\}, b \in \{0,1\}})$$

exactly as described in the real scheme $\mathbf{BE.Enc}$, with inputs $\mathbf{pp}$ (as provided to $\mathcal{A}$ above), $h_{\mathbf{x}} = (E_1, R_1, S_1)$ (received by $\mathcal{B}$ as part of the CDHwMT challenge), and the challenge index $i^* = i$. Note that generating the above sub-ciphertext component does not require inputting any messages.

- Next, $\mathcal{B}$ generates the other sub-ciphertext component $\mathbf{subct}_{1,1-x_i}$ using its input CDHwMT challenge as follows (this is the crux of the proof):
 - For each $j \in [n]$ such that $j \neq i$, $\mathcal{B}$ sets

$$Q_{j,x_j,1-x_i} = V_{j,0}, \quad Q_{j,1-x_j,1-x_i} = V_{j,1}$$

- Finally, it sets

$$\mathbf{subct}_{1,1-x_i} = (F_{i,1-x_i} = F'_0, \{Q_{j,b,1-x_i}\}_{j \in [n] \setminus \{i\}, b \in \{0,1\}})$$

- Eventually, the adversary $\mathcal{A}$ outputs γ .
- $\mathcal{B}$ also outputs γ as its output in the CDHwMT security game.

Probability of Abort. We first argue that the probability that $\mathcal{B}$ aborts is $1 - 1/n$. Since the index i is sampled uniformly at random by the challenger in the CDHwMT game, we have

$$\Pr[i = i^*] = 1/n$$

where $i^* \in [n]$ is the index chosen by the BE adversary. Hence, the probability that $\mathcal{B}$ aborts is $1 - 1/n$.

$\mathcal{A}$'s view. We now argue that, assuming $\mathcal{B}$ does not abort, $\mathcal{B}$'s simulation of $\mathcal{A}$'s view is identical to that in the auxiliary experiment described above. We argue this as follows:

- To begin with, it is easy to see that $\mathbf{pp}$ is distributed exactly as the output of $\mathbf{BE.Setup}(1^\lambda, n)$ in the auxiliary experiment.
- Next, we claim that

$$\mathbf{Hash}(\mathbf{pp}, x) = (E_1, R_1, S_1).$$

Indeed,

$$K_x = \bigoplus_{j \in [n]} \langle P_{j,x_j} \rangle = \bigoplus_{j \in [n]} \langle U_{j,0} \rangle = K_\phi.$$

Thus the hash isogeny and ϕ have the same kernel and hence define isomorphic marked quotient triples. Since both are normalized according to the canonicalization convention of [Algorithm 1](#), their canonical representatives coincide, giving $\mathbf{Hash}(\mathbf{pp}, x) = (E_1, R_1, S_1)$, as desired.

- Since subct_{1,x_i} is generated exactly as per **BE.Enc**, it is distributed as in the real auxiliary experiment.
- Next, we claim that the tuple

$$(F'_0, \{Q_{j,b,1-x_i}\}_{j \in [n] \setminus \{i\}, b \in \{0,1\}})$$

is distributed exactly as $\text{subct}_{1,1-x_i}$ in **BE.Enc**. To see this, observe that the CDHwMT game generates F'_0 and the points $V_{j,b}$ on F'_0 using the isogeny η , whose kernel is determined by the point $U_{i,1}$, while **BE.Enc** should generate $F_{i,1-x_i}$ and the points $Q_{j,b,1-x_i}$ on $F_{i,1-x_i}$ using the isogeny η' , whose kernel is determined by the point $U_{i,1}$. Since $P_{i,1-x_i} = U_{i,1}$, the distributions are identical, as desired.

This concludes the argument that $\mathcal{B}$'s simulation is perfect. Finally, we note that, by construction,

$$j(F_1) = j(F_{\mathbf{x},1-x_i})$$

Hence, in the event that $\mathcal{B}$ does not abort, it outputs $j(F_1)$ correctly, assuming that $\mathcal{A}$ outputs $\gamma = j(F_{\mathbf{x},1-x_i}) = j(F_1)$ correctly in the auxiliary experiment.

Let $\mathcal{A}$'s probability of winning the auxiliary experiment be $\epsilon > \text{negl}(\lambda)$. Let $\epsilon_{\mathcal{B}}$ be the probability that $\mathcal{B}$ outputs the correct $j(F_1)$. Then, we have

$$\epsilon_{\mathcal{B}} = \frac{\epsilon}{n}$$

Since $\epsilon > \text{negl}(\lambda)$ and $n = \text{poly}(\lambda)$, we have $\epsilon_{\mathcal{B}} > \text{negl}(\lambda)$. This completes the proof of [Lemma 5](#), and hence, the proof of [Theorem 11](#) for the security of our BE scheme.

6.5 Blindness

Theorem 12 (Blindness of BE). *Our BE construction satisfies blindness as per [Definition 12](#) information-theoretically.*

Proof. Consider the blindness experiment $\text{Expt}_{\beta, \mathcal{A}}^{\text{BE-BLIND}}(1^\lambda, n)$ in [Definition 12](#) for $\beta \in \{0, 1\}$, and observe the following:

- Note that in our BE construction, the first ciphertext sub-component

$$\text{subct}_1 = (\text{subct}_{1,0}, \text{subct}_{1,1})$$

is computed solely from pp , the index i , and the encryption randomness, and is entirely independent of both the hash value $h_{\mathbf{x}}$ and the message-pair (μ_0, μ_1) . Hence, our BE construction satisfies the decomposition property required by [Definition 12](#). Additionally, observe that, since subct_1 is independent of (μ_0, μ_1) , its marginal distribution is identical regardless of the value of $\beta \in \{0, 1\}$.

– Next, for our BE construction, consider the second ciphertext sub-component

$$\text{subct}_2 = (\text{subct}_{2,0}, \text{subct}_{2,1}) = (\mathbf{e}_0, \mathbf{e}_1)$$

From the properties of gl-enc (see [Section 3.8](#) and [Definition 14](#)), if μ_0 and μ_1 are uniformly random bits, then the distributions of both $\mathbf{e}_0$ and $\mathbf{e}_1$ are (information-theoretically) uniform.

It follows immediately that the distributions corresponding to $\beta = 0$ and $\beta = 1$ in $\text{Expt}_{\beta, \mathcal{A}}^{\text{BE-BLIND}}(1^\lambda, n)$ (as in [Definition 12](#)) are identical. Hence we have

$$\text{Adv}_{\mathcal{A}}^{\text{BE-BLIND}}(\lambda, n) = 0$$

even for an unbounded adversary $\mathcal{A}$. This completes the proof of [Theorem 12](#) for the blindness property of our BE scheme.

6.6 Applications of BE from CDHwMT

For the sake of completeness, we state as corollaries the various applications of (blind) BE that we obtain as a result from the CDHwMT assumption.

Anonymous IBE from CDHwMT. The following corollary follows immediately from [Theorem 8](#) and [Theorem 2](#).

Corollary 1 (Anonymous IBE from CDHwMT). *Under the CDHwMT assumption ([Definition 15](#)), there exists an anonymous IBE scheme satisfying [Definition 3](#).*

Leakage-resilient and KDM-secure PKE from CDHwMT. The following corollary follows immediately from [Theorem 8](#) and [Theorem 3](#).

Corollary 2 (Leakage-resilient and KDM-secure PKE from CDHwMT). *Under the CDHwMT assumption ([Definition 15](#)), there exists a public-key encryption scheme that simultaneously satisfies: (i) leakage resilience as in [Definition 6](#) for leakage rate $(1 - o(1))$, and (ii) KDM security as in [Definition 7](#) with respect to affine functions of the secret key.*

Further, by prior works [[12](#), [11](#)], the CDHwMT assumption also implies the existence of a PKE scheme satisfying KDM security as in [Definition 7](#) with respect to a much larger class of *arbitrary* functions of the secret key that can be represented via circuits with a priori bounded size.

Laconic OT from CDHwMT. Finally, the following corollary follows immediately from [Theorem 8](#) and [Theorem 4](#).

Corollary 3 (Laconic OT from CDHwMT). *Under the CDHwMT assumption ([Definition 15](#)), there exists a laconic oblivious transfer (OT) scheme satisfying [Definition 8](#).*

Finally, [39] demonstrates a generic transformation from IBE to hierarchical IBE (HIBE) [51, 20, 21]. Informally speaking, HIBE is a generalization of IBE to hierarchically structured identities that delegates to entities the ability to derive keys for identities below them in the hierarchy without contacting the central authority holding the master secret key. This yields the following corollary.

Corollary 4 (Hierarchical IBE from CDHwMT). *Under the CDHwMT assumption (Definition 15), there exists a hierarchical IBE scheme as defined in [51, 20, 21, 39].*

7 Single-Server PIR from CDHwMT

In this section, we show that our blind batch encryption scheme from Section 6 implies single-server private information retrieval (PIR) scheme with polylogarithmic (in the database size) communication.

7.1 Laconic PSI with Sender Security

Our starting point is a *restricted* version of the notion of laconic PSI (LPSI) from [7], wherein we *do not* require receiver security.

Definition 16 (Laconic PSI with Sender Security). *An LPSI scheme is a tuple of PPT algorithms $(\text{GenCRS}, R_1, S, R_2)$:*

$\text{crs} \leftarrow \text{GenCRS}(1^\lambda)$. *It takes as input the security parameter λ and outputs a common reference string crs .*
 $(\text{psi}_1, \text{st}) \leftarrow R_1(\text{crs}, S_R)$. *On input crs and the receiver's set $S_R \subseteq \{0, 1\}^\lambda$, it outputs a receiver-message psi_1 and a state st .*
 $\text{psi}_2 \leftarrow S(\text{crs}, \text{id}, \text{psi}_1)$. *On input crs , the sender's element $\text{id} \in \{0, 1\}^\lambda$ and psi_1 , it outputs a sender-message psi_2 .¹⁷*
 $I \leftarrow R_2(\text{crs}, \text{st}, \text{psi}_2)$. *On input crs , the receiver's state st , and the sender-message psi_2 , it outputs a set $I \subseteq S_R$ with $|I| \leq 1$.*

We require the following properties to hold for an LPSI scheme (where $M = |S_R| = \text{poly}(\lambda)$).

Correctness. for all S_R and id , with probability $1 - \text{negl}(\lambda)$ over all coins, R_2 outputs $\{\text{id}\}$ if $\text{id} \in S_R$ and $\emptyset$ otherwise.

Laconic Efficiency. $|\text{psi}_1| \leq \text{poly}(\lambda, \log M)$, $|\text{psi}_2| \leq \text{poly}(\lambda, \log M)$, and the running time of S is $\text{poly}(\lambda, \log M)$.

¹⁷ A sender holding a set would run the algorithm S once per element, reusing psi_1 .

Sender Security against Semi-honest Receivers. There exists a PPT simulator $\text{Sim}_{\ell\text{PSI}}$ such that for every $S_R \subseteq \{0,1\}^\lambda$ with $|S_R| = M$ and every $\text{id} \in \{0,1\}^\lambda \setminus S_R$,

$$(\text{crs}, \rho_R, S(\text{crs}, \text{id}, \text{psi}_1)) \approx_c (\text{crs}, \rho_R, \text{Sim}_{\ell\text{PSI}}(1^\lambda, \text{crs}, \rho_R, S_R)),$$

where ρ_R denotes the receiver's random coins and psi_1 is the first message determined by $(\text{crs}, S_R, \rho_R)$. That is, the receiver's entire view can be simulated without knowledge of id whenever $\text{id} \notin S_R$.

Note that when $\text{id} \in S_R$, the receiver is entitled to learn id , so no hiding is required in that case; this matches the PSI functionality.

7.2 Blind Garbled Circuits with Invalid-Label Propagation

We use the blind garbling scheme of [27] for our PIR construction. A garbling scheme $(\text{Garb}, \text{GCEval}, \text{Sim}_{\text{GC}})$ for circuits $C : \{0,1\}^m \rightarrow \{0,1\}^\ell$ satisfies the usual correctness and simulation security (for any C and input $\mathbf{w}$, $(\tilde{C}, \{\text{lb}_{i,w_i}\}) \approx_c \text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(C), C(\mathbf{w}))$, where $\text{topo}(C)$ denotes the topology of C), and additionally:

Definition 17 (Blindness of Garbled Circuits [27, Definition 4.2]). *A garbling scheme is blind if for every circuit topology, sampling $y \leftarrow \{0,1\}^\ell$ uniformly at random, the simulator output $\text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(C), y)$ is computationally indistinguishable from a uniformly random string.*

Lemma 6 (Imported from [27, Lemma 4.1]). *Assuming any one-way function, a blind garbling scheme exists.*

Pseudorandomness under Invalid-Label Evaluation. In our application, a (blind) garbled circuit is sometimes evaluated on labels that do not correspond to a consistent valid input: while some input wires may carry an honest label perturbed by a fixed mask, at least one input wire carries an independently uniform string unrelated to either of its valid wire labels. We require that such an invalid label propagates through the garbled evaluation: whenever this wire influences an output, the corresponding carried labels become pseudorandom, and if its influence reaches every output, the entire decoded output should be computationally indistinguishable from uniform.

It turns out that we can obtain the above property from the PRF-based point-and-permute blind garbling scheme from [27, Section 4.2], albeit by applying standard domain separation of the PRF queries by gate, input position, and row-selection bits. More concretely, we replace the PRF input at the p -th input of gate g by (g, p, β_1, β_2) for domain separation. To ensure that an invalid input indeed influences every output, we additionally compile the circuit being garbled into a diffusion normal form: for every relevant input bit a and output bit y , we append a fresh functionally neutral dependency gadget computing $(y \oplus (a \wedge 0))$, using a constant-zero wire. This preserves the function computed by the circuit, while guaranteeing a syntactic path from a to y . The following lemma formalizes

the resulting invalid-label propagation property of the point-and-permute blind garbling scheme from [27, Section 4.2].

Lemma 7 (Pseudorandomness under Invalid-Label Evaluation). *Instantiate the point-and-permute blind garbling scheme from [27, Section 4.2] with the domain-separated PRF inputs specified above. Let Z be arbitrary PPT-sampleable state and, conditional on $Z = z$, let $C_z : \{0, 1\}^m \rightarrow \{0, 1\}^\ell$ be a polynomial-size circuit. Sample*

$$(\tilde{C}, \{\text{lb}_{i,b}\}_{i \in [m], b \in \{0,1\}}) \leftarrow \text{Garb}(C_z)$$

using fresh coins independent of Z . Let H, J be a partition of $[m]$, and let $(b_i, \delta_i)_{i \in H}$ with $\delta_i \in \{0, 1\}^{\ell_{\text{lb}}}$, where the partition, bits, and masks are determined by Z . Supply

$$L_i = \text{lb}_{i,b_i} \oplus \delta_i \quad (i \in H), \quad L_i \leftarrow \{0, 1\}^{\ell_{\text{lb}}} \quad (i \in J),$$

where the latter strings are mutually independent and independent of Z and the fresh garbling coins. Assume that C_z uses the diffusion compilation above, and that some $a \in J$ reaches every output through the fresh, pairwise distinct terminal dependency gadgets of that compilation. Then

$$(Z, \text{GCEval}(\tilde{C}, (L_1, \dots, L_m))) \approx_c (Z, U_\ell), \quad U_\ell \leftarrow \{0, 1\}^\ell.$$

Proof. Expose only the garbled-table rows read by the evaluation and generate all unused rows lazily. Mark wire a tainted and thereafter mark the output of every gate having a tainted input. At the first tainted gate, choose a tainted input occurrence. In particular, although the permutation bit contained in an invalid carried label may select an arbitrary garbled-table row, the honest seed corresponding to that selected row remains fresh and independent of the invalid carried string, except on the stated collision event. The honest seed associated with the row selected by its carried permutation bit is uniform and appears neither in Z nor among the supplied strings on the other input wires, except with negligible collision probability.

We note that the above remains true when an H -wire carries a masked honest label: its mask is fixed independently of the current garbling, and labels on distinct wires are independent. Replace, in a multi-query PRF hybrid, the selected honest seed's values at all outgoing-gate queries by independent uniform strings. Domain separation by the gate, input position, and two row bits makes these queries distinct, including under fan-out or when one wire occupies both input positions.

After this replacement, the newly tainted carried labels are jointly uniform and mutually independent: every newly tainted gate output is masked using a fresh PRF value corresponding to a distinct domain-separated query. Moreover, these values are independent of Z and of all honest wire labels whose outgoing queries have not yet been hybridized. This gives the following inductive invariant: after processing any prefix of the tainted frontier, the carried labels on the

processed tainted wires are jointly uniform and independent of Z and of the unprocessed honest-wire seeds. Domain separation by the gate, input position, and row-selection bits ensures that the PRF queries used at distinct occurrences remain distinct even under fan-out or when the same wire feeds both inputs of a gate. We now propagate this invariant gate-by-gate along the forward cone of a . By the diffusion compilation, for every output coordinate there is a fresh terminal dependency gadget lying in this forward cone, and these terminal gadgets are pairwise distinct across output coordinates. Hence the final tainted contribution to each output depends on a distinct fresh domain-separated PRF value. The resulting decoded output bits are therefore jointly uniform in the final hybrid.

Collisions between a carried seed and an honest seed contribute $O(|C|)2^{-\lambda}$, and a hybrid over the $O(|C|)$ domain-separated PRF queries contributes only the corresponding PRF advantage. Since Z is sampled before the current garbling coins, every PRF seed changed by this argument is hidden from Z , proving the asserted joint marginal distribution. This completes the proof of [Lemma 7](#).

Remark 3. Note that [Lemma 7](#) does not claim indistinguishability jointly with $\tilde{C}$, the supplied labels, or another evaluation of the same garbling.

7.3 BE Properties for PIR

In this section, we make explicit two structural features of our BE construction from [Section 6](#) that allow us to bootstrap it into a single-server PIR scheme via an adaptation of the approaches in [\[7, 25\]](#).

- **Interpretation as Hash Encryption.** We first explain how BE can be used to encrypt pairs of garbled wire labels, following the hash-encryption paradigm of [\[41, 7\]](#). For our BE construction from [Section 6](#), we additionally identify a class of well-formed hash values on which our encryption algorithm remains well-defined even without an explicitly known preimage.
- **PIR-friendly Blind Decomposition.** Recall that the standard blindness definition of BE ([Definition 12](#)) already requires the BE ciphertext to decompose into a component independent of the target hash value and plaintext messages (which we call the *offline* component henceforth) and a remaining message-dependent component (which we call the *online* component henceforth). We make this decomposition explicit for our construction and observe two stronger structural facts that are useful for realizing single-server PIR:
 - All of the large algebraically structured ciphertext components in our BE construction (in particular, elliptic curves and points) can be placed in the offline part, while the online part consists of only two bits.
 - Moreover, for independently uniform plaintext rows, this online component is information-theoretically uniform *even conditioned on the offline component*.

Looking ahead, this strengthened form of the blind decomposition is particularly useful for our PIR construction: instead of asking a garbled circuit to output an

entire BE ciphertext as in [Section 6.1](#), we generate the substantially larger offline component externally and have the garbled circuit compute only the short online component.

Our BE as a Hash Encryption Scheme. We will use our BE scheme to encrypt garbled wire labels: for each input wire, the two messages encrypted by BE are the two labels associated with that wire. Since our BE scheme natively encrypts pairs of bits, we encrypt the labels bit-by-bit using independent BE invocations. More formally, let $\ell_{\text{lb}} = \lambda + 1$ denote the total wire-label length in the point-and-permute garbling scheme from [Section 7.2](#) (any $\Theta(\lambda)$ label length gives the same asymptotics). Following the *hash encryption paradigm* from [\[41, 7\]](#), for a hash key pp with input length n , a hash value h , and a tuple of label pairs $\{\text{lb}_{i,b}\}_{i \in [n], b \in \{0,1\}}$ with $\text{lb}_{i,b} \in \{0,1\}^{\ell_{\text{lb}}}$, we write

$$\text{HEnc}(\text{pp}, h, \{\text{lb}_{i,b}\}) = \left\{ \text{ct}_{i,t} \leftarrow \text{BE.Enc}(\text{pp}, (h, i), (\text{lb}_{i,0}[t], \text{lb}_{i,1}[t])) \right\}_{i \in [n], t \in [\ell_{\text{lb}}]}.$$

In other words, each bit position t of each label pair is encrypted by an independent invocation of BE.Enc . Given any preimage $\mathbf{z} \in \{0,1\}^n$ of h , BE correctness ([Definition 9](#)) allows $\text{BE.Dec}(\text{pp}, (\mathbf{z}, i), \text{ct}_{i,t})$ to recover exactly the labels $\{\text{lb}_{i,z_i}\}_{i \in [n]}$, by [Theorem 9](#). Conversely, BE security ([Definition 11](#)) implies that the complementary labels $\{\text{lb}_{i,1-z_i}\}$ are computationally hidden (this follows from a standard hybrid argument over the $n\ell_{\text{lb}}$ positions). Finally, invoking BE blindness ([Definition 12](#)) allows extending the above property to the multi-ciphertext setting.

Well-formed Hash Values. When realizing PIR from our blind BE construction, it will be useful to evaluate the BE encryption algorithm on hash values without assuming that we explicitly know a preimage under BE.Gen . We therefore identify the syntactic conditions on a hash value that suffice for BE.Enc to be well-defined.

We say that a claimed hash value $h = (E, R, S)$ is *well-formed* with respect to pp , written $\text{ValidDigest}(\text{pp}, h) = 1$, if: E is a supersingular elliptic curve over $\mathbb{F}_{p^2}$ with $\#E(\mathbb{F}_{p^2}) = (p+1)^2$; $R, S \in E[N]$ both have exact order N ; and the Weil pairing $e_N(R, S)$ has order N (so that (R, S) is a basis of $E[N]$). Obviously, every honestly computed hash value is well-formed. Additionally, observe the following:

- All of the above well-formedness conditions are checkable in time $\text{poly}(\lambda)$: supersingularity by the deterministic test of Sutherland [\[75\]](#), and the order and pairing conditions directly, since $N = 2^{2\lambda}$.
- Further, BE.Enc is well-defined on every well-formed h (irrespective of whether h is honestly generated): the encryptor’s kernel point $R + [r]S$ always has exact order N because its first coordinate with respect to the basis (R, S) is 1, a unit modulo N .

Decomposable Blindness. The second crucial feature of our BE construction from [Section 6](#) is that the computation of a ciphertext using BE.Enc can be naturally

split into two parts. The first, offline part is independent of both the target hash value h and the plaintext messages, and can therefore be generated before either is known. All dependence on h and the plaintext messages is confined to a very short online part consisting of only two bits.

We additionally need a blindness property tailored to this decomposition. If the two plaintext rows are independently uniform, then the corresponding online component should itself be uniformly random, even conditioned on the offline component. The following lemma formalizes these two properties; the proof follows directly from the structure of our BE construction and the blindness argument of [Theorem 12](#).

Lemma 8 (PIR-Friendly Blind Decomposition of Our BE Scheme).

Let $\text{BE} = (\text{BE.Setup}, \text{BE.Gen}, \text{BE.Enc}, \text{BE.Dec})$ be the BE construction from [Section 6](#). There exist PPT algorithms Enc^{off} and Enc^{on} such that for every pp , every well-formed $h = (E, R, S)$, every index i , every message pair $(\mu_0, \mu_1) \in \{0, 1\} \times \{0, 1\}$, and every choice of encryption randomness ρ :

1. **Decomposition:** $\text{BE.Enc}(\text{pp}, (h, i), (\mu_0, \mu_1); \rho)$ can be written as the pair

$$\left(\underbrace{\text{Enc}^{\text{off}}(\text{pp}, i; \rho)}_{= (\text{subct}_{1,0}, \text{subct}_{1,1}, \alpha_0, \alpha_1)}, \underbrace{\text{Enc}^{\text{on}}(\text{pp}, h, i, (\mu_0, \mu_1); \rho)}_{= (\sigma_0, \sigma_1) \in \{0, 1\}^2} \right),$$

where $F_{h,c}$ is the codomain of the N -isogeny from E with kernel $\langle R + [r_c]S \rangle$ determined by ρ , and $\text{subct}_{2,c} = (\alpha_c, \sigma_c)$ with $\sigma_c = \langle \alpha_c, j(F_{h,c}) \rangle \oplus \mu_c$. In particular, Enc^{off} does not take h or (μ_0, μ_1) as input, and the full ciphertext is publicly reassembled from the two components.

2. **Independent-row Blindness:** For every well-formed h (not necessarily in the image of Gen), if $(\mu_0, \mu_1) \leftarrow \{0, 1\}^2$ consists of two independent uniform bits unknown to the distinguisher, then

$$(\text{Enc}^{\text{off}}(\text{pp}, i; \rho), \sigma_0, \sigma_1) \equiv (\text{Enc}^{\text{off}}(\text{pp}, i; \rho), u), \quad u \leftarrow \{0, 1\}^2,$$

where the two distributions are identical (information-theoretically), even given pp , h , i , and arbitrary side information independent of (μ_0, μ_1) .

Proof. Decomposition is immediate from the description of BE.Enc in [Section 6.1](#): the components $\text{subct}_{1,c} = (F_{i,c}, \{Q_{j,b,c}\})$ and the Goldreich-Levin seeds α_c are computed from pp , i and ρ alone; only the masked bits σ_c depend on the target h (through $F_{h,c}$) and on the plaintext messages (μ_0, μ_1) .

For independent-row blindness, the proof of [Theorem 12](#) applies verbatim: the offline components are independent of (h, μ_0, μ_1) by construction, and each bit $\sigma_c = \langle \alpha_c, j(F_{h,c}) \rangle \oplus \mu_c$ is one-time padded by the independent uniform bit μ_c . Thus (σ_0, σ_1) is exactly uniform, jointly and conditioned on the offline component. Since the argument never evaluates Gen , it holds for every well-formed h on which the two pushforwards are defined, which is all of them as noted above. The independence of μ_0 and μ_1 is essential; Item 2 does not apply to an equal-row pair (μ, μ) . This completes the proof of [Lemma 8](#).

7.4 Laconic PSI with Sender Security from CDHwMT

We now construct an LPSI protocol with sender security from the CDHwMT assumption (Definition 15). Our construction builds on the Merkle-tree-based LPSI framework of [7, Construction 1], instantiated using our collision-resistant hash function from Section 4 and our blind BE scheme from Section 6.

Importantly, however, this instantiation is inherently non-black-box: its correctness and security exploit concrete structural features of our hash and blind-BE constructions beyond merely invoking their security guarantees in a black-box way. In particular, we use the concrete form of well-formed hash outputs and the PIR-friendly shape of the blind decomposition established in Section 7.3, in which all large algebraic components are offline and the online component is only two bits. Together with the invalid-label propagation property of Section 7.2, these features allow us to adapt the framework of [7] and establish sender security under CDHwMT.

Notation. We first fix some notation. Throughout, we denote by $(\text{Setup}, \text{Hash})$ our hash function from Section 4, and by $\text{BE} = (\text{BE.Setup}, \text{BE.Gen}, \text{BE.Enc}, \text{BE.Dec})$ our BE construction from CDHwMT in Section 6, where

$$\text{BE.Enc}(\text{pp}, (h, i), (\mu_0, \mu_1); \rho) = \left(\text{Enc}^{\text{off}}(\text{pp}, i; \rho), \text{Enc}^{\text{on}}(\text{pp}, h, i, (\mu_0, \mu_1); \rho) \right),$$

as in Lemma 8. Recall that BE.Setup is essentially identical to the Setup in our hash function, so we use these interchangeably. Let $(\text{Garb}, \text{GCEval}, \text{Sim}_{\text{GC}})$ be a garbling scheme, instantiated with the point-and-permute garbling scheme as in Lemma 7, with domain-separated PRF inputs and the diffusion compilation described in Section 7.2.

By Theorem 10, there is a fixed polynomial upper bound, independent of the hash-input length, on the canonical encoding length of a hash value $(E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}})$. We pad canonical encodings to this bound and denote their common length by $\ell_h = \text{poly}(\lambda)$.

We handle every receiver-set size explicitly. The empty set has the trivial protocol in which R_1 publishes a fixed marker, S returns a fixed response, and R_2 always outputs $\emptyset$, so suppose $M_0 = |S_R| \geq 1$. Set $\ell_{\text{id}} = \lambda + 1$ and encode a real identity $\text{id} \in \{0, 1\}^\lambda$ internally as $\widehat{\text{id}} = 0 \parallel \text{id} \in \{0, 1\}^{\ell_{\text{id}}}$. Let $M = 2^{\lceil \log M_0 \rceil}$ and pad $\{0 \parallel \text{id} : \text{id} \in S_R\}$ to M elements using distinct dummy identities $1 \parallel u$ from the disjoint tagged copy of the universe. Sort this padded internal set as $\widehat{\text{id}}_1 < \dots < \widehat{\text{id}}_M$, and set $d = \log M$. A sender query is always real-tagged and therefore never equals a dummy; R_2 filters dummy leaves and removes the tag from a real output. Since $M < 2M_0$ unless M_0 is already a power of two, the padding changes neither the asymptotics nor the polynomial-regime union bounds. Moreover $d \leq \lambda$, so the CRS below contains every required level key. We use M for the padded size in the displayed construction below. For later reference, define the hash-input lengths

$$n_0 = \ell_{\text{id}} + 2\lambda, \quad n_w = 2\ell_h + \ell_{\text{id}} \quad (w \geq 1).$$

Circuits. For $\text{id} \in \{0, 1\}^{\ell_{\text{id}}}$ and $r, r' \in \{0, 1\}^\lambda$, define

$$F[\text{id}, r, r'](\text{id}', x, x') = \begin{cases} r & \text{if } \text{id} = \text{id}', \\ r' & \text{otherwise,} \end{cases}$$

a circuit on inputs of length $\ell_{\text{id}} + 2\lambda$ with output length λ . For a BE key pp' with input length m' , label pairs $\{\text{lb}_{i,b}\}_{i \in [m'], b \in \{0,1\}}$, and encryption randomness $\rho = \{\rho_{i,t}\}_{i \in [m'], t \in [\ell_{\text{lb}}]}$, first set

$$\tau^* = \begin{cases} h & \text{if } \text{id} \leq \text{id}'', \\ h' & \text{otherwise,} \end{cases}$$

$$\omega_{i,t}(\tau) = \text{Enc}^{\text{on}}(\text{pp}', \tau, i, (\text{lb}_{i,0}[t], \text{lb}_{i,1}[t]); \rho_{i,t}).$$

Define the circuit

$$V[\text{pp}', \text{id}, \{\text{lb}_{i,b}\}, \rho](h, h', \text{id}'')$$

$$= \begin{cases} \{\omega_{i,t}(\tau^*)\}_{i \in [m'], t \in [\ell_{\text{lb}}]} & \text{if } \text{ValidDigest}(\text{pp}', \tau^*) = 1, \\ 0^{2m'\ell_{\text{lb}}} & \text{otherwise.} \end{cases}$$

on inputs of length $2\ell_h + \ell_{\text{id}}$, with output in $\{0, 1\}^{2m'\ell_{\text{lb}}}$. Note that V computes only the online components; by Lemma 8 these are single bits per invocation, and V is implementable as a circuit of size $\text{poly}(\lambda)$ since Enc^{on} runs in time $\text{poly}(\lambda)$ (the efficiency analysis of Section 6 applies, as Enc^{on} performs one pushforward isogeny computation and inner products).

Construction. The construction of laconic PSI from our hash function and blind BE is as follows:

- $\text{GenCRS}(1^\lambda)$: For every $w \in [0, \lambda]$, sample $\text{pp}^{(w)} \leftarrow \text{BE.Setup}(1^\lambda, n_w)$ and output the following common reference string:¹⁸

$$\text{crs} = (\text{pp}^{(0)}, \text{pp}^{(1)}, \dots, \text{pp}^{(\lambda)}).$$

- $\text{R}_1(\text{crs}, S_R)$: Form and sort the padded internal set $\widehat{S}_R = \{\widehat{\text{id}}_1, \dots, \widehat{\text{id}}_M\}$ as above, where $M = 2^d$. For each $i \in [M]$ sample $x_i, x'_i \leftarrow \{0, 1\}^\lambda$ and set the leaf values $h_i^{(0)} = \text{Hash}(\text{pp}^{(0)}, (\widehat{\text{id}}_i, x_i, x'_i))$.¹⁹ For $w \in [d]$ and $j \in [2^{d-w}]$, set

$$h_j^{(w)} = \text{Hash}(\text{pp}^{(w)}, (h_{2j-1}^{(w-1)}, h_{2j}^{(w-1)}, \text{id}_{[j,w]})),$$

¹⁸ We use one independent hash key per tree level. Note that Definitions 11 and 12 are selective (the adversary announces the hash input before seeing pp), whereas the hash inputs at level w of the tree are themselves hash values, and thus depend on the hash keys. With per-level keys, the level- w input $\mathbf{z}_w$ depends only on $\text{pp}^{(0)}, \dots, \text{pp}^{(w-1)}$, so a reduction can sample those keys itself, announce $\mathbf{z}_w$, and only then receive the challenge key $\text{pp}^{(w)}$, exactly as required by the selective definitions. The same device appears in the tree-based IBE of [27].

¹⁹ The leaf randomness (x_i, x'_i) is inherited from [7], where it feeds the robustness-based receiver-privacy argument. We make no receiver-privacy claim and retain it only for structural fidelity; correctness and sender security hold with or without it.

where $\text{id}_{[j,w]}$ is the largest internal identity under the left child of the j -th node at level w (i.e., $\text{id}_{[j,w]} = \widehat{\text{id}}_f$ with $f = (2j-1)2^{w-1}$). Output

$$\text{psi}_1 = (d, h_{\text{root}}),$$

where:

- $h_{\text{root}} = h_1^{(d)}$, and
 - the state st consists of the original set S_R , the padded list $\widehat{S}_R$, the (x_i, x'_i) , and all node values.
- $S(\text{crs}, \text{id}, \text{psi}_1)$: Set $\widehat{\text{id}} = 0 \parallel \text{id}$ and parse $\text{psi}_1 = (d, h_{\text{root}})$. If $\text{ValidDigest}(\text{pp}^{(d)}, h_{\text{root}}) = 0$, output $\perp$. Sample $r \leftarrow \{0, 1\}^\lambda$ and $r' \leftarrow \{0, 1\}^\lambda \setminus \{r\}$, and garble

$$(\widetilde{C}_0, \{\text{lb}_{i,b}^{(0)}\}) \leftarrow \text{Garb}(F[\widehat{\text{id}}, r, r']).$$

For each $w \in [d]$, do the following:

1. Sample encryption randomness $\rho_w = \{\rho_{w,i,t}\}$ for input length n_{w-1} and publish the offline components

$$\text{off}_w = \left\{ \text{Enc}^{\text{off}}(\text{pp}^{(w-1)}, i; \rho_{w,i,t}) \right\}_{i \in [n_{w-1}], t \in [\ell_b]}.$$

2. Garble $(\widetilde{C}_w, \{\text{lb}_{i,b}^{(w)}\}) \leftarrow \text{Garb}(V[\text{pp}^{(w-1)}, \widehat{\text{id}}, \{\text{lb}_{i,b}^{(w-1)}\}, \rho_w])$.
Compute the top-level (full) ciphertexts as

$$\text{ct}^{\text{top}} \leftarrow \text{HEnc}(\text{pp}^{(d)}, h_{\text{root}}, \{\text{lb}_{i,b}^{(d)}\})$$

where

$$\begin{aligned} & \text{HEnc}(\text{pp}^{(d)}, h_{\text{root}}, \{\text{lb}_{i,b}^{(d)}\}) \\ &= \left\{ \text{ct}_{i,t}^{(d)} \leftarrow \text{BE.Enc}(\text{pp}^{(d)}, (h_{\text{root}}, i), (\text{lb}_{i,0}^{(d)}[t], \text{lb}_{i,1}^{(d)}[t])) \right\}_{i \in [n], t \in [\ell_b]}. \end{aligned}$$

Finally, output the second message:

$$\text{psi}_2 = (\widetilde{C}_0, \dots, \widetilde{C}_d, \text{off}_1, \dots, \text{off}_d, \text{ct}^{\text{top}}, r).$$

- $R_2(\text{crs}, \text{st}, \text{psi}_2)$: For each $i \in [M]$, let $\text{pth}_i = (\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_d)$ be the leaf-to-root path for leaf i , where:
- $\mathbf{z}_0 = (\widehat{\text{id}}_i, x_i, x'_i)$ and,
 - for $w \geq 1$, $\mathbf{z}_w = (h_i^{(w-1)}, h_{i'}^{(w-1)}, \text{id}_{[\cdot, w]})$ is the preimage of the level- w node on the path.

Next, run a procedure $\text{DecPath}(\text{pth}_i, \text{psi}_2)$ that operates as follows:

1. Decrypt ct^{top} with $(\mathbf{z}_d)$ to obtain labels $\{\text{lb}_{i, \mathbf{z}_d[i]}^{(d)}\}$.
2. Then, for $w = d, d-1, \dots, 1$: evaluate $\Sigma_w = \text{GCEval}(\widetilde{C}_w, \cdot)$ on the current labels, reassemble the full level- w ciphertexts from (off_w, Σ_w) as in [Lemma 8](#), and decrypt them with $\mathbf{z}_{w-1}$ to obtain the next labels.

3. Finally, set $r_i = \text{GCEval}(\tilde{C}_0, \cdot)$ on the level-0 labels. If there is a unique index i such that $r_i = r$ and $\hat{\text{id}}_i = 0 \parallel \text{id}_i$ is real-tagged, output $\{\text{id}_i\}$; otherwise, output $\emptyset$.

Lemma 9 (Wrong-Path Soundness). *Instantiate BE with our blind BE construction from Section 6 that satisfies BE security (Definition 11) under the CD-HwMT assumption (Definition 15). Also, instantiate $(\text{Garb}, \text{GCEval}, \text{Sim}_{\text{GC}})$ with the point-and-permute garbling scheme as in Lemma 7, with domain-separated PRF inputs and the diffusion compilation described in Section 7.2. For an honestly generated tree and sender message, every leaf other than the one selected by the comparisons in the V circuits satisfies $\Pr[r_i = r] \leq 2^{-\lambda} + \text{negl}(\lambda)$. Consequently, no non-routed leaf passes the check except with negligible probability when $M = \text{poly}(\lambda)$.*

Proof. If $d = 0$, the sole leaf is routed and there is nothing to prove. Assume $d \geq 1$. Fix a non-routed leaf and scan the routed and non-routed paths from the root downwards. At their first divergence, the circuit at some level $q \in [d]$ produces ciphertexts targeted at $h = \text{Hash}(\text{pp}^{(q-1)}, \mathbf{z})$, where $\mathbf{z}$ is the preimage of the routed child, while the non-routed path next decrypts them with a distinct preimage $\mathbf{z}'$. Let

$$J = \{j : \mathbf{z}[j] \neq \mathbf{z}'[j]\}.$$

The identity or separator fields of $\mathbf{z}$ and $\mathbf{z}'$ are distinct, so J contains a coordinate a in such a field. By the per-level-key convention, both strings are fixed before $\text{pp}^{(q-1)}$ is sampled.

Except with negligible probability, $\text{Hash}(\text{pp}^{(q-1)}, \mathbf{z}) \neq \text{Hash}(\text{pp}^{(q-1)}, \mathbf{z}')$. Otherwise BE security is violated as follows. Commit selectively to $(\mathbf{z}, a)$ and, after receiving the key, publicly test whether the two hashes collide. Choose two message pairs that agree in row $\mathbf{z}[a]$ and differ in row $1 - \mathbf{z}[a]$. Note that the collision event $\text{Hash}(\text{pp}^{(q-1)}, \mathbf{z}) = \text{Hash}(\text{pp}^{(q-1)}, \mathbf{z}')$ depends only on $(\text{pp}^{(q-1)}, \mathbf{z}, \mathbf{z}')$ and is therefore independent of the challenge bit β . Conditioned on this event, decrypting the challenge ciphertext with $(\mathbf{z}', a)$ recovers $\mu_{\beta, \mathbf{z}'[a]} = \mu_{\beta, 1 - \mathbf{z}[a]}$, which reveals β since the two challenge messages differ in this row. Off the collision event, output a fair coin. Thus, a non-negligible collision probability gives a non-negligible BE distinguishing advantage.

Starting from the real experiment, proceed down the routed path from level d through level q . At each level, first replace the current garbling by its simulation on the routed output, and then use BE security position-by-position to replace every complementary plaintext row by an independent uniform bit. The standard hybrid argument used later in Lemma 10 shows that this changes the distribution only negligibly. In other words, this partial hybrid is indistinguishable from the real experiment and leaves $\tilde{C}_{q-1}, \dots, \tilde{C}_0$ honestly garbled.

Write C_{q-1} for the ungarbled circuit underlying $\tilde{C}_{q-1}$. At the divergence, for every $j \in J$ and $t \in [\ell_{\text{ib}}]$, keep row $\mathbf{z}[j]$ unchanged and replace row $\mathbf{z}'[j] = 1 - \mathbf{z}[j]$ by an independent $u_{j,t} \leftarrow \{0, 1\}$. For every coordinate set $c_j = \mathbf{z}'[j]$ and define

$$\Delta_j[t] = \left\langle \alpha_{j,t,c_j}, j(F_{h,c_j}) \oplus j(F_{c_j}^{\mathbf{z}'}) \right\rangle,$$

where F_{h,c_j} is the pushforward curve used by encryption and $F_{c_j}^{\mathbf{z}'}$ is the curve reconstructed by decryption with $\mathbf{z}'$; dependence of these curves on (j, t) is suppressed. The explicit Goldreich–Levin rule gives

$$\widehat{\text{lb}}_j[t] = \begin{cases} \text{lb}_{j,\mathbf{z}[j]}^{(q-1)}[t] \oplus \Delta_j[t], & j \notin J, \\ u_{j,t} \oplus \Delta_j[t], & j \in J. \end{cases}$$

Thus the labels on J are mutually independent uniform strings, while every other input is an honest label xor a mask fixed independently of the current garbling. This is the point at which the explicit Goldreich–Levin form of our BE, rather than generic BE security alone, is used.

Let Z_{q-1} contain crs , r , h , $\mathbf{z}$, $\mathbf{z}'$, the tree data needed below level $q-1$, the offline components $\text{off}_1, \dots, \text{off}_q$ and their sampling coins (and hence the masks Δ_j), the ungarbled description of C_{q-1} , and all garblings and data strictly below level $q-1$. It excludes the fresh bits $u_{j,t}$, $\tilde{C}_{q-1}$, its wire labels and garbling coins, and every other upper-level value already consumed. All included quantities are independent of the fresh garbling coins of C_{q-1} and may therefore be sampled first. With $H = [n_{q-1}] \setminus J$, $b_j = \mathbf{z}[j]$, and $\delta_j = \Delta_j$ for $j \in H$, the input labels just described satisfy the hypotheses of Lemma 7, and wire $a \in J$ reaches every output through the fresh terminal gadgets of the diffusion compilation. Write Σ_{q-1} for this evaluation output (so $\Sigma_0 = r_i$). If ℓ_{q-1}^{out} denotes λ when $q = 1$ and $2n_{q-2}\ell_{\text{lb}}$ when $q \geq 2$, that lemma gives

$$(Z_{q-1}, \Sigma_{q-1}) \approx_c (Z_{q-1}, U_{\ell_{q-1}^{\text{out}}}).$$

Once Σ_{q-1} has been computed, neither the current garbling nor the upper transcript is needed to determine the remainder of this leaf evaluation, so marginalising them is legitimate.

If $q = 1$, then $\Sigma_0 = r_i$, and the last display already gives $(r, r_i) \approx_c (r, U_\lambda)$. If $q \geq 2$, reassembly and Goldreich–Levin decryption at the next level select one independent uniform online bit per ciphertext and xor deterministic pads determined by the lower-level state. Consequently, the complete input-label vector supplied to $\tilde{C}_{q-2}$ is jointly uniform and independent even of that lower garbling. For the next application, form Z_{q-2} by discarding the consumed upper data and the current garbling $\tilde{C}_{q-2}$ together with its labels and coins, while retaining its ungarbled description and every strictly lower-level value. Conditional on this state, $\tilde{C}_{q-2}$ is a fresh garbling and its entire supplied input is uniform. We may therefore apply Lemma 7 with the partition reset to $H = \emptyset$ and $J = [n_{q-2}]$, using any identity or separator coordinate as a . Repeating this marginal argument down to level 0 yields $(r, r_i) \approx_c (r, U_\lambda)$. Hence

$$\Pr[r_i = r] \leq 2^{-\lambda} + \text{negl}(\lambda).$$

There are at most $M = \text{poly}(\lambda)$ non-routed leaves; a union bound over these non-routed leaves therefore shows that no non-routed leaf passes the check except with negligible probability.

Theorem 13 (Correctness and Efficiency). *Instantiate BE with our blind BE construction from Section 6 that satisfies BE security (Definition 11) under the CDHwMT assumption (Definition 15). Also, instantiate (Garb, GCEval, Sim_{GC}) with the point-and-permute garbling scheme as in Lemma 7, with domain-separated PRF inputs and the diffusion compilation described in Section 7.2. Then the above construction satisfies correctness and laconic efficiency as in Definition 16.*

Proof. We prove correctness and efficiency separately below.

Correctness. Suppose $\text{id} \in S_R$, say $\widehat{\text{id}} = \widehat{\text{id}}_{i^*} = 0\|\text{id}$. By construction, z_d is a preimage of h_{root} under $\text{pp}^{(d)}$. Hence, BE correctness applied to ct_{top} yields the labels $\left\{ \text{lb}_{i, z_d[i]}^{(d)} \right\}_i$ required to correctly evaluate $\widetilde{C}_d$. Additionally, every child digest on the routed path is honestly generated and hence satisfies $\text{ValidDigest} = 1$, so V evaluates its encryption branch at every level.

Now, along pth_{i^*} , at every level w the comparison inside V routes the encryption towards the child containing $\widehat{\text{id}}$: by definition of $\text{id}_{[j,w]}$, $\widehat{\text{id}} \leq \text{id}_{[j,w]}$ exactly when $\widehat{\text{id}}$ lies under the left child. Hence the online components produced by evaluating $\widetilde{C}_w$ on the path labels target precisely the hash value $h^{(w-1)}$ on pth_{i^*} ; by Lemma 8 the reassembled ciphertexts equal honestly generated BE.Enc ciphertexts under that hash value with the sender's randomness ρ_w , so Theorem 9 and garbling correctness yield the labels of the next level, and finally $r_{i^*} = F[\widehat{\text{id}}, r, r'](\widehat{\text{id}}_{i^*}, x_{i^*}, x'_{i^*}) = r$. By Lemma 9, no other leaf produces r except with negligible probability, so R_2 removes the tag and outputs $\{\text{id}\}$.

If $\text{id} \notin S_R$, the comparison gates still select one routed leaf in the padded internal list. The correctly evaluated routed path ends in $F[\widehat{\text{id}}, r, r'](\widehat{\text{id}}_j, x_j, x'_j) = r' \neq r$, whether that routed leaf is real or dummy. Every other path is non-routed and hence fails the check except with negligible probability by Lemma 9. Thus R_2 outputs $\emptyset$ except with negligible probability. This completes the proof of correctness.

Efficiency. Note that $|\text{psi}_1| = \ell_h + O(\log \lambda)$. Each garbled circuit has size $\text{poly}(\lambda)$ (the circuits F and V have size $\text{poly}(\lambda)$, the latter by the efficiency of Enc^{on}), each off_w has size $\text{poly}(\lambda)$, and there are $d+1 = 1 + \log M$ levels. Hence there is a fixed polynomial p_{psi} , independent of M , for which $|\text{psi}_2| \leq p_{\text{psi}}(\lambda)(1 + \log M)$, and S runs in time $\text{poly}(\lambda, 1 + \log M)$. (R_2 runs in time $M \cdot \text{poly}(\lambda, 1 + \log M)$, which is permitted.)

Theorem 14 (Sender Security). *Instantiate BE with our blind BE construction from Section 6 that satisfies BE security (Definition 11) under the CDHwMT assumption (Definition 15). Also, instantiate (Garb, GCEval, Sim_{GC}) with the point-and-permute garbling scheme satisfying Definition 17. Then the above LPSI construction satisfies sender security against semi-honest receivers as in Definition 16.*

Proof. Fix an external set S_R and an identity $\text{id} \notin S_R$, and apply the padding described above. Let $\widehat{S}_R = \{\widehat{\text{id}}_1, \dots, \widehat{\text{id}}_M\}$ denote the sorted padded internal list,

where $\widehat{\text{id}} = 0\|\text{id}$ is the real-tagged sender input and $M = 2^d$. The comparisons in the circuits V route $\widehat{\text{id}}$ to the leaf

$$\widehat{\text{id}}_j = \min\{\widehat{\text{id}}_k : \widehat{\text{id}} \leq \widehat{\text{id}}_k\},$$

if this set is nonempty, and to $\widehat{\text{id}}_M$ otherwise. Let $\text{pth} = (z_0, z_1, \dots, z_d)$ denote the corresponding leaf-to-root path, and define $h_w = \text{Hash}(\text{pp}^{(w)}, z_w)$ for $w \in [0, d]$, so that $h_d = h_{\text{root}}$. Observe that (crs, ρ_R) determines the entire path pth . Moreover, for every $w \in [0, d]$, the preimage z_w depends only on $(\text{pp}^{(0)}, \dots, \text{pp}^{(w-1)})$ (together with S_R and ρ_R), but not on $\text{pp}^{(w)}$. This ordering of dependencies will be important when invoking the security of BE below.

All hybrids below describe the distribution of the sender message ψ_2 ; the pair (crs, ρ_R) is common to all experiments. Our proof proceeds in two stages. First, we replace the real sender message by a distribution Hyb_1 in which the garbled circuits along the active path are simulated and, for every BE encryption, the plaintext row selected by the active-path preimage is preserved while the complementary row is replaced by fresh randomness. We then remove the remaining dependence on the active path and obtain a fixed distribution that is independent of id .

We write $\text{topo}(F)$ and $\text{topo}(V)$ for the fixed topologies of the corresponding circuit families. In particular, hardwiring the sender input, wire labels, public parameters, or encryption randomness does not alter these topologies.

Hyb_0 . This is the real distribution of $S(\text{crs}, \text{id}, \psi_1)$.

Hyb_1 . We define this distribution bottom-up, since the simulated output at level $w-1$ determines the selected plaintext row encrypted at level w . First sample $r \leftarrow \{0, 1\}^\lambda$ and $r' \leftarrow \{0, 1\}^\lambda \setminus \{r\}$, and set

$$(\tilde{C}_0, \{\text{lb}_i^{(0)}\}_i) \leftarrow \text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(F), r').$$

For every $w \in [d]$, sample the encryption randomness ρ_w and publish off_w exactly as in the real construction. For every position (i, t) , let $b_{w-1,i} = z_{w-1}[i]$. Set the plaintext row selected by the active-path preimage to the corresponding simulated wire-label bit $m_{w,i,t,b_{w-1,i}} = \text{lb}_i^{(w-1)}[t]$, and sample the complementary row independently and uniformly, $m_{w,i,t,1-b_{w-1,i}} \leftarrow \{0, 1\}$. Then compute

$$\Sigma_w = \left\{ \text{Enc}_{\text{on}} \left(\text{pp}^{(w-1)}, h_{w-1}, i, (m_{w,i,t,0}, m_{w,i,t,1}); \rho_{w,i,t} \right) \right\}_{i,t},$$

and sample

$$(\tilde{C}_w, \{\text{lb}_i^{(w)}\}_i) \leftarrow \text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(V), \Sigma_w).$$

Finally, for every top-level position (i, t) , encrypt the selected label bit $\text{lb}_i^{(d)}[t]$ in row $z_d[i]$, and an independent uniform bit in the complementary row. Denote the resulting collection of top-level ciphertexts by ct_{top} , and output

$$\psi_2 = (\tilde{C}_0, \dots, \tilde{C}_d, \text{off}_1, \dots, \text{off}_d, \text{ct}_{\text{top}}, r).$$

We refer to the above replacement as a *fresh-complement encryption*: the plaintext row selected by the active-path preimage is left unchanged, while the complementary row is replaced by an independent uniform bit. Crucially, once the selected label itself becomes uniform, the two plaintext rows of every fresh-complement encryption are independent uniform bits. This property will allow us to invoke decomposable blindness in [Lemma 11](#).

The following lemma shows that the above transformation from the real sender message to Hyb_1 is computationally indistinguishable.

Lemma 10. *Instantiate BE with our blind BE construction from [Section 6](#) that satisfies BE security ([Definition 11](#)) under the CDHwMT assumption ([Definition 15](#)). Also, instantiate $(\text{Garb}, \text{GCEval}, \text{Sim}_{\text{GC}})$ with the point-and-permute garbling scheme satisfying [Definition 17](#). Then, $\text{Hyb}_0 \approx_c \text{Hyb}_1$.*

Proof. This is the descending argument of [[7](#), Lemmas 6, 8 and 9], with an important ordering detail made explicit. A BE challenge cannot be embedded into values hardwired inside an honestly garbled circuit. At each internal level we therefore first simulate the garbling on its real active-path output and then switch the complementary plaintext rows in that explicit simulator input.

Top fresh-complement switch. Begin with ct^{top} . For every position (i, t) , keep the row selected by $\mathbf{z}_d[i]$ equal to $\text{lb}_{i, \mathbf{z}_d[i]}^{(d)}[t]$ and replace the other row by an independent uniform bit. A hybrid over the $n_d \ell_{\text{lb}}$ positions reduces this change to [Definition 11](#). Indeed, the reduction samples every key except $\text{pp}^{(d)}$, computes $\mathbf{z}_d$ before receiving $\text{pp}^{(d)}$, announces $(\mathbf{z}_d, i)$ to its selective challenger, and embeds the returned ciphertext at (i, t) . The selected plaintext row is unchanged, as required by the game. This notation also covers $d = 0$, when $n_d = n_0$ is the leaf-input length.

For some level $q \leq d$, throughout the descending hybrid, after treating levels $d, \dots, q$, the active-path labels entering level $(q - 1)$ remain exactly unchanged, since every BE-security switch modifies only the complementary plaintext row. Thus these labels can be used in the next application of garbling simulation security. Concretely, let

$$C_q = V[\text{pp}^{(q-1)}, \widehat{\text{id}}, \{\text{lb}_{i,b}^{(q-1)}\}, \rho_q]$$

be the honestly hardwired level- q circuit. At this point only the labels $\{\text{lb}_{i, \mathbf{z}_q[i]}^{(q)}\}$ selected by the active preimage are needed above level q . Garbling simulation security therefore replaces

$$(\tilde{C}_q, \{\text{lb}_{i, \mathbf{z}_q[i]}^{(q)}\}) \quad \text{by} \quad \text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(C_q), C_q(\mathbf{z}_q)).$$

By the routing comparison, $C_q(\mathbf{z}_q)$ is the tuple of online components targeting h_{q-1} and encrypting the full label pairs $\{\text{lb}_{i,0}^{(q-1)}, \text{lb}_{i,1}^{(q-1)}\}$. Now, and only now, apply BE security position by position to replace each complementary row by a fresh uniform bit while keeping the row selected by $\mathbf{z}_{q-1}$ fixed.

The reduction embeds the challenge key as $\text{pp}^{(q-1)}$, the selective input as $\mathbf{z}_{q-1}$, the challenge offline component in off_q , and its online component in the

input to Sim_{GC} . Although the challenge encryption randomness is unknown to the reduction, this causes no difficulty: by [Lemma 8](#) the returned BE ciphertext can be parsed syntactically into its offline and online components, which are inserted respectively into off_q and the explicit output string supplied to Sim_{GC} .

Moreover, $\mathbf{z}_{q-1}$ depends only on lower-level keys and is fixed before the challenger supplies $\text{pp}^{(q-1)}$. This produces exactly the level- q distribution specified in Hyb_1 and exposes only the selected labels needed for the next descending step.

Base circuit. After level 1 has been treated, simulation security replaces the honestly garbled pair $(\tilde{C}_0, \{\text{lb}_{i, \mathbf{z}_0[i]}^{(0)}\})$ by

$$\text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(F), F[\widehat{\text{id}}, r, r'](\mathbf{z}_0)) = \text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(F), r'),$$

because the routed leaf identity differs from id . The case $d = 0$ consists only of the top fresh-complement switch followed by this base-circuit switch. There are polynomially many positions and $O(d)$ levels, so the accumulated loss is negligible; [Theorem 11](#) reduces all BE steps to CDHwMT.

Observe that Hyb_1 still depends on id through the choice of the active path (in the general case, through the rank of id in S_R). The remaining hybrids remove this dependence. This is where our argument departs from [\[7\]](#): rather than proving psi_2 pseudorandom, we prove it indistinguishable from a fixed distribution.

Hyb₂: Sample psi_2 from the following distribution $\mathcal{D}^*$, which is samplable from (crs, d) : $r \leftarrow \{0, 1\}^\lambda$; each $\tilde{C}_w$ is a uniformly random string of the appropriate length; each off_w is generated honestly (its distribution depends only on $\text{pp}^{(w-1)}$); and the top-level ciphertexts are sampled as $(\text{Enc}^{\text{off}}(\text{pp}^{(d)}, i; \rho), u)$ with fresh ρ and independent uniform $u \leftarrow \{0, 1\}^2$ per position. The value d is part of psi_1 and is determined by the receiver's input and coins.

Lemma 11. *Instantiate BE with our blind BE construction from [Section 6](#) that satisfies [Lemma 8](#). Also, instantiate $(\text{Garb}, \text{GCEval}, \text{Sim}_{\text{GC}})$ with the point-and-permute garbling scheme satisfying [Definition 17](#). Then, $\text{Hyb}_1 \approx_c \text{Hyb}_2$. Moreover Hyb_2 is independent of id .*

Proof. We replace components bottom-up, maintaining the invariant that after step w , everything at levels $\leq w$ is (indistinguishable from) uniform or offline-only. In each step below, all components above the level currently being replaced are viewed as efficient randomized post-processing and are regenerated from the modified intermediate values; hence statistical identity or computational indistinguishability of the intermediate distributions is preserved for the complete transcript.

Step 0. In Hyb_1 , $(\tilde{C}_0, \{\text{lb}_i^{(0)}\}) = \text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(F), r')$, where, conditioned on the public value r , the value r' is uniform on $\{0, 1\}^\lambda \setminus \{r\}$. Replacing r' by an independent uniform λ -bit string changes the distribution by statistical distance exactly $2^{-\lambda}$. By [Definition 17](#), the resulting simulator output is computationally

indistinguishable from a uniform string. In particular, after this replacement, the labels $\{\text{lb}_i^{(0)}\}$ are mutually uniform and independent of both $\tilde{C}_0$ as well as of all state fixed before this replacement. They are not themselves exposed in the final transcript; their only subsequent use is as plaintext inputs to the computation of Σ_1 , from which the remaining higher-level components can be regenerated as efficient post-processing. The distinguisher for this step regenerates all higher levels itself, which it can, knowing crs , ρ_R , S_R and the sender randomness of levels ≥ 1 .

Step w , online part ($w \in [d]$). At this point the selected labels $\{\text{lb}_i^{(w-1)}\}$ encrypted in Σ_w are mutually uniform and independent of all state generated prior to level w ; their subsequent influence on the transcript occurs only through Σ_w and the higher-level components generated from it. Their complementary rows were sampled independently and uniformly in the fresh-complement switch defining Hyb_1 . Thus the two plaintext rows at every position (i, t) are jointly independent uniform bits. By Lemma 8, applied per position (with independent messages and randomness, so the per-position replacements compose statistically), the joint distribution (off_w, Σ_w) is identical to (off_w, u_w) with u_w uniform. Note that this step is information-theoretic and holds for the arbitrary (here: honest) well-formed target h_{w-1} ; in particular it does not depend on which node of level $w-1$ the active path visits, and this is precisely the point at which the dependence of Hyb_1 on the path is erased, level by level.

Step w , garbling part. Now $(\tilde{C}_w, \{\text{lb}_i^{(w)}\}) = \text{Sim}_{\text{GC}}(1^\lambda, \text{topo}(V), \Sigma_w)$ where its input Σ_w has just been replaced by a uniform string. By Definition 17, the pair is indistinguishable from uniform, and we continue upward.

Top level. After step d , the selected labels $\{\text{lb}_i^{(d)}\}$ are uniform and unknown, while the complementary rows in the fresh-complement ciphertexts are independent uniform bits. Hence Lemma 8, applied to the full ciphertexts ct^{top} (a full ciphertext is the reassembly of its offline and online components), shows that ct^{top} is distributed identically to fresh offline components together with uniform bits, matching $\mathcal{D}^*$.

Each replacement is either statistical (Lemma 8) or reduces to Definition 17 with a reduction that samples every other component itself; there are $O(d) = O(\log M)$ steps. Finally, $\mathcal{D}^*$ makes no reference to id or to the active path, so Hyb_2 is independent of id .

The simulator $\text{Sim}_{\ell\text{PSI}}$ of Definition 16 obtains d from (S_R, ρ_R) and outputs a sample of $\mathcal{D}^*$ from (crs, d) , without reference to id . Combining Lemmas 10 and 11 completes the proof of Theorem 14.

Remark 4 (Sender-security vs Pseudorandomness). Our sender-security guarantee is weaker than the pseudorandomness property established in [7, Theorem 1]. In particular, we do not claim that the sender message ψ_2 is indistinguishable from a uniformly random bit string: the offline components off_w , as well as the structured components of ct^{top} , contain elliptic curves and torsion points and are therefore efficiently distinguishable from uniform strings. Instead, Theorem 14

shows that ψ_2 is computationally indistinguishable from a fixed, efficiently sampleable distribution $\mathcal{D}^*$ that is independent of the sender’s input id. This is precisely the simulation-based sender-security guarantee required by our half-PIR construction below, where the LPSI receiver is the PIR server and already knows its own set. Consequently, the stronger pseudorandomness guarantee of [7] is unnecessary for our application. On the other hand, applications of [7] that inherently rely on pseudorandomness of the entire sender message, such as the stronger anonymity notion for their self-detecting encryption, do not follow from our construction without additional techniques.

Remark 5 (Need for Decomposable Blindness). The role of the decomposition established in Lemma 8 is now apparent. In the sender-security hybrids above, the outputs supplied to the GC simulator consist only of the online components of BE ciphertexts. Once the two plaintext rows are replaced by independent uniform bits, Lemma 8 guarantees that these online components are themselves uniformly distributed, allowing us to invoke the blindness property of Definition 17. This argument would fail if the garbled circuits instead output complete BE ciphertexts: even for uniformly random plaintexts, such ciphertexts contain structured objects (in particular, elliptic curves and torsion points) and therefore cannot be uniformly distributed bit strings. Thus, separating the structured, plaintext-independent offline components from the short online components is precisely what makes the blind-garbling hybrid, and hence our sender-security proof, go through.

7.5 Bootstrapping to Half-PIR and PIR

Let $(\text{GenCRS}, R_1, S, R_2)$ be the construction of LPSI with server security from the CDHwMT assumption, as described in Section 7.4. In this section, we show that it can be bootstrapped to half-PIR by the same transformation as in [25, Section 5.2].

Overview. We first informally recall the transformation. For every database position j , the server associates two pseudorandom values with the possibilities $z_j = 0$ and $z_j = 1$, and includes in its LPSI set only the value corresponding to the actual database bit z_j . To query position i , the client chooses a uniform bit b and uses the corresponding pseudorandom value as its LPSI input. If $b = z_i$, this value belongs to the server’s set; otherwise, it does not, allowing the client to recover $z_i = 1 - b$. The latter event occurs with probability $1/2$ and, crucially, sender security of LPSI hides the client’s input whenever it is not contained in the server’s set. Thus, a non-membership outcome simultaneously allows the client to recover the desired bit and hides its query from the server.

The above transformation only uses the correctness, efficiency, and sender-security properties established in Theorems 13 and 14. In particular, it does not require receiver security or pseudorandomness of the LPSI transcript. We now formalize the transformation below for the sake of completeness.

Half-PIR from LPSI. Let $\text{PRF} : \mathcal{K} \times ([M] \times \{0, 1\}) \rightarrow \{0, 1\}^\lambda$ be a pseudorandom function (which exists from any one-way function [58, 52], and hence, from the CDHwMT assumption).²⁰ The construction is as follows:

- **Setup:** Set $\text{crs} = (\text{crs}_{\text{psi}}, K)$ where $\text{crs}_{\text{psi}} \leftarrow \text{GenCRS}(1^\lambda)$ and $K \leftarrow \mathcal{K}$.
- **Server encodes the database:** On input a database $z \in \{0, 1\}^M$, the server computes the set $S_R = \{\text{PRF}(K, (j, z_j))\}_{j \in [M]}$ with duplicate values removed, runs $(\text{psi}_1, \text{st}) \leftarrow R_1(\text{crs}_{\text{psi}}, S_R)$, and sends psi_1 to the client.
- **Client queries a random candidate bit:** On input an index $\iota \in [M]$, the client samples $b \leftarrow \{0, 1\}$, sets $\text{el} = \text{PRF}(K, (\iota, b))$, and sends $\text{psi}_2 \leftarrow S(\text{crs}_{\text{psi}}, \text{el}, \text{psi}_1)$ to the server.
- **Server tests membership:** The server computes $I \leftarrow R_2(\text{crs}_{\text{psi}}, \text{st}, \text{psi}_2)$ and sends $\beta = |I| \in \{0, 1\}$ to the client.
- **Client determines success and outputs:** if $\beta = 0$, the client outputs $(1 - b, \text{success})$; if $\beta = 1$, the client outputs (b, fail) .

Theorem 15 (Half-PIR from CDHwMT). *Assuming that the CDHwMT assumption (Definition 15) holds, the above construction is a half-PIR scheme as per Definition 5, with total communication at most $p_{\text{half}}(\lambda)(1 + \log M)$ for a fixed polynomial p_{half} independent of M .*

Proof. We argue correctness with noticeable success, correctness on success, and client privacy on success, and then present the communication overhead analysis.

Correctness with Noticeable Success. Fix a database z and a query index ι . Consider first the event $b \neq z_\iota$, which occurs with probability $1/2$. Since the input $(\iota, b) = (\iota, 1 - z_\iota)$ is distinct from every input (j, z_j) used to construct S_R , a standard PRF-to-random-function hybrid followed by a union bound gives

$$\Pr [\text{PRF}(K, (\iota, 1 - z_\iota)) \in S_R] \leq M/2^\lambda + \text{negl}(\lambda) = \text{negl}(\lambda).$$

Hence, except with negligible probability, $\text{el} \notin S_R$, in which case, by LPSI correctness, the server then returns $\beta = 0$, and the client outputs

$$(1 - b, \text{success}) = (z_\iota, \text{success}).$$

Since $b \neq z_\iota$ with probability $1/2$, the client therefore outputs the correct database bit with flag **success** with probability at least $1/2 - \text{negl}(\lambda)$.

For completeness, if $b = z_\iota$, then $\text{el} \in S_R$ by construction, and LPSI correctness causes the client to output $(b, \text{fail}) = (z_\iota, \text{fail})$, except with negligible probability.

²⁰ Following [25], the PRF maps database positions into the λ -bit element space of the LPSI. We do not condition on all $2M$ outputs being distinct; correctness only needs the client's unused value to avoid the at most M values in the server's set. The LPSI layer internally encodes every such real PRF value as $0 \parallel \text{PRF}(\cdot)$, while its padding values carry tag 1.

Correctness on Success. It remains to rule out an incorrect output carrying the **success** flag. Such an event can occur only when $b = z_\iota$ and the server nevertheless returns $\beta = 0$: indeed, whenever $b \neq z_\iota$, the value output upon success is $1 - b = z_\iota$. When $b = z_\iota$, however, we have $\text{el} = \text{PRF}(K, (\iota, z_\iota)) \in S_R$ by construction, and LPSI correctness therefore gives $\beta = 1$ except with negligible probability. Consequently,

$$\Pr[v \neq z_\iota \wedge \text{flag} = \text{success}] \leq \text{negl}(\lambda).$$

Client Privacy on Success. The key observation is that, up to negligible probability, success occurs precisely in the branch $b \neq z_\iota$. Indeed, the client declares success exactly when the server sends $\beta = 0$. Conditioned on $b \neq z_\iota$, the collision bound above implies $\text{el} \notin S_R$, and LPSI correctness then gives $\beta = 0$ except with negligible probability. Thus, conditioned on success, the client's LPSI input is a non-member of S_R , except with negligible probability.

We may therefore invoke the sender-security guarantee of [Theorem 14](#). For any two query indices ι_0, ι_1 , the corresponding server views satisfy the following:

$$\begin{aligned} (\text{crs}, \rho_R, \text{psi}_2(\iota_0)) &\approx_c (\text{crs}, \rho_R, \text{Sim}_{\ell\text{PSI}}(1^\lambda, \text{crs}_{\text{psi}}, \rho_R, S_R)) \\ &\approx_c (\text{crs}, \rho_R, \text{psi}_2(\iota_1)), \end{aligned}$$

where all three distributions are conditioned on success. Further, observe that the distribution $(\text{crs}, \rho_R, \text{Sim}_{\ell\text{PSI}}(1^\lambda, \text{crs}_{\text{psi}}, \rho_R, S_R))$ depends on the server's set S_R , but not on the client's LPSI input, and hence not on ι or b . This establishes client privacy conditioned on success.

Next, observe that LPSI correctness makes $\beta = 0$ an overwhelming event for a nonmember input, and sender-security implies the same for the simulated distribution. Conditioning on this event therefore preserves computational indistinguishability. The server's final message β is fixed to 0 under the conditioning and reveals no additional information.

We emphasize that PRF security is *not* used to hide the client's query from the server, since K is part of the CRS and hence known to the server. It is used only to bound the probability that the client's nonmember candidate collides with an element of S_R .

Communication. By [Theorem 13](#), $|\text{psi}_1| + |\text{psi}_2| + 1 \leq p_{\text{half}}(\lambda)(1 + \log M)$ for a fixed polynomial p_{half} independent of M .

[Theorem 15](#) completes the main construction-specific step: our isogeny-based blind BE from [Section 6](#) yields a half-PIR protocol with communication $p_{\text{half}}(\lambda)(1 + \log M)$. We can now invoke the generic half-PIR-to-PIR transformation of [Theorem 1](#), which incurs an additional $O(\lambda(1 + \log M))$ factor in communication. Together with [Theorem 15](#) and [Theorem 8](#), this yields our final PIR result.

Corollary 5 (Single-Server PIR from CDHwMT). *Under the CDHwMT assumption ([Definition 15](#)), there is a fixed polynomial p , independent of M ,*

such that for every polynomially bounded database-size function $M = M(\lambda)$ there exists a semi-honest single-server PIR protocol as per [Definition 4](#) with total communication at most $p(\lambda)(1 + \log M)^2$.

Discussion. We conclude this section by discussing two limitations of our construction and the extent to which they are inherent to our current approach. First, unlike the LPSI construction of [7], our construction does not provide receiver privacy; we explain why the robustness property used to establish this guarantee appears incompatible with our isogeny-based hash function. Second, our formal results are restricted to semi-honest adversaries. We discuss what additional ingredients would be needed to strengthen sender security of LPSI against maliciously generated digests, and why this alone would still not suffice to obtain (single-server) PIR security against malicious servers.

On Receiver Privacy and Robustness. The receiver-privacy proof of [7] relies on a *robustness* property [7, Lemma 4]: whenever sufficiently many input bits are uniformly random, the hash output is statistically close to a uniform string. This property is applied recursively across the levels of the tree—the near-uniform outputs at one level provide the randomness needed at the next—and ultimately ensures that the receiver’s digest statistically hides its set S_R .

Our hash function cannot satisfy this robustness property. Indeed, its outputs $(E_{\mathbf{x}}, R_{\mathbf{x}}, S_{\mathbf{x}})$ lie in a sparse and efficiently recognizable algebraic subset of the ambient bit-string space, and hence are at statistical distance $1 - \text{negl}(\lambda)$ from uniform. This rules out the particular robustness-based argument of [7] unconditionally.

One could nevertheless ask for a weaker property tailored specifically to receiver privacy: whether, after randomizing the remaining input slots, the distribution of the hash output hides the identity occupying a designated slot. We do not prove that our hash fails to satisfy such a property. There is, however, heuristic evidence against it already at the leaves. The real-leaf inputs $(0 \parallel \text{id}, x, x')$ range over a domain of size $2^{3\lambda}$, whereas a birthday estimate in an output space of effective size approximately $2^{8\lambda}$ suggests that the leaf map should be close to injective. This is consistent with the collision-resistance analysis of [Section 4.4](#), where any collision yields a **OneEnd** solution of the constrained form $\psi \equiv [D] \bmod N$ with $\text{nrd}(\psi) = D^2$. Under this near-injectivity heuristic, different identities would induce essentially disjoint output supports, making the receiver’s set recoverable by exhaustive enumeration. We stress that this is only heuristic evidence and not an impossibility result.

Fortunately, receiver privacy is unnecessary for our PIR application. In the half-PIR construction, the LPSI receiver is the PIR server, and S_R is derived from the server’s own database; the PIR definition therefore places no requirement on hiding this set. Obtaining receiver-private LPSI from our hash would require a new decisional, per-level “prefix-hiding” assumption, where the non-identity slots are distributed according to the outputs of the preceding tree level rather than as uniform bits. Formulating and analyzing such an assumption is an interesting

direction for future work, but it is qualitatively stronger than CDHwMT and is not needed for any of the results in this paper.

Sender Security for Maliciously Formed Digests. Our proof of [Theorem 14](#) assumes a semi-honest receiver. In particular, the hybrids of [Lemma 10](#) follow an honestly generated Merkle path and invoke BE security relative to honestly generated hash values for which the corresponding preimages are known. This argument no longer applies if the receiver supplies an arbitrary psi_1^* .

Nevertheless, two features of our construction appear useful for extending sender security to this setting. First, well-formedness of a claimed digest is publicly checkable via `ValidDigest` (using [\[75\]](#)), and [Lemma 8](#) guarantees blindness for any well-formed target, even when no preimage of that target is known: the offline ciphertext components and Goldreich–Levin seeds are independent of the target. Second, if an adversary can successfully decrypt at some position under two distinct openings of the same hash value, it obtains a collision in `Hash`, which in turn yields a `OneEnd` solution by the reduction of [Section 4.4](#).

These observations suggest a possible route to sender security against maliciously formed digests under the additional `EndRing` assumption: rather than maintaining the garbled-circuit hybrids by following a known honest path, one could use collision resistance to rule out inconsistent openings. We leave a complete treatment to future work, however, and state all of our results in the semi-honest model.

Functionality-level Barrier to Malicious PIR Security. Even if the sender-security issue above were resolved, our half-PIR transformation would still not provide client privacy against a malicious server. This is a separate, functionality-level obstacle. For any index ι , a cheating server can include both $\text{PRF}(K, (\iota, 0))$ and $\text{PRF}(K, (\iota, 1))$ in its LPSI set. Regardless of the client’s random choice b , its element is then a member of the set, allowing the LPSI execution to reveal ι . The server can subsequently return $\beta = 1$, causing the client to declare failure; since half-PIR guarantees privacy only conditioned on success, its privacy requirement imposes no constraint on this execution.

Thus, upgrading [Corollary 5](#) to malicious-server security requires more than strengthening the underlying LPSI sender-security proof: one must additionally enforce that the server’s set contains exactly one value associated with each database index, and hence encodes a well-defined database. This could potentially be achieved by augmenting the protocol with an appropriate proof of well-formedness, but doing so is beyond the scope of this work. Our semi-honest formulation therefore matches both the functionality supported by the present construction and the security model of [\[25\]](#).

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